

PHYSICS GRADE 10: PROPERTIES OF WAVES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 2.0: Waves and Nuclear Physics
Sub-Strand	Sub-Strand 2.1: Properties of Waves
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Rectilinear propagation of light (shadows, pinhole camera, eclipses) – Reflection of waves (laws of reflection, echoes, plane and curved surfaces) – Refraction of waves (refractive index, Snell's law, apparent depth, lenses) – Diffraction of waves (sound and light around obstacles and through gaps) – Interference of waves (constructive and destructive, ripple tank and sound patterns) – Stationary (standing) waves and resonance (nodes, antinodes, strings and air columns) – Doppler effect (change in frequency due to relative motion) – Applications of waves in communication and medicine (radio and FM, mobile signals, ultrasound) – Noise pollution: sources, effects on health and methods of control
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe the reflection of waves and relate it to echoes in the environment, - investigate refraction of waves in different media and apply it to lenses and everyday observations, - explain diffraction of waves around obstacles and through gaps, - investigate interference of waves and the formation of interference patterns, - describe the formation of stationary waves and their relationship to resonance and musical pitch, - explain the Doppler effect and its applications, - relate wave properties to technologies in communication and medicine, - appreciate the need to reduce noise pollution in the environment. Key concepts: waves transfer energy without transferring matter; reflection, refraction, diffraction and interference are wave behaviours that can be predicted with models; frequency, wavelength, speed and amplitude link to what we hear and see; the Doppler shift depends on relative motion.
Core Competencies	– Critical thinking and problem solving: learners analyse evidence from simulations and experiments to explain echoes, bending, interference and frequency shifts. – Communication and collaboration: learners work in groups during investigations, share findings and defend explanations using evidence.

	– Digital literacy: learners use PhET simulations and online videos to investigate wave behaviour. – Learning to learn: learners revise their models and use the Driving Question Board to track their growing understanding. – Creativity and imagination: learners design simple devices such as a pinhole camera, a string telephone and sound-proofing for a classroom.
Core Values	– Responsibility: learners take care of equipment and play assigned roles in group investigations. – Respect: learners listen to and value the ideas of others when discussing and revising explanations. – Unity: learners work as teams to collect, compare and combine data. – Integrity: learners report observations honestly even when they differ from predictions.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about echoes and add them to the Driving Question Board. – Developing and using models: learners build and revise a wave model that explains reflection, refraction, diffraction, interference and the Doppler effect. – Planning and carrying out investigations: learners plan experiments on echo timing, refraction in water and standing waves on a string. – Analyzing and interpreting data: learners interpret ripple patterns, simulation data and frequency measurements. – Using mathematics and computational thinking: learners use $v = f\lambda$, Snell's law and echo distance $d = vt/2$. – Constructing explanations and designing solutions: learners explain echoes and design ways to reduce noise. – Engaging in argument from evidence: learners debate why some places give stronger echoes. – Obtaining, evaluating and communicating information: learners read about and present uses of waves in radio and ultrasound.
Pertinent & Contemporary Issues (PCIs)	– Health education: learners examine the effects of noise pollution on hearing and well-being and the safe use of ultrasound in medicine. – Environmental education: learners relate noise from traffic in Nairobi and from industry to its effects and ways of reducing it. – Safe use of ICT: learners use digital devices and simulations responsibly. – Gender equity: learners of all genders take equal leadership roles in practical work.
Career Connections	– Audio and sound engineer: uses reflection, interference and resonance in studios, music halls and noise control. – Radiographer or sonographer: uses ultrasound waves to image internal organs in hospitals such as Kenyatta National Hospital. – Telecommunications engineer: uses radio waves and FM modulation in Safaricom, KBC and radio networks. – Optometrist and optical technician: uses refraction and lenses in spectacles and instruments. – Meteorologist and aviation technician: uses Doppler radar for weather and aircraft monitoring. – Architect or acoustic designer: designs auditoria and buildings to manage echoes and noise. – Musician and instrument maker: applies stationary waves and resonance to tune instruments such as the nyatiti and guitar.
Focus for Lessons	This unit uses the puzzle of echoes at Kenyan landmarks to build a connected understanding of how waves behave. Learners predict, investigate with simulations and hands-on activities, explain their findings and keep revising a class model. The unit ends with applications in communication and medicine and a design challenge on reducing noise pollution.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do sound waves reflect, and why do some places produce stronger echoes than others? KICD KEY INQUIRY QUESTIONS: 1. How do waves behave when they meet obstacles or change medium?

	2. How are the properties of waves applied in day-to-day life?
Anchoring Phenomenon	A group of Grade 10 learners on a trip to Hell's Gate National Park in the Rift Valley shout towards the cliff face and, after a short delay, hear their own voices come back. Near Ngong Hills, a learner standing in a valley hears a clear echo, while in the school hall in Kisumu the echo is muddy and the voice sounds loud and drawn out. On the open grounds of a Nairobi school the shout produces no echo at all. It is puzzling because the sound seems to leave and then return. Why do some places and surfaces send sound back strongly while others do not, why is there a delay, and what is happening to the sound wave when it meets a wall or a mountain?
Storyline Thread	Lesson 1: Anchoring phenomenon. Learners shout at a wall or cliff and record what they hear, predict why echoes occur, create the initial Driving Question Board and draw an initial model of the echo. Lesson 2: Rectilinear propagation. Using the PhET Geometric Optics simulation, pinhole cameras and shadows, learners show that light travels in straight lines and compare how waves travel. Lesson 3: Reflection of waves, part 1. Learners investigate the laws of reflection with plane mirrors and ripple or simulation demonstrations, and add a reflection component to the model. Lesson 4: Reflection of sound and echoes. Learners measure the echo delay across a school field, calculate distance using $d = vt/2$ and compare hard and soft surfaces to explain strong and weak echoes. Lesson 5: Refraction of waves. Learners watch YouTube demonstrations and investigate the bent straw and apparent depth, relating speed change in a new medium to the change in direction. Lesson 6: Refraction, Snell's law and lenses. Learners use refractive index calculations and ray diagrams for lenses, linking to spectacles, cameras and microscopes, and revise the model. Lesson 7: Diffraction of waves. Using the PhET Wave Interference simulation, learners explore waves passing through gaps and around edges, explaining why sound turns corners more readily than light. Lesson 8: Interference of waves. Learners observe constructive and destructive interference in ripples and simulation, and explain noise-cancelling headphones. Lesson 9: Stationary waves and resonance. Using the PhET Standing Waves on a String simulation and a string or air-column experiment, learners identify nodes and antinodes and link them to pitch in the nyatiti and flute. Lesson 10: Doppler effect. Learners watch a YouTube Doppler demonstration, model moving sources and explain the changing pitch of an ambulance siren and its use in radar and astronomy. Lesson 11: Waves in communication and medicine. Learners use Khan Academy resources on FM modulation to explain radio and mobile signals, and investigate ultrasound imaging. Lesson 12: Noise pollution and the final model. Learners analyse sources and health effects of noise, design noise-reduction solutions, return to the Driving Question Board and present a final model explaining echoes and other wave behaviours.
Supporting Phenomena	– Refraction in water and lenses: a straw in a glass of water appears bent, and lenses focus light to form images in glasses, cameras and microscopes, as when a fisherman at Lake Victoria misjudges the depth of a fish. – Diffraction of sound and light: sound is heard around corners of a classroom block and light bends slightly around edges. – Interference patterns: ripples on water or overlapping sound waves create patterns of reinforcement and cancellation, as in noise-cancelling headphones. – Stationary waves and resonance: vibrating strings or air columns produce standing waves with nodes and antinodes, as in the nyatiti, guitar and flute. – Doppler effect: the pitch of an ambulance siren on Thika Road changes as it passes by. – Wave applications in communication and medicine: radio waves carry FM radio and mobile phone signals, and ultrasound produces images of internal organs. – Noise pollution: traffic in Nairobi, matatu hooters and industrial sounds affect human health.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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**LESSON 1 (40 minutes (single lesson); the school-ground shouting activity can be done in the first 10 minutes outside):
Anchoring Phenomenon: Echoes at Hell's Gate, Ngong, Kisumu and Nairobi**

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by having learners observe and describe the echo phenomenon at four Kenyan sites, make first predictions about why sound comes back, and draw a first echo model that the class will revise through the unit.
Knowledge	Learners recognise that an echo is sound coming back after meeting a surface. They describe, from the Echo Table, what is heard at four sites: Hell's Gate (cliff 85 m away, clear echo after 0.50 s), Ngong Hills (valley wall 170 m away, clear echo after 1.00 s), the Kisumu school hall (wall 12 m away, delay 0.07 s, loud, muddy and drawn-out sound) and the Nairobi school open grounds (nearest hard wall 300 m away, no echo heard). Learners notice that a delay exists and that places differ, without calculating anything yet.
Skills	Observing and describing a sound event carefully; making and recording predictions before instruction; comparing data across four sites; asking testable questions; developing and drawing an initial model (NGSS SEP: Asking Questions, Developing and Using Models); communicating ideas in pairs and to the class.
Attitudes	Curiosity about everyday sounds; willingness to commit to an idea that may later be revised; respect for peers' different ideas (Respect); responsibility for roles in group work (Responsibility); communication and collaboration, since learners share predictions and build the DQB together.
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 1 launches the anchoring phenomenon and opens the Driving Question Board (DQB) and the class echo model. It builds on nothing taught earlier, only on learners' everyday experience of shouting at walls. Later lessons will return to the four sites: Lesson 2 draws the sound path as a ray, Lesson 3 gives the laws of reflection, and Lesson 4 gives the echo-distance formula and the surface table. This lesson deliberately leaves out those ideas and does no calculations.
Safety Notes	Do not let learners shout at full volume close to each other's ears, and keep shouting short (3 to 4 calls per group). If going outside, stay within the school compound, supervise near roads and keep learners together. Keep the audio or story clip volume at a comfortable level (below about 85 dB, well under matatu music level). Hands off any wall during the shout; keep clear of loose stones or bricks. If a learner has hearing sensitivity, let them observe and time instead of shouting.

B. LESSON OVERVIEW

Mr. Mwangi's Grade 10 class hears the story of Wanjiru, Otieno, Kiprono and Amina, who shout at a cliff at Hell's Gate and hear their voices come back after a short delay. At Ngong Hills the echo is clear, in the Kisumu school hall it is loud, muddy and drawn-out, and on the open Nairobi grounds there is no echo at all. Learners first predict what causes an echo, then observe by shouting or clapping towards a school wall and studying the four-site Echo Table. They explain what they noticed in pairs, build the first Driving Question Board with their questions, and draw an initial echo model that will be revised in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Characteristics of Sound Waves - Overview Source: CK-12  Search ARES for similar videos  HTML: Sound Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners listen to the opening story: Wanjiru, Otieno, Kiprono and Amina shout at the cliff at Hell's Gate and hear their voices return. Individually, learners write in their notebooks: (1) What do you think an echo is? (2) Where does the sound go after it leaves your mouth, and what brings it back? (3) Which of these places will give the best echo: a cliff 85 m away, a valley wall at Ngong Hills, a school hall in Kisumu, or an open grass field in Nairobi, and why? They then share predictions with a partner and sketch a quick prediction diagram of the shout's journey. They are told there are no wrong answers yet.	Read the story with a pause after the line 'and then their own voices came back.' Ask: "Where did that voice come from?" Give WAIT TIME of 10-15 seconds in silence while learners write. Say: "Write what you really think, not what you think I want to hear. We will check these ideas later." Circulate and note common ideas (sound bounces like a ball, the wall makes a new sound, the mountain is repeating us, the wind carries it back). Cold-call 4 learners from different parts of the room to share, without confirming or correcting. Record 3 to 4 predictions on the board under the heading 'Our first ideas'. Do not introduce the word 'reflection' or any formula yet.	Predict-Share-Record (Eliciting Prior Ideas): writing a personal prediction before any teaching makes learners commit to an idea, so that later evidence can confirm, challenge or revise it. Public recording of varied ideas shows that scientific explanations begin as competing hypotheses and shapes the rest of the unit.	Check notebooks for: (a) a written statement of what an echo is, (b) any mention of sound travelling to a surface and returning, (c) a reason given for choosing a best-echo site. Observable indicators: learners commit to a prediction in writing within 3 minutes; at least 4 different ideas appear on the board; learners who use words like 'bounces' or 'comes back' are noted for later. Note misconceptions such as 'the wall makes the sound' or 'sound disappears over open ground' for follow-up.
Observe Phase  VIDEO: Characteristics of Sound Waves - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Sound Waves Practice Source: CK-12  Search ARES for similar readings	Part A (about 8 minutes): In groups of four (roles: Shouter, Listener-Recorder, Timer-Counter, Reporter), learners go to the school wall, classroom block or an open area chosen by the teacher, and shout 'Hello Kenya!' or clap once, in turn, towards the wall. They record whether they hear anything return, how it sounds (clear, muddy, loud, faint, none) and roughly how long it takes to come back, counting 'one-and-two' or using a phone or watch if available. Where possible they try two places, such as facing the stone or concrete wall and facing the open field. Part B (about 7 minutes): Back in class, groups study the Echo Table for the four trip sites: A Hell's Gate cliff, 85 m, delay 0.50 s, clear echo; B Ngong Hills valley wall, 170 m, delay 1.00 s, clear echo; C Kisumu school hall, 12 m to far wall, delay 0.07 s, loud, muddy, drawn-out sound; D	Before going out, say: "Your job is to be scientists: listen carefully and write exactly what you hear, not what you expect." Demonstrate once with one volunteer group, safely and briefly. Remind: "One shout per person, then listen in silence." Move between groups asking: "What did you hear? Where did it seem to come from? Was it the same as your shout?" Back in class, project or draw the Echo Table. Ask: "Which sites gave an echo you could hear clearly? Which did not?" Give WAIT TIME of 10-15 seconds after the question. Cold-call 3 groups to report on one site each. Then ask: "What is different about site C compared with A and B?" and "What is different about site D?" Accept all observations; do not give the explanation or the formula. If someone mentions that delay depends on distance, say: "Good noticing. Hold that thought,	Structured Observation with Data Table Comparison: first-hand experience of shouting at a wall gives every learner a shared experience of the phenomenon, then the Echo Table lets them see patterns across four sites (distance, delay, what is heard). Describing before explaining builds the habit of using evidence as a base for reasoning.	Observable indicators: each group records at least 2 observations (e.g. echo heard or not, clear or muddy, delay noticed); learners use descriptive words correctly for the four sites (clear, muddy, drawn-out, none); at least one group notes that the delay at Ngong is longer than at Hell's Gate; at least one notes that the hall (C) sounds different from the cliffs. Use a quick checklist for Reporter statements. Look for learners who only say 'it echoes' without describing, and prompt them to add detail.

	Nairobi school open grounds, nearest hard wall 300 m, no echo heard. Learners copy the table and add a 'What I notice' column, describing patterns in what is heard at each place without calculating.	we will test it later."		
Explain Phase  VIDEO: Introduction to waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) Source: CK-12  Search ARES for similar readings	In pairs, learners use their observations and the Echo Table to talk through three prompts: (1) What do you think happens to the sound between the shout and the echo? (2) Why do you think there is a delay before the echo is heard? (3) Why might the Kisumu hall sound muddy and the Nairobi field give no echo at all? Each pair writes a short 'Our best idea so far' statement of 2 to 3 sentences, using words from the observations, and notes one thing they are unsure about. Pairs then join another pair to compare ideas and mark where they agree and disagree.	Say: "Turn to your partner. Use what you heard outside and what you see in the table. Tell each other your best explanation." Allow 4 minutes of talk. Circulate listening for ideas such as 'the sound hits the wall and returns', 'it takes time to travel', 'grass swallows sound', 'the hall has many walls'. Ask probing questions: "What do you think is travelling from you to the wall?" and "What might the wall be doing?" Do not confirm answers. Then ask: "Who has an idea that is different from your partner's?" Give WAIT TIME of 10-15 seconds. Cold-call 3 pairs. Write key phrases on the board and mark the ones that disagree with a star. Say: "These stars are the exciting parts. We will test them over the next lessons."	Talk Moves (Think-Pair-Share with Pair-Square): learners put ideas into words, listen to a competing view and justify their thinking with evidence from the site data. Highlighting disagreements, rather than resolving them, creates a need to know and drives investigation.	Listen for: use of evidence such as 'at Hell's Gate we heard it clearly' to support claims; mention that sound travels or moves to the surface and back; recognition that the sites differ; at least one unresolved question per pair. Collect or check 'Our best idea so far' statements: a strong statement links the sound's journey, the surface or place, and the delay. Note learners who think sound is only made by the wall, or that distance does not matter, to address in Lessons 2 to 4.
Driving Question Board (DQB) Creation  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos	Learners see the Driving Question displayed: 'How do sound waves reflect, and why do some places produce stronger echoes than others?' Each learner writes 2 or 3 questions of their own on sticky notes or paper slips, such as 'Why is there a delay?', 'Why does the hall sound muddy?', 'Does the shape of the mountain matter?', 'Why no echo on the open field?'. Groups sort their questions into themes (e.g. the delay, the surface, the place, what travels) and place them on the class DQB, explaining each to the class in one	Write the Driving Question at the top of a chart paper or wall space. Say: "What do you still wonder? Write questions that start with Why or How." Give 2 minutes of silent writing, using WAIT TIME of 10-15 seconds before anyone speaks. Have each group choose their best 3 questions and place them. Cold-call 5 learners to read out and explain one question. Cluster similar questions with learners, asking: "Which of these belong together?" Ask: "Which question could we test with an investigation?" Point out which	Driving Question Board (Question Clustering): collecting and grouping learners' own questions gives them ownership of the unit and a visible map of what is still unknown. Later lessons link back to the board so learners see how each investigation answers real questions.	Observable indicators: each learner contributes at least 2 questions; at least 3 questions are about causes (why/how) rather than facts; groups can explain how their questions connect to at least one site; categories on the board include the delay, surfaces and places. Check that questions refer to the phenomenon (e.g. Hell's Gate, Kisumu hall). Learners who only ask 'What is an echo?' can be pushed with: "What do you want to know about why it happens?"

 HTML: Sound And Waves Source: PhET  Search ARES for similar readings	sentence. The class agrees on 2 or 3 big categories and links them to the Driving Question.	questions link to the KICD inquiry question, 'How do waves behave when they meet obstacles or change medium?' Keep the DQB visible for the whole unit and tell learners they will add and answer questions each lesson. Avoid answering questions now.		
Model Building Phase  VIDEO: Sound Properties: Amplitude, period, frequency, wavelength <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Sound Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Individually, each learner draws an Initial Echo Model on a page showing: the person shouting, the wall or cliff, what they think travels, and how the sound returns, with arrows and short labels. They then add a small second sketch to show how the model would differ for site C (hall) or site D (open field). Learners write a one-line caption: 'In my model, an echo happens because...'. Groups of four then compare models, and Amina's style of annotated drawing is highlighted as a good example. Each group chooses one model to display. Finally, learners complete a summary table in their books (What we observed / What we think / What we still need to find out).	Say: "Draw what you think is happening. Show where the sound goes. Label everything. It's fine if you're not sure." Give a prompt: "Include what you can't see." Circulate and ask: "What does this arrow stand for?" and "What happens at the wall in your drawing?" Allow WAIT TIME of 10-15 seconds before responding to hesitant learners. Cold-call 3 learners to explain their model to the class. Ask: "What parts do we all agree on? What parts are different?" Create a class model on the board from the common parts (shout, travel, surface, return) and mark uncertain parts with question marks (e.g. 'what happens at the wall?', 'why delay?', 'why no echo on grass?'). Keep the models safely for later revision. Close with: "In the next lesson we will think about the path the sound takes." Do not mention ray diagrams in detail or the law of reflection.	Initial Model Drawing and Gallery Comparison: sketching a causal model makes hidden thinking visible and testable. A shared class model with question marks shows exactly which ideas future evidence must support, and later lessons revise it so learners can see their own conceptual change.	Look for: a model with at least a source, a surface and a returning path; labels or captions explaining the cause of the echo; some attempt to show differences between sites C or D and the clear-echo sites; recognition of uncertainty. Use a simple 3-point check: (1) sound path shown, (2) surface involved, (3) place difference addressed. Observable evidence of misconceptions (e.g. wall creates a new sound, sound bounces like a ball with no mention of waves) should be noted and revisited in Lessons 2 to 4.

D. TEACHER REFLECTION

Which of my learners' predictions and DQB questions showed everyday experience with echoes, and which showed misconceptions I should address in Lessons 2 to 4? Did every learner, including quieter ones, have a chance to speak and commit to an idea? Did I resist explaining the echo too early? Were the Echo Table observations clear enough to spark curiosity about differences among Hell's Gate, Ngong, Kisumu and Nairobi? What changes would help the shouting activity run more safely and quietly next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	At Hell's Gate (85 m) and Ngong Hills (170 m) we heard clear echoes after a delay (0.50 s and 1.00 s). In the Kisumu hall (12 m) the sound was loud, muddy and drawn-out. On the open Nairobi grounds (nearest hard wall 300 m) we heard no echo. At the school wall we also listened for our own sound coming back and described what we heard.
What did I learn?	An echo is sound coming back to us after it meets a surface. Different places give different results, and there is a delay between the shout and the echo. We do not yet know exactly why some places give strong echoes and others do not.
How does this explain the phenomenon?	Our first ideas: the sound travels from us to the wall or cliff and comes back, and the delay may be the time it takes to travel there and back. The hall may sound muddy because of many walls, and the open field may give no echo because the sound spreads out or is absorbed. We will test these ideas in later lessons.

LESSON 2 (40 minutes (one single period; extend to a double period of 80 minutes if shadows and the pinhole camera are built from scratch)): Rectilinear Propagation: Light and Wave Paths from Source to Cliff and Back

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To give learners evidence that light travels in straight lines and to introduce the ray model, so they can draw a wave's path from source to surface as a straight ray. This gives the class a way to picture the path of the sound that leaves the shouting learner at Hell's Gate, reaches the cliff and returns. It prepares for the laws of reflection in Lesson 3.
Knowledge	Learners will state that light travels in straight lines (rectilinear propagation) in a uniform medium. They will explain how shadows and pinhole-camera images support this. They will describe a ray as a straight line showing the direction in which wave energy travels. They will compare light (3.0×10^8 m/s) with sound (340 m/s) and draw the path of sound to the Hell's Gate cliff (85 m away) as a straight ray out and a straight ray back.
Skills	Making and testing predictions; planning and carrying out simple investigations (shadows, pinhole camera, PhET Geometric Optics simulation); analysing and interpreting data from observations; developing and using models (ray diagrams); using mathematical thinking to compare speeds (light speed is about 10^6 times sound speed); constructing explanations from evidence; communication and collaboration in groups of four.
Attitudes	Responsibility: learners take different roles (holder, measurer, recorder, reporter) in the group. Respect: learners listen to and build on the ideas of others, including ideas that turn out to be incomplete. Curiosity and appreciation of how simple investigations with locally available materials (a torch, cardboard, a tin) explain everyday experiences.
Key Inquiry Question	How do waves behave when they meet obstacles or change medium? (Lesson 2 focus: How do we know the path a wave takes before it meets an obstacle, and how can we draw it?)
Purpose in Storyline	Lesson 1 launched the phenomenon: at the four echo sites something travels from the source to a surface and comes back, so it needs a path. Lesson 2 gives learners a first tool for describing that path, the ray, using light, which is easy to see. Learners then transfer the ray model to the sound path at Hell's Gate (site A). Lesson 3 will use these ray diagrams to study the laws of reflection.
Safety Notes	Do not look directly at the Sun or shine torches or laser pointers into anyone's eyes. Use only low-power torches or a candle placed on a stable tray under adult supervision, and keep long hair and paper away from the flame (a torch is preferred). Use blunt pins or pencil tips to make pinholes and take care with scissors. Make the card room dark only if the classroom is safe to move in; otherwise use a cloth or box cover instead. Keep bags and cables off the floor when using a computer or tablet for PhET.

B. LESSON OVERVIEW

Learners start from Wanjiru's question after the trip: how does the voice get from the cliff at Hell's Gate and back? They predict how light and sound travel and what path a shout takes. In the Observe phase they gather evidence in rotating stations: (1) shadows formed by a torch and an object (a cup or a mandazi tin), including a test with a bent tube or card with aligned holes, (2) a pinhole camera built from a tin or box, and (3) the PhET Geometric Optics simulation (or the teacher demonstration if only one device is available). They use the evidence to conclude that light travels in straight lines, and define a ray as a straight line showing the direction of travel of wave energy. In the Explain phase they compare light speed (3.0×10^8 m/s) with sound speed (340 m/s), and draw the sound path from a learner to the Hell's Gate cliff (85 m) as a straight ray, noting that the 0.50 s delay is the time for the sound to travel 170 m. They add new questions to the DQB and add the first ray drawing to the class model, keeping in mind that the ray is a model, not the wave itself.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Lunar Eclipses: Umbra and Penumbra (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners recall Wanjiru's question from Lesson 1: the sound seemed to leave and then return, so what path does it follow? Individually, they sketch on a sticky note or in their notebook (3 min) how they think light from a torch travels across a room and what path a shout takes from a learner at Hell's Gate to the cliff and back. They write a one-line prediction: 'Light travels _____ because _____.' They share in pairs and then add their prediction to a class chart with two columns: 'Straight' and 'Curved/Spreads in all ways'. Learners also predict what will happen if a card with a hole is placed in front of a torch: will the light bend to reach the wall? Predictions are expected to be varied and incomplete, and that is acceptable.	Open with: 'Remember Wanjiru's question at Hell's Gate. Her voice left her mouth and came back after 0.50 seconds. If something travelled out and back, what path did it take? Draw it.' Give 2 minutes of silent drawing, then WAIT TIME of 10 seconds after asking: 'Does light go in a straight line, or does it curve? What makes you think so?' Cold-call 4 learners (not volunteers) to read predictions, and record exactly what they say on the chart, with no correction. Say: 'I am not looking for the right answer yet, I am looking for your reasons. Today we collect evidence.' Circulate and note learners who draw curved or wavy sound paths, and those who mention light coming 'instantly'. Say: 'Hold on to the word instantly, we will return to it.'	Predict-Observe-Explain (POE) with a public prediction chart: committing to a prediction and a reason before seeing evidence makes ideas visible and gives learners a reason to test them. Revisiting the chart later lets learners see their own thinking change, which builds ownership of the final explanation.	Look for: (1) at least 70% of learners can give a reason (not just a guess) for their prediction; (2) number of learners who draw the sound path as a straight line, a curve, or spreading in all directions (tally on the teacher sheet); (3) whether any learner links light travelling and sound travelling by using the word 'path' or 'direction'. Note learners who say light 'just appears' as those who need more support with the idea of travelling energy.
Observe Phase  VIDEO: Mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: mechanical waves and light <i>Related topic: not an exact match for this lesson.</i>	Groups of four rotate through three 8-minute stations, each with a recorder, a materials handler, a measurer and a reporter. Station 1 (Shadows): learners shine a torch on a cup or a mandazi tin and a card screen, trace the shadow edges, move the object closer to and farther from the torch, and measure shadow size. They then line up three cards with holes and test whether light passes when the holes are in a straight line versus when one card is shifted, or when a bent tube (a flexible hose) is used. They draw the shadow with straight lines from torch to the edges of the object. Station 2 (Pinhole camera): learners use a tin or box with a pinhole at one end	Brief the stations in 3 minutes: 'At each station your job is to answer one question: how does light travel? Write what you SEE first, and what you THINK it means second.' Circulate with a checklist. At Station 1 ask: 'If light could bend around the cup, what would happen to the shadow? What do you see?' WAIT TIME of 10 seconds before follow-up. At Station 2 ask: 'Why is the image upside down? Trace the light from the top of the object through the pinhole. Where does it land?' At Station 3 ask: 'What happens to the ray when it hits the screen? Does it change direction before that?' Cold-call 3 groups to report one observation each, using 'We	Evidence table with observation versus inference columns (Claim-Evidence-Reasoning scaffold): separating what was seen from what it means stops learners jumping to conclusions and helps them build a claim from several independent pieces of evidence, namely shadows, aligned holes, the pinhole image and the simulation.	Observable indicators: (1) learners use a ruler to draw straight lines in their shadow and pinhole diagrams; (2) the recorder's 'What we saw' column contains observations such as 'the shadow had sharp edges' or 'the image was upside down', not just conclusions; (3) at least 3 of 4 learners in each group can say what happened when the middle card was moved off line; (4) the group can explain the inverted image using straight lines crossing at the pinhole. Intervene with groups who draw curved rays by asking them to test with a string pulled tight.

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and tracing paper or greaseproof paper at the other, point it at a bright window or a candle-like torch and draw the image seen (upside-down). They draw lines from the top and bottom of the object through the pinhole to the screen. Station 3 (PhET Geometric Optics): learners use the simulation (or the teacher's demonstration) to place a light source and a screen with a slit, observe the ray travelling from the source, and record that the ray is straight until it meets an obstacle or a boundary. They switch the ray display on and off to see that the ray shows direction. Each group completes an Observation Table with columns: Station, What we did, What we saw, What this tells us about how light travels. Early finishers make a ray drawing of the shout at Hell's Gate.	saw...' sentences, and have another group add 'We agree because...'. Keep a running list of observations on the board, separating observations from interpretations. If a group claims light 'bent around' the cup, ask them to check the edge of the shadow and the light source size (a wide source gives a fuzzy edge or penumbra). Introduce the terms umbra and penumbra only if groups notice them.		
Explain Phase VIDEO: Mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos H5P: Mr. Spooky <i>Related topic: not an exact match for this lesson.</i> Source: Global Digital Library - Book Catalog Search ARES for similar readings	Whole class sensemaking (about 10 minutes). Using the combined observations, learners write a Claim-Evidence-Reasoning statement in groups: Claim (light travels in ...), Evidence (three observations from the stations), Reasoning (why straight-line travel explains the shadow shape, the aligned holes and the upside-down pinhole image). They agree on the term rectilinear propagation and define a ray as a straight line with an arrow showing the direction in which wave energy travels. They then compare light and sound using data: light 3.0×10^8 m/s and sound 340 m/s. In pairs they calculate how many times faster light is (about 880 000 times, roughly 10^6) and the time light takes to cross 85 m (about 2.8×10^{-7} s, essentially instant) versus	Ask: 'Looking at all three stations, what do they have in common? What claim can we make about how light travels?' WAIT TIME of 15 seconds, then cold-call 4 learners to build the claim sentence by sentence on the board. Challenge: 'What would the shadow look like if light travelled in curves? Which of your observations rules that out?' Introduce the ray: 'A ray is not a thing you can touch. It is a straight line we draw to show where the energy is going. Why is that useful?' Then ask: 'Light is 3.0×10^8 m/s and sound is 340 m/s. Work with your partner: how many times faster is light?' Allow 3 minutes, circulate, then cold-call 3 pairs for answers, handling errors with powers of ten. Ask: 'Remember Kiprono timing the	Claim-Evidence-Reasoning (CER) combined with modelling: learners connect several observations to one explanatory claim and then represent the idea as a ray diagram, a simplified model. Applying the model to a different wave (sound at Hell's Gate) is a transfer task that tests whether the idea of a straight path generalises while the speed comparison highlights what differs between light and sound.	Observable indicators: (1) CER statements name at least two different pieces of evidence and link them to straight-line travel; (2) ray diagrams use a ruler, arrows showing direction, and correct labels of 85 m each way; (3) learners correctly calculate that light is about 10^6 times faster than sound and that the 0.50 s delay at site A is the sound's travel time over 170 m; (4) learners can say a ray shows direction of energy travel, not the wave itself. Exit-ticket style check: 'Draw the path of Wanjiru's shout to the cliff and back, and state why it is drawn straight.' Learners who draw the ray bending, or who confuse the delay with the cliff 'storing' the sound, are noted for follow-up in Lesson 3.

	sound ($85 / 340 = 0.25$ s). They use this to explain why we see a distant lightning flash before hearing the thunder, and why a ray of light seems to arrive 'instantly' while sound shows a delay. Finally they draw the Hell's Gate situation to scale as a ray diagram: a learner 85 m from the cliff, a straight ray out to the cliff, an arrow, a straight ray back, labelled $85 \text{ m} + 85 \text{ m} = 170 \text{ m}$, and $170 \text{ m} / 340 \text{ m/s} = 0.50 \text{ s}$ as the round trip time. They discuss that a ray is a model: sound is a longitudinal mechanical wave needing a medium (air), while light is a transverse electromagnetic wave that does not need a medium, but both can be shown by rays when we care about direction. They revisit their predictions chart and mark which ideas the evidence supports.	echo at Hell's Gate? Why did he hear a delay of 0.50 s but never see a delay in the cliff?' Guide the drawing of the ray diagram, insisting on a ruler, arrows and labels. Address the misconception that sound travels like a wave of light by stating: 'Both travel as waves, and both can be drawn as rays, but sound needs air and is much slower. Today we only claim that the PATH is straight in a uniform medium.' Do not teach the laws of reflection: say 'Next lesson we will find out what happens at the cliff, the angle it comes back at.' Return to the prediction chart: 'Which of our predictions does the evidence support? Move a note to show it.'		
Driving Question Board (DQB) Creation  VIDEO: Mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Lunar Eclipses: Umbra and Penumbra (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12	Learners return to the Driving Question Board started in Lesson 1 (How do sound waves reflect, and why do some places produce stronger echoes than others?). In groups they write at least one new question on a sticky note from today's evidence, for example: 'Does sound also travel in a straight line?', 'What happens to the ray when it hits the cliff?', 'Why does the echo come back at the same angle?', 'Why is there no echo on the open Nairobi grounds if the sound still travels straight?', 'Why is the Kisumu hall sound muddy if rays go straight?'. They place each note under a category on the board (Path of the wave, What happens at the surface, Speed and delay, Strength of the echo) and star questions they think they can now partly answer. They	Say: 'You now know something new about the path. What new questions has it raised? Write one question per sticky note.' Allow 3 minutes of silent writing, then WAIT TIME of 10 seconds before sharing. Cold-call 3 groups to read one question each, asking: 'Which site does this question connect to, A, B, C or D?' Sort questions with the class, grouping similar ones. Highlight a question about what happens at the cliff: 'This one will be our focus in Lesson 3.' Ask: 'Which earlier question can we now partly answer? What is our evidence?' Mark partially answered questions with a half-star, keeping any questions about strength and the open grounds open for later lessons. Take a photo or write the DQB question list in the class record for offline	Driving Question Board (DQB) with categories and evidence tracking: turning today's findings into new questions links each lesson to the unit's phenomenon and shows learners how their questions are driving the investigation. Recording what is partly answered models how scientists keep a growing record of evidence.	Observable indicators: (1) every group posts at least one new question that is linked to the phenomenon (an echo site, the path, or the surface); (2) at least half of the questions go beyond 'what' to 'why' or 'how'; (3) learners can name the evidence that partly answers a question, such as the shadow or pinhole results; (4) learners distinguish questions answerable by investigation from those that are not. Note groups who copy Lesson 1 questions without new thinking.

Search ARES for similar readings	also move any answered Lesson 1 questions into an 'Answered so far' area with the evidence listed, such as 'light travels in straight lines (shadows, pinhole camera, PhET)'.	continuity.		
Model Building Phase  VIDEO: Mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Understand: mechanical waves and light <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise the class echo model started in Lesson 1. Each group receives a large sheet (or the shared model chart) showing the four echo sites. Amina's role is taken by a group member who draws: the source (learner), a straight ray to the surface (cliff at Hell's Gate, 85 m, then a ray returning), and labels for distance 85 m and delay 0.50 s. They use the same ray idea for Ngong Hills (170 m, 1.00 s), and place a question mark on the surface for what happens there. For Kisumu (12 m) and the Nairobi grounds (300 m) they draw only the outgoing ray and mark with a '?' what they still do not know about strength and overlapping. A box labelled 'What we now know' lists: 1. Light travels in straight lines in a uniform medium. 2. A ray shows the direction of wave energy travel. 3. Sound takes time to travel (340 m/s), light is about 10^6 times faster. A second box 'What we still need to find out' lists: what happens at the surface, the angle of return, why strength differs. Groups do a gallery walk (3 minutes) and leave one comment sticker on another group's model.	Say: 'Our model needs a path. Show the path of the sound at Hell's Gate with a ruler and arrows. Add only what our evidence supports today.' Insist on honest question marks: 'If we have not tested it yet, mark it with a question mark.' Circulate and ask: 'Why did you draw that ray straight? Which observation supports that?' WAIT TIME of 10 seconds, then cold-call 3 groups to explain their model in one sentence. During the gallery walk ask: 'What is one thing another group did that improves our model?' Close with: 'Next lesson we will look at what happens when the ray reaches the cliff, a mirror or a barrier. Keep your models, we will revise them.' Collect models or photograph them for the next lesson.	Cumulative modelling with gallery walk and peer feedback: revising a shared model each lesson shows how scientific understanding grows by adding evidence-backed elements, and question marks mark the limits of the current model. Peer comments encourage justification with evidence.	Observable indicators: (1) the model shows straight rays with arrows and the 85 m and 0.50 s labels for site A; (2) the 'What we now know' list uses evidence from today's stations; (3) the model avoids claims about reflection angles or surface strength that have not yet been investigated; (4) at least one peer comment refers to evidence. Check learners can explain the model using a sentence frame: 'The sound travels along a straight ray from ... to ... and returns, taking ... seconds.'

D. TEACHER REFLECTION

After the lesson, note: Did learners separate observations from interpretations at the stations? Which groups drew curved rays or struggled to explain the inverted pinhole image, and what helped (a string pulled tight, the PhET ray view)? Did learners see the ray as a model of direction rather than a physical line? How many learners could compute the speed comparison between light and sound without help? Did the Hell's Gate ray diagram make the 0.50 s delay meaningful (170 m at 340 m/s)? Were there misconceptions to address in Lesson 3, such as sound 'spreading' versus a single ray, or the cliff 'creating' the sound? Record DQB questions that

learners care about most, and consider how to adjust station time or group roles if the rotation felt rushed. If the PhET simulation or computers were unavailable, note how the demonstration worked and what learners missed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We saw that shadows have sharp edges shaped like the object, that light passes through aligned holes but not when one hole is moved, that a pinhole camera makes an upside-down image, and in the PhET Geometric Optics simulation that the ray from the source stays straight until it meets an obstacle. We saw that light seems instant while a shout at Hell's Gate takes 0.50 s to return.
What did I learn?	Light travels in straight lines in a uniform medium (rectilinear propagation). A ray is a straight line with an arrow showing the direction in which wave energy travels. Light travels at 3.0×10^8 m/s, about a million times faster than sound at 340 m/s, so sound shows a delay over distance.
How does this explain the phenomenon?	The shout at Hell's Gate can be drawn as a straight ray from the learner to the cliff 85 m away and a straight ray back, a total of 170 m. At 340 m/s this takes 0.50 s, which is the delay Wanjiru and Kiprono noticed. We still need to find out what happens at the cliff, why the sound returns, and why the echo is strong at some places and weak at others.

LESSON 3 (40 minutes (one lesson)): Reflection of Waves: The Laws of Reflection

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how waves bounce off surfaces and establish the laws of reflection using a plane mirror, ripple (or simulation) reflection from a barrier, and a ray diagram of a sound ray hitting the Hell's Gate cliff. They add a reflection component to the class echo model.
Knowledge	State the laws of reflection: (1) the angle of incidence i equals the angle of reflection r , both measured from the normal; (2) the incident ray, the reflected ray and the normal lie in the same plane. Know that a plane mirror with $i = 40$ degrees gives $r = 40$ degrees, and that the same laws apply to water waves and sound.
Skills	Planning and carrying out investigations (measure angles from the normal with a protractor); developing and using models (ray diagrams, class echo model); analysing and interpreting data (compare i and r in a table); constructing explanations using evidence; using mathematical thinking to compare angles.
Attitudes	Curiosity about why a shout returns from a cliff; precision and honesty when measuring and recording angles; teamwork and respect when sharing results; appreciation that the same wave rules apply to light, water and sound.
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 2 gave learners straight-line rays for light and waves, and the sound path to the Hell's Gate cliff drawn as a straight ray. Lesson 3 asks what happens to that ray when it reaches the cliff: it reflects according to precise laws. This explains HOW the sound returns, which is the first half of the driving question. Lesson 4 will then use the reflection idea to explain the delay ($d = vt/2$), the 17 m condition and why some surfaces give stronger echoes than others. Today we do not calculate delays or compare surfaces.
Safety Notes	Keep water away from electrical sockets and the laptop or projector when using the ripple tank, and wipe spills at once to prevent slipping. Handle plane mirrors carefully; do not use cracked ones. Sharp pencil points and pins should be pointed downwards. Do not shout loudly close to others' ears during the prediction activity; use a normal voice.

B. LESSON OVERVIEW

Back at school, Mr. Mwangi asks Wanjiru, Otieno, Kiprono and Amina: 'The shout travels in a straight line to the Hell's Gate cliff (85 m away). What does it do when it gets there, and does it come back along the same path or another one?' Learners predict, then gather evidence from three sources: a plane mirror with a ray (a torch or pin method) to measure i and r , with $i = 40$ degrees giving $r = 40$ degrees; ripple tank or PhET wave simulation showing straight wavefronts bouncing off a barrier; and a ray diagram of the sound ray hitting the cliff. They compare data, state the two laws of reflection and apply them to a sound ray at the cliff. The lesson ends with the DQB update and a first reflection component added to the class echo model. Timing: Predict 6 min, Observe 12 min, Explain 10 min, DQB 5 min, Model 7 min.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the Lesson 2 ray diagram of the sound travelling in	Draw the cliff and the incoming ray on the board. Say: 'Wanjiru	Predict-Observe-Explain (POE) with Public Prediction Gallery:	Observable indicators: (a) learner draws the reflected ray leaving the

 VIDEO: 06 Laws Of Reflection Source: Seavuria  Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	a straight line to the Hell's Gate cliff. On a half-page they draw the cliff as a vertical line and the incoming sound ray, then predict with an arrow where the ray goes after it hits the cliff, and write one sentence: 'I think the sound ray ... because ...'. They also predict: if a ray hits a mirror at 40 degrees to the normal, what angle will the reflected ray make? Pairs compare their predictions and place an X on any idea they disagree on. Link to the phenomenon: Wanjiru and Otieno heard their voices return from the cliff.	shouted at the Hell's Gate cliff. The ray reaches the cliff. Draw what happens next. There is no wrong prediction today.' Explain a normal: draw a dashed line at 90 degrees to the cliff where the ray hits and say 'This dashed line is the normal. We always measure angles from the normal, not from the surface.' Give 2 minutes of silent drawing, then WAIT TIME of 10 seconds before any talk. Cold-call 4 learners (not volunteers) to show their drawing under the document camera or on the board; ask 'What made you draw it that way?' Record 3 different predictions on the board (e.g., straight back, same angle on the other side, absorbed) without judging them. Circulate and note learners who place the reflected ray on the same side as the incident ray.	Making a prediction first commits learners to an idea, creating cognitive conflict or confirmation when evidence appears. Seeing the spread of predictions on the board shows learners that evidence, not opinion, will settle the question.	cliff on the same side as the incoming ray; (b) learner uses the normal rather than the cliff surface when predicting angles; (c) learner predicts the mirror reflected angle as 40 degrees, 50 degrees or something else, giving a reason. Tally on a class list how many predicted a reflected angle equal to the incident angle versus other values, and note common misconceptions (sound is absorbed and a new sound is made at the cliff; angle measured from the surface).
Observe Phase  VIDEO: Specular and diffuse reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Law of Reflection (Flexbook) Source: CK-12  Search ARES for similar readings	Station A, plane mirror (about 5 min): in groups of 4 with roles (Wanjiru-style questioner, Kiprono-style measurer, Amina-style recorder/drawer, Otieno-style reporter), learners stand a plane mirror upright on a sheet of paper, draw its line and a normal, and send a thin ray (a torch through a slit, or two pins and sight-lines) at the mirror at $i = 20, 40$ and 60 degrees from the normal. They mark the reflected ray, measure r with a protractor and record i and r in a table. They check that the incident ray, reflected ray and normal all lie flat on the paper (the same plane), and lift the torch slightly to see that the ray no longer stays on the paper. Station B, ripples (about 4 min): in a ripple tank or the PhET Wave Interference simulation, learners	Distribute materials and say: 'Your job is to find a pattern. Measure carefully from the normal, not from the mirror surface.' Circulate and ask each group: 'What do you notice about i and r?' and 'Where does your reflected ray start from?' Remind groups: 'If your angles differ by more than 2 degrees, remeasure, and check the protractor is on the normal.' Provide a model measurement at the board: $i = 40$ degrees measured at the normal. For the ripple demonstration (if no ripple tank is available, project the PhET simulation), pause the ripples and ask: 'Which way is the wave travelling now? Which way after the barrier?' WAIT TIME of 10 seconds. Cold-call 3 groups to report their i and r values for 40 degrees and write them on a class	Data Table Pattern Finding with Multiple Representations: collecting i and r values across three angles and three contexts (mirror, water ripples, sound ray) lets learners see the same pattern and generalise it. Moving among a table, a diagram and a moving simulation builds a stronger concept than any one alone.	Observable indicators: (a) learner draws the normal at 90 degrees to the mirror before measuring; (b) table shows r within 2 degrees of i for all three angles; (c) learner states that the reflected ray lies on the opposite side of the normal from the incident ray; (d) learner notices that the ripple direction changes after the barrier but the wave keeps its straight pattern; (e) learner draws the sound ray reflection at the cliff with $r = 40$ degrees. Check the class table for outlying results and ask the group to retry.

	watch straight ripples travel to a straight barrier, then bounce off. They sketch an arrow for the direction of travel before and after, and note how the angle before compares with the angle after when the barrier is tilted. Station C, sound ray at the cliff (about 3 min): learners use a protractor to draw a normal on the cliff diagram and a sound ray hitting it at 40 degrees, then draw the reflected sound ray. They compare it with the mirror result. Data collected: mirror $i = 40$ degrees gives r about 40 degrees; ripples reflect from the barrier at an equal angle; sound ray drawn with $r = 40$ degrees.	table on the board. If time allows, ask: 'If a ray hits the mirror along the normal ($i = 0$), what happens?' Prompt Station C: 'This is the path of the sound from Wanjiru to the Hell's Gate cliff you drew in Lesson 2. What does it do at the cliff now?' Watch for learners who measure from the mirror surface (angles of 50 degrees instead of 40) and redirect them to the normal.		
Explain Phase  VIDEO: 06 Laws Of Reflection Source: Seavuria  Search ARES for similar videos  EXERCISE: Law of Reflection Practice Source: CK-12  Search ARES for similar readings	Learners use their tables and diagrams to write a Claim-Evidence-Reasoning statement in their notebooks: Claim: the angle of reflection equals the angle of incidence, and the rays and normal lie in the same plane. Evidence: their measured values ($i = 40$ degrees gives $r = 40$ degrees, plus two other angles), the ripple observation and the paper-plane check. Reasoning: both light and water waves obey the same rule when reflected from a smooth surface, so sound waves hitting the cliff should too. They then apply the laws to a new case: a sound ray from Kiprono hits the Hell's Gate cliff at $i = 30$ degrees; they sketch the reflected ray and state r. They also answer: 'Does the sound need to come back along the same line it went?' and explain that only a ray hitting the cliff along the normal ($i = 0$) returns along its own path. Finally they compare their pre-lesson prediction with the evidence and write a sentence on what changed.	Pose: 'Look at your three i and r pairs. What is your claim? What evidence supports it?' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds. Use a talk move: ask 'Who can add on to what Amina's group said?' and cold-call 4 learners to share. Write the two laws on the board in learners' own words first, then refine into the standard wording: (1) angle of incidence $i =$ angle of reflection r, both measured from the normal; (2) the incident ray, reflected ray and normal lie in the same plane. Say: 'Wanjiru asked why her voice comes back. Today we can say: the sound ray hits the cliff and reflects obeying these laws.' Probe the misconception about angles measured from the surface: 'If the ray hits the cliff at 30 degrees to the surface, what is i?' ($i = 60$ degrees from the normal). Give the application question and, after 10 seconds of WAIT TIME, cold-call 3 learners for the reflected angle. Clarify the boundary of today's lesson: 'We	Claim-Evidence-Reasoning (CER) with Prediction Revision: structuring the explanation forces learners to connect numerical evidence to a general law and then extend it to sound. Comparing the final claim with the original prediction makes the change in thinking visible and builds scientific argumentation.	Observable indicators: (a) CER claim states both laws correctly, including 'from the normal' and 'same plane'; (b) evidence cites at least two measured pairs of i and r; (c) reasoning links mirror, ripple and sound as obeying the same rule; (d) learner correctly gives $r = 60$ degrees when $i = 30$ degrees is measured from the normal and corrects the surface-angle error; (e) learner revises the earlier prediction in writing. Use a 3-finger confidence check ('3 = I could teach it, 1 = I am lost') and follow up with learners showing 1 or 2.

		are not yet explaining the delay or why Ngong is clearer than Nairobi. That comes in Lesson 4.' Do not teach $d = vt/2$ today.		
Driving Question Board (DQB) Creation VIDEO: Specular and diffuse reflection Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	Learners revisit the Driving Question Board: 'How do sound waves reflect, and why do some places produce stronger echoes than others?' In groups they write on sticky notes (a) what they now know about how sound reflects (e.g., 'sound bounces off the cliff at an equal angle to the normal'), and (b) new questions raised by today's evidence (e.g., 'Why is there a delay?', 'Why does the Nairobi field give no echo if the sound still reflects?', 'Does a rough wall reflect the same way as a mirror?', 'Why is Kisumu hall muddy?'). They place notes under headings: Answered (partly), Still to investigate: delay, Still to investigate: surfaces and places. Link to phenomenon: each note refers to one of the four sites.	Say: 'Look at our Driving Question. Which part did we help answer today? Which parts are still open?' Give learners 3 minutes to write and WAIT TIME of 10 seconds before sharing. Cold-call 3 groups to read a note and place it; ask 'Which site does your question come from: Hell's Gate, Ngong, Kisumu or Nairobi?' Highlight that today's evidence covers how the sound returns (reflection) but not the delay or the differences between places. Group similar questions and point to the lessons ahead: 'Many of you asked about the delay and the surfaces. Next lesson we will measure and calculate them.' Move any note answered today to the 'Partly answered' column and record the supporting evidence (the $i = r$ table) beside it.	Driving Question Board Gallery with Question Sorting: turning today's findings into new, categorised questions keeps the phenomenon central and gives learners ownership of the learning path. Sorting by site helps learners see which parts of the phenomenon the reflection laws explain and which need other ideas.	Observable indicators: (a) each group writes at least one 'what we know' note citing evidence and one new question; (b) questions refer to a specific site (A to D); (c) at least one group asks about the delay or the differences between surfaces, showing awareness that reflection alone does not explain everything; (d) learners place notes in the correct category. Collect the notes to plan Lesson 4.
Model Building Phase VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos HTML: Law of Reflection (Flexbook)	Learners open the class echo model from Lesson 1 and, individually and then in groups, add a reflection component: they redraw the Hell's Gate scene (learner, 85 m to the cliff face) showing a sound ray travelling to the cliff, a dashed normal at the point of impact, equal angles i and r , and the reflected ray going back toward the listener. Amina-style drawers label the incident ray, reflected ray, normal, i and r . Learners write a caption: 'Sound reflects from the cliff obeying the law of reflection: $i = r$.' Below the model they add a 'Still missing' box: delay, why the echo is strong at Hell's Gate and Ngong, why	Show the Lesson 1 model and say: 'What does our model show about the sound so far? What needs adding?' Give 3 minutes for individual drawing and 3 minutes for group comparison. Model labelling of the normal and the angles on the board, but let learners add the details themselves. Circulate and ask: 'Where is the normal on your diagram? Are your two angles equal?' and 'Which part of the phenomenon does your model not explain yet?' WAIT TIME of 10 seconds after each question. Cold-call 2 groups to show their model and explain the added reflection part. Collect the best version on	Cumulative Model Revision: adding a new component to the evolving class model shows how each piece of evidence builds the explanation. The 'Still missing' box makes the limits of the model visible and motivates the next investigation.	Observable indicators: (a) model shows a normal drawn at 90 degrees to the cliff; (b) i and r are drawn equal and measured from the normal; (c) the reflected ray returns on the opposite side of the normal; (d) labels are correct (incident ray, reflected ray, normal); (e) the 'Still missing' box lists at least two of: delay, surface type, hall, open ground. Use these as a quick checklist and record names of learners needing support.

Source: CK-12 Search ARES for similar readings	Kisumu is muddy, why Nairobi has none. They also note one question the model cannot yet answer.	the class model chart. Close with: 'Wanjiru asked why her voice comes back. We can now say that the sound ray reflects from the cliff. Next time we ask: how long does it take, and why is the echo strong in some places and weak in others?' Do not add calculation of delay today.		
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D. TEACHER REFLECTION
After the lesson, consider: Did learners measure angles from the normal rather than from the surface? Which learners still think sound is absorbed and 'a new sound' made at the cliff, rather than reflected? Were the ripple or simulation observations clear enough for learners to see equal angles? How well did learners connect the mirror and ripple evidence to the sound ray at Hell's Gate? Did the DQB questions show curiosity about the delay and the surfaces that will lead into Lesson 4? What would you change about group roles, timing or materials (e.g., if mirrors or a ripple tank were limited)? Note which learners need extra practice with the protractor and the normal before Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We shone a ray at a plane mirror and found that when $i = 40$ degrees, $r = 40$ degrees (measured from the normal). We watched ripples bounce off a barrier at an equal angle, and drew a sound ray reflecting from the Hell's Gate cliff.
What did I learn?	The two laws of reflection: the angle of incidence equals the angle of reflection (both measured from the normal), and the incident ray, reflected ray and normal lie in the same plane. These laws apply to light, water waves and sound.
How does this explain the phenomenon?	When Wanjiru shouts at the Hell's Gate cliff, the sound travels in a straight line, reflects from the cliff obeying the law of reflection ($i = r$) and returns to the listeners. This explains how the voice comes back, but not yet the delay or why some places give stronger echoes than others.

LESSON 4 (80 minutes (double lesson: 40 min classroom + 40 min field and debrief)): Reflection of Sound: Echo Measurement and Surfaces

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the echo-distance formula $d = vt/2$ to calculate distances from echo delays, explain why a distinct echo needs a reflector at least 17 m away, and compare hard and soft surfaces to explain why some places give strong echoes and others weak or none.
Knowledge	Sound is a longitudinal mechanical wave that reflects from surfaces following the law of reflection (Lesson 3). Speed of sound in air is 340 m/s. Echo-distance formula: $d = vt/2$, where t is the total time for sound to leave and return and d is the one-way distance. Ear separates sounds at least 0.1 s apart, so $0.1 \times 340 / 2 = 17$ m is the minimum distance for a distinct echo. Reverberation is overlapping reflections that merge and prolong sound. Hard smooth surfaces reflect most sound energy (cliff rock 90%, painted concrete 85%, wooden board 60%); soft surfaces absorb it (curtains 15%, thick grass or foam mattress 10%).
Skills	Measuring distance with a tape measure and time with a stopwatch; calculating with $d = vt/2$ and $t = 2d/v$; averaging by timing 10 clap-echo intervals; analysing and interpreting data in tables; constructing explanations from evidence; using mathematical thinking (NGSS SEP: Using Mathematics and Computational Thinking; Planning and Carrying Out Investigations; Constructing Explanations).
Attitudes	Curiosity about everyday sounds, precision and honesty in recording measurements, teamwork and responsibility when sharing roles in the field, and appreciation of how physics explains places learners know.
Key Inquiry Question	How do waves behave when they meet obstacles or change medium? How are the properties of waves applied in day-to-day life? (Echo timing gives distance; surface choice controls echoes in halls and open fields.)
Purpose in Storyline	Lesson 1 launched the four echo sites A to D and Lesson 3 showed that waves reflect following the law of reflection. This lesson turns the idea 'sound bounces back' into numbers: it explains the delays at the four sites with $d = vt/2$, tests the 17 m condition, and uses surface data to explain strong versus weak echoes. It supplies the quantitative evidence the class model needs, and the echo timing idea will be reused later in the unit.
Safety Notes	Field activity: stay with the teacher on the school field, keep to the marked 50 m line parallel to a safe distance from the wall, and avoid roads and uneven ground. Clap at normal strength and do not shout directly into anyone's ear. Keep the stopwatch carriers away from the wall base to avoid loose stones. Use hats and water in hot weather.

B. LESSON OVERVIEW

Mr. Mwangi's learners return from Hell's Gate asking Wanjiru's question: why does the voice come back after a delay, and why not everywhere? Learners first predict the delay at each site, then gather evidence in two ways: calculating echo delays from the echo table using $d = vt/2$, and measuring the echo distance themselves by standing about 50 m from the school's long stone wall and timing 10 clap-echo intervals (2.94 s gives 0.294 s per interval and $d = 340 \times 0.294 / 2 = 49.98$ m, about 50 m). They then compare the surface table (cliff rock 90%, painted concrete 85%, wooden board 60%, curtains 15%, grass or foam 10%) to explain strong versus weak echoes. Learners explain all four sites using delay, distance and surface, add new questions and answers to the Driving Question Board, and revise the class echo model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Characteristics of Sound Waves - Overview Source: CK-12  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the four sites from Lesson 1: Hell's Gate cliff (learners 85 m away), Ngong Hills valley wall (170 m), Kisumu school hall (speaker 12 m from far wall) and the open grounds of a Nairobi school. In pairs they individually write on a sheet: (1) Which site has the longest delay and which the shortest? Rank A, B, C and D. (2) Is there a minimum distance below which you cannot hear a separate echo? Guess a number. (3) Which would give the stronger echo: a painted concrete wall or a curtain? Why? Learners sketch a quick ray diagram of sound going from the shouting learner to the cliff and back, using the law of reflection from Lesson 3.	Display the four site names on the board. Say: 'Remember Wanjiru's question at Hell's Gate: the voice leaves and then comes back. Before I give you any numbers, rank the four sites by delay and write down your reasoning.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and compare, then I want your best guess for the smallest distance that gives an echo.' WAIT TIME 10-15 seconds after each question before taking answers. Cold-call 4 pairs for their ranking and 3 pairs for a minimum-distance guess; record every guess on the board without judging (guesses likely range from 1 m to 100 m). Ask: 'What is your evidence for that guess, from a time you shouted at a wall?' Cold-call 2 learners for everyday experiences, for example shouting in an empty classroom versus at a school fence.	Predict-Observe-Explain with public recording of ideas. Writing predictions before any data creates a commitment that learners later test, which makes the gap between prior ideas (for example 'a nearby wall always gives an echo') and evidence visible and motivates the investigation.	Listen for: ranking that connects delay to distance (far wall means longer delay) versus ranking by loudness; whether learners mention surfaces (hard, soft) in their reasoning; whether ray sketches show an incident and reflected path. Indicator: at least 3 of 4 pairs rank D or B as longer than C, and sketches show a path that returns to the source. Note common ideas such as 'the sound bounces only from big walls' or 'echoes need silence' for revisiting in the Explain phase.
Observe Phase  VIDEO: Characteristics of Sound Waves - Overview Source: CK-12  Search ARES for similar videos  HTML: Sound Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Part A (15 min, classroom): In groups of four, with roles (timekeeper, recorder, calculator, reporter), learners receive the echo table with distances and use $t = 2d/v$ with $v = 340$ m/s to calculate the delay at each site: A 85 m gives round trip 170 m and delay 0.50 s; B 170 m gives 340 m and 1.00 s; C 12 m gives 24 m and 0.07 s; D 300 m gives 600 m and 1.76 s. They compare each delay with the 0.1 s ear limit and derive the minimum distance $0.1 \times 340 / 2 = 17$ m. Part B (30 min, field): The class goes to the school's long stone wall. Using the tape measure, learners mark 50 m from the wall. One learner claps and listens, and adjusts the clapping	Part A: Say: 'Use $t = 2d/v$. Why do we divide by 2 in $d = vt/2$? Talk with your group first.' WAIT TIME 10-15 seconds. Cold-call 3 groups to explain that t is the round trip, so one-way distance is half. Circulate and ask: 'Site C has a delay of 0.07 s. Is that more or less than 0.1 s? What does that mean for what your ear hears?' Do not give the answer; ask a group to show the reasoning. Part B: Brief on safety, then say: 'Timing one interval is hard because your reaction time is about 0.2 s, so we time 10 intervals and divide by 10. Why does that reduce the error?' WAIT TIME 10-15 seconds, cold-call 2 learners. Walk along the line and check that clap and echo are	Data table analysis with structured group roles and error discussion. Calculating all four delays from distances links numbers to the remembered experiences, and the field measurement lets learners check the formula against a known distance, which builds trust in $d = vt/2$. Timing 10 intervals teaches how repeated measures reduce uncertainty.	Observable indicators: groups correctly calculate 0.50 s, 1.00 s, 0.07 s and 1.76 s; recorders write the 0.1 s threshold and 17 m; field groups obtain an interval near 0.294 s and a distance near 50 m (within about 5 m) and explain the use of 10 intervals; learners order surfaces by reflected percentage. Check for errors: forgetting to halve the time, using the round trip as the distance, or measuring time in the wrong units. Use a checklist with three marks: formula used, units correct, interpretation stated.

	rhythm until each clap coincides with the returning echo of the previous clap. Kiprono's role is played by a timekeeper who starts the stopwatch on a clap and stops it after 10 clap-echo intervals. Groups record time for 10 intervals (class example: 2.94 s), compute one interval (0.294 s) and calculate $d = 340 \times 0.294 / 2 = 49.98$ m, about 50 m, then compare with the tape measure value and discuss sources of error such as reaction time and wind. Part C (8 min): Learners compare the 'surface table' (cliff rock 90%, painted concrete 85%, wooden board 60%, curtains 15%, thick grass or foam mattress 10%) and optionally test by clapping near a classroom curtain, a wooden board and a foam mattress, listening for loudness of the return.	synchronised ('listen: does your clap land exactly on the echo?'). Ask groups with a 50 m result far from 50 m to repeat. Part C: Ask: 'Which surface gives the loudest returning clap, and which almost none? Rank them and link to the percentages.' Cold-call 4 groups for ranking and record on a class chart.		
Explain Phase  VIDEO: Introduction to waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Sound Waves Practice Source: CK-12  Search ARES for similar readings	Back in class, groups build a claim-evidence-reasoning explanation for each site: why is the echo clear (A, B), merged and muddy (C) or absent (D)? They use three pieces of evidence: delay compared with 0.1 s, distance compared with 17 m, and surface or spreading (cliff rock 90% reflects; the Kisumu hall has many hard walls and roof so reflections keep overlapping, giving reverberation of about 1.2 s, loud, muddy and drawn out; the Nairobi grassy field has grass and bushes that absorb 10% reflection and the sound spreads over a huge area, and the only big wall is 300 m away so the 1.76 s return is too faint). Learners explain why a long delay does not guarantee an audible echo (site D) and why a short delay does not mean no sound comes back (site C). They	Say: 'Choose one site. Make a claim, give the evidence from our numbers, and give the reasoning using what we know about reflection.' Provide the sentence frame: 'The echo at ____ is ____ because the delay is ____ s, the distance is ____ m and the surface is ____.' WAIT TIME 10-15 seconds before groups begin. Then run a 'whip around': cold-call 4 groups, one for each site. Challenge with: 'Site D has the longest delay, 1.76 s. Why is there no echo?' WAIT TIME 10-15 seconds, then cold-call 2 learners. Press on precision: 'Is the sound being destroyed or is its energy absorbed and spread? Where does the energy go?' Revisit the Predict ideas: 'Who changed their minimum-distance guess? What evidence changed it?' Cold-call 3 learners. Correct misconceptions such as 'soft	Claim-Evidence-Reasoning (CER) with sentence frames. It forces learners to connect numerical evidence (delay, distance, percentage reflected) to the wave idea of reflection and absorption, and comparing four contrasting cases helps them separate the roles of distance, surface and spreading.	Indicators: explanations name all three factors (delay compared with 0.1 s, distance compared with 17 m, surface and spreading) for at least two sites; learners use correct terms echo and reverberation; they state that the echo-distance formula uses the round trip time. Look for a correct statement that site C has delay 0.07 s so the echo merges with the voice, and that site D fails because of weak returning energy rather than time. Quick exit check: 'A learner hears an echo 0.40 s after a clap. How far is the wall?' Expected answer: $340 \times 0.40 / 2 = 68$ m.

	compare with Wanjiru's original question and use Otieno's idea that the hall sounds muddy and Amina's sketch to complete the explanation. They also contrast echo (distinct reflected sound heard at least 0.1 s later) with reverberation (overlapping reflections merging and prolonging sound).	surfaces make no reflection at all' by pointing to 10% and 15% values.		
Driving Question Board (DQB) Creation  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Sound And Waves Source: PhET  Search ARES for similar readings	Learners revisit the Driving Question Board: 'How do sound waves reflect, and why do some places produce stronger echoes than others?' In groups they place sticky notes in two columns: 'Questions we can now answer' (for example: Why is there a delay? Why does the Kisumu hall sound muddy? Why is there no echo on the Nairobi field? How far is the cliff?) and 'New questions' (for example: Does the shape of the hall matter? How do bats and ultrasound use echoes? Does reflection from water behave the same way? What happens when sound goes into water? Why does sound bend around a building?). They write the evidence beside each answered question, such as '0.50 s gives 85 m' or '10 intervals in 2.94 s gives about 50 m'.	Say: 'Look at our board. Which questions can we cross off or answer today, and what evidence will you cite?' WAIT TIME 10-15 seconds. Cold-call 3 groups to move a question to the answered column and explain it with data. Then ask: 'What new question came up in the field that we cannot yet answer?' Cold-call 3 learners. Group new questions into themes (for example 'sound entering a new medium' for the next lesson, 'bending around corners', 'overlapping sounds'), and say: 'Some of these will be tackled in the coming lessons; keep them on the board.' Point out that Lesson 5 will look at what happens when a wave enters a second medium.	Driving Question Board as a public record of learning. Linking answered questions to specific evidence makes progress visible, and student-generated follow-up questions keep the unit coherent and give purpose to later lessons.	Indicators: at least 4 questions are moved to the answered column with a cited number or observation; at least 3 new meaningful questions are posted that relate to wave behaviour (for example refraction or diffraction), not only to facts; learners can explain why a question stays open. Note which questions show lingering confusion about reverberation versus echo.
Model Building Phase  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12	Learners revise the class echo model started in Lesson 1. Using the Lesson 3 idea of reflection following the law of reflection, each group draws a labelled diagram (Amina leads the sketching) with: the shouting learner, the straight sound ray to the wall, the reflected ray, the normal, the one-way distance d, and the round trip 2d. Next to the diagram they add the new rules: $d = vt/2$; echo heard only when t is at least 0.1 s and	Say: 'Revise your model so that a classmate who was absent could use it to predict whether a new place gives an echo.' Give a test case: 'A hard wall is 10 m away. Will you hear a separate echo? What about a wall 40 m away covered with thick curtains?' WAIT TIME 10-15 seconds, then cold-call 3 groups to use their model. Circulate and ask: 'Where on your model does the surface type appear? What is missing for the	Cumulative model revision with prediction tests. Using the model to predict new cases tests whether the ideas are usable, and annotating gaps shows what later lessons (refraction, diffraction, interference) must add.	Indicators: models include a normal and equal angles of incidence and reflection, the formula $d = vt/2$ with 340 m/s, the 0.1 s and 17 m condition, and surface reflection percentages; learners predict correctly that a wall 10 m away gives no distinct echo and the curtained wall gives a weak echo; at least one gap is identified. Use a 4-point rubric: ray and reflection shown, formula and threshold stated, surface effect

Search ARES for similar videos HTML: Reflection of Mechanical Waves (Flexbook) Source: CK-12 Search ARES for similar readings	the reflector is at least 17 m away; reverberation when reflections overlap; strong echo from hard smooth surfaces (rock 90%, concrete 85%) and weak from soft surfaces (curtains 15%, grass or foam 10%). They annotate the four sites A to D with delay and surface and mark what remains unexplained.	Kisumu hall?' Have 2 groups present with a gallery walk, and ask others to add one question or suggestion on a sticky note. Close by asking: 'What part of the model do we still not understand, such as what happens when the wave moves into a new medium?'	explained, test cases predicted correctly.
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D. TEACHER REFLECTION

After the lesson reflect: Did learners treat t as the round trip and halve it correctly? Which groups confused echo with reverberation, and what evidence helped them separate the two? Were the field results close to 50 m, and did learners explain differences through reaction time, wind or the position of the wall? Did the surface table help learners explain site D, or did they still believe a long delay guarantees an echo? Which learners relied on partners during calculations, and how can roles be rotated next time? Record the questions left on the DQB about refraction and diffraction to shape Lesson 5. If no suitable wall exists, plan to use a school building or a hillside, adjusting the distance but keeping the 340 m/s figure.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that sound clapped at about 50 m from the school wall returned after about 0.294 s per interval (10 intervals in 2.94 s), that the delays at the four sites were 0.50 s (Hell's Gate), 1.00 s (Ngong Hills), 0.07 s (Kisumu hall) and 1.76 s (Nairobi grounds), and that hard surfaces such as rock (90%) and painted concrete (85%) return much more sound than curtains (15%) or grass and foam (10%).
What did I learn?	We learned that $d = vt/2$ gives the one-way distance from the round trip time, that a distinct echo needs a delay of at least 0.1 s which means a reflector at least 17 m away, and that strong echoes need hard smooth surfaces while soft surfaces absorb sound. Reverberation is overlapping reflections that merge and prolong the sound.
How does this explain the phenomenon?	We can explain that the voice returns after a delay of $2d/v$ because sound reflects from the cliff, valley wall or building. Echoes are clear at Hell's Gate and Ngong Hills because the surfaces are far (more than 17 m) and hard; the Kisumu hall is muddy because the wall is only 12 m away (0.07 s) and many hard surfaces keep reflecting; and the Nairobi field gives no echo because the sound spreads out, grass absorbs it and the only big wall is 300 m away, so the return is too faint.

LESSON 5 (40 minutes (one single lesson; a double period can be used if the practical is done in full)): Refraction of Waves: Why Waves Change Direction When Their Speed Changes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain that a wave changes direction (refracts) when its speed changes as it enters a new medium, using the bent straw, apparent depth and water waves moving from deep to shallow. They then link this to the echo phenomenon: reflection sends a wave back into the same medium, whereas refraction describes the part of a wave that goes on into a second medium at a different speed.
Knowledge	Learners will: (1) describe refraction as the change in direction of a wave when it passes from one medium to another because its speed changes; (2) state that light travels at 3.0×10^8 m/s in air, 2.26×10^8 m/s in water ($3.0 \times 10^8 / 1.33$) and 2.0×10^8 m/s in glass, so it slows down going from air into water or glass; (3) state that sound travels at 340 m/s in air and about 1500 m/s in water, so sound speeds UP going from air into water; (4) explain apparent depth: a Lake Victoria shallow pool of real depth 1.33 m looks 1.00 m deep because rays from the bottom bend away from the normal as they leave the water; (5) state that a wave entering along the normal is not bent but still changes speed; (6) distinguish refraction from reflection (reflection: wave stays in the first medium, $i = r$; refraction: wave enters the second medium and its speed changes).
Skills	NGSS Science and Engineering Practices embedded: (SEP 3) Planning and carrying out investigations: learners trace the bent straw and apparent-depth observations with a ruler and pencil; (SEP 4) Analyzing and interpreting data: learners compare the speeds of light and sound in different media and relate them to the direction of bending; (SEP 2) Developing and using models: learners draw wave-front and ray diagrams for deep-to-shallow water and air-to-water, and add refraction to the class echo model; (SEP 6) Constructing explanations: learners write a claim-evidence-reasoning statement explaining why the straw looks broken. KICD skills: observing, measuring angles from the normal, drawing ray diagrams, communicating scientifically.
Attitudes	Core competencies: Communication and collaboration (teamwork in the bent-straw and apparent-depth stations, sharing findings); Learning to learn (revising earlier ideas about reflection); Digital literacy (using offline downloaded video demonstrations and the PhET Bending Light or ripple tank simulation). Values: Responsibility (each learner holds a role: observer, recorder, measurer, reporter); Respect (listening to peers' predictions, respecting different ideas). PCI: Safety and Security Education (careful handling of glass beakers and water). Appreciation of how everyday observations at Lake Victoria and Kenyan streams are explained by physics.
Key Inquiry Question	KICD Key Inquiry Question 1: How do waves behave when they meet obstacles or change medium? (Today we focus on the change-of-medium part.) Link to the driving question: How do sound waves reflect, and why do some places produce stronger echoes than others? Lesson question: What happens to a wave when it does not bounce back from a boundary but goes into a second medium, and why does it change direction?
Purpose in Storyline	Lessons 3 and 4 showed that waves reflect from a surface (the law of reflection $i = r$, and the echo-distance formula $d = vt/2$ with hard surfaces reflecting about 85-90% of sound energy and soft ones about 10-15%). Lesson 5 asks the learners what happens to the part of the wave that is not reflected: it enters the second medium and changes speed and direction. This completes the picture of a wave meeting a boundary (reflection plus refraction) and prepares Lesson 6 (Snell's law $n_1 \sin i = n_2 \sin r$ and lenses) and Lesson 7 (diffraction). Learners see that sound speed depends on the medium (340 m/s in air versus 1500 m/s in water), which is why a wave behaves differently in different media, whereas the echo at the cliff face stays in the air.
Safety Notes	Use clear plastic cups or beakers on a flat, dry table, away from the edge. Wipe up spilled water immediately to prevent slipping. Do not use glass that is chipped or cracked. Do not look at the sun through water or glass; use torch or classroom light only. Laser pointers are NOT used; use a torch with a slit made from card. Learners keep drinking straws out of mouths and do not share them. Wash hands after handling the water containers. Keep electrical devices (tablet, projector) away from water.

B. LESSON OVERVIEW

Back at school after the Hell's Gate trip, Mr. Mwangi's class has explained echoes by reflection: the shout hits the cliff face 85 m away and returns after 0.50 s. Kiprono then asks, 'When Amina put a straw in her glass of water at lunch it looked broken. Is that another way waves meet a boundary?' This lesson uses that question to investigate refraction. Learners first predict what a straw in water will look like and how deep a Lake Victoria shallow pool will appear. They then observe a straw in a cup, trace the apparent depth of a coin, and watch downloaded video demonstrations of water waves passing from deep to shallow water. They compare the speed of light in air (3.0×10^8 m/s), water (2.26×10^8 m/s) and glass (2.0×10^8 m/s), and the speed of sound in air (340 m/s) and water (about 1500 m/s). In the Explain phase they build the idea that a change in speed at a boundary causes a change in direction, and that a pool of real depth 1.33 m looks only 1.00 m deep. They then add the new idea to the Driving Question Board and the class echo model: at a boundary some wave energy is reflected (making the echo) and some goes into the second medium and refracts.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Refraction and frequency Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Refraction: Light Entering Water (PLIX) Source: CK-12  Search ARES for similar readings	(8 minutes) Wanjiru begins the lesson by recalling the Hell's Gate echo: 'The sound hit the cliff and came back. But what about the part that does not come back?' Each learner receives a clear cup half filled with water and a straw standing in it, but they are told NOT to move or look sideways yet. On a Predict sheet learners (1) sketch what they think the straw will look like when viewed from the side, (2) predict whether a coin at the bottom of a 1.33 m deep Lake Victoria pool will look deeper than, shallower than, or at the same depth as its real depth, and (3) predict whether sound will travel faster or slower in water than in air, and light faster or slower in water than in air. They give a reason using 'I think... because...' Pairs then share with a neighbour.	Show a cup with a straw covered by a cloth so the straw bend is not seen. Say: 'Yesterday we explained how the shout bounces off the cliff at Hell's Gate. Today a straw in water looks broken, just like a wave doing something new at a boundary. Before we look, what do you think will happen? Write your prediction and a reason.' Give silent writing time (2 minutes), then WAIT TIME of 10-15 seconds after asking: 'Wanjiru said some of the wave does not come back. Where does it go?' Cold-call 4 learners (not volunteers) to share predictions. Record on the board without judging: 'Straw looks bent / straight / broken', 'Coin deeper / shallower'. Probe: 'Otieno, what made you think the coin will look deeper?' Ask: 'Does anybody think the speed of the wave changes in water?' Note the number of learners who predict faster, slower or the same speed for sound in water and for light in water. Do NOT correct predictions.	Predict-Observe-Explain (POE) with a commitment sheet: writing a prediction and a reason before seeing evidence makes learners' prior ideas (for example 'the straw really breaks' or 'water is thicker so everything is slower') visible, so that the later evidence creates cognitive conflict. Cold-calling and recording all predictions on the board builds a shared record to return to in the Explain Phase.	Collect Predict sheets. Observable indicators: (1) learner gives a reason, not just a guess; (2) learner mentions speed, medium or direction; (3) learner links the straw to waves/boundaries. Tally on the board how many predict the coin looks shallower. Common misconceptions to listen for: 'the straw really bends in water', 'refraction is the same as reflection', 'sound is slower in water because water is denser'. Note them for the Explain Phase.

Observe Phase  VIDEO: Refraction in a glass of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Refraction of Mechanical Waves (Flexbook) Source: CK-12  Search ARES for similar readings	(14 minutes) Learners rotate through three stations in groups of four (Wanjiru, Otieno, Kiprono and Amina's roles are used as role cards: questioner, measurer, recorder, reporter). Station 1, Bent straw: view the straw in the cup from the side, from above and from directly above (along the normal), and sketch each view. Station 2, Apparent depth: place a coin in an opaque cup, move back until the coin just disappears behind the rim, then have a partner pour water slowly and watch the coin reappear. Trace the sight line with a ruler on the cup sketch. Station 3, Offline video/simulation: watch the downloaded YouTube demonstrations of (a) a ripple tank where water waves move from deep to shallow water and the wave fronts bend while wavelength becomes shorter, and (b) a torch beam passing from air into water in a glass tank, plus the PhET Bending Light simulation if a tablet is available. Learners fill in an observation table: Medium 1 / Medium 2 / Did the wave bend? / Toward or away from the normal / Speed bigger or smaller? Then they study the data card: light in air 3.0×10^8 m/s, in water 2.26×10^8 m/s ($3.0 \times 10^8 / 1.33$), in glass 2.0×10^8 m/s; sound in air 340 m/s, in water about 1500 m/s; and a Lake Victoria pool whose real depth is 1.33 m but looks 1.00 m deep ($1.33/1.33$).	Circulate with a checklist. Say at Station 1: 'Look straight down the straw from above. Is it bent now? What changed about your viewing direction?' (Looking along the normal shows no bend.) At Station 2: 'Mark where the coin looks to be, then mark where it really is. Which one is nearer the surface?' At Station 3 pause the video at the deep-to-shallow transition and ask: 'What do you notice about the spacing of the wave fronts?' WAIT TIME of 10-15 seconds before taking answers. Ask each group's reporter (cold-call 4 groups) to give ONE observation using the stem 'We observed... when the wave went from ___ to ___.' Prompt measurers to use a protractor to measure angles from the normal, not from the surface (convention from reflection in Lesson 3). Ask: 'Kiprono, if the pool in Kericho is 1.33 m deep, how deep will it look?' (1.00 m.) Remind learners that glass cups must be handled carefully and spills wiped at once.	Gallery observation with an Observation-Data table: rotating through three different phenomena (straw, coin, water waves) lets learners notice the same pattern in different contexts. The data card (light and sound speeds in different media) pushes them to look for what changes when the wave bends (speed), not just that it bends. Pairing visual evidence with numbers helps learners move from 'it looks bent' to 'the speed has changed'.	Observable indicators: (1) learners draw the straw/coin sight line with the bend at the water surface, not at the straw; (2) learners measure angles from the normal; (3) learners record that no bend occurs along the normal; (4) learners record that wavelength shortens in shallow water; (5) learners fill in the observation table with 'slower in water/glass' for light and 'faster in water' for sound. Teacher checklist: tick each group that correctly states the apparent depth as 1.00 m for a real depth of 1.33 m. Listen for the misconception 'the water magnifies so it looks closer' and redirect with the sight-line trace.
Explain Phase  VIDEO: Refraction in water Source: Khan Academy (English	(10 minutes) In groups of four, learners analyse their evidence to answer: 'What happens to a wave's direction when its speed changes as it enters a new medium, and why does the straw	Say: 'Use your evidence, not your memory. What is the claim? What is the evidence? What is the reasoning that links them?' Use an analogy: 'Imagine a matatu driving from tarmac onto a muddy patch at	Claim-Evidence-Reasoning (CER) with an analogy-and-limits check: CER requires learners to link observations to a scientific principle (speed change causes direction change). The matatu-on-	Observable indicators: (1) CER statement names the speed change as the cause of bending; (2) the ray diagram shows the normal, a ray bending away from the normal as it leaves water, and

- US curriculum) Search ARES for similar videos EXERCISE: Refraction of Mechanical Waves Practice Source: CK-12 Search ARES for similar readings	look broken?' They (1) write a Claim-Evidence-Reasoning (CER) statement about the bent straw; (2) draw a ray diagram of the coin in water showing the real ray bending away from the normal as it leaves the water into air, with the eye tracing it back to a shallower apparent position; (3) draw wave fronts for deep-to-shallow water showing that the part of the wave front that enters the shallow water first slows down, so the wave front swings and the direction changes; (4) use the data card to complete: 'Light is slower in water (2.26×10^8 m/s) than in air (3.0×10^8 m/s), so it bends ___ the normal when entering water.' (toward). They also compare sound: 'Sound goes from 340 m/s in air to 1500 m/s in water, so it speeds up and bends ___ the normal' (away). Finally, learners write one sentence distinguishing reflection from refraction using the Hell's Gate cliff: 'At the cliff the sound reflects with $i = r$ and stays in air; refraction would happen only if the sound entered a different medium, such as water.'	an angle. The wheel that hits the mud first slows down, so the whole matatu turns. That is what happens to a wave front.' Then check the analogy: 'Where does the analogy fail?' WAIT TIME of 10-15 seconds. Cold-call 5 learners to give parts of the explanation (claim, evidence, reasoning, deep-to-shallow, light vs sound). Ask: 'Amina, show us your wave-front sketch. Which side slowed down first?' Clarify that refraction depends on a change in SPEED, so a wave crossing a boundary along the normal changes speed but not direction. Revisit predictions on the board and ask learners to mark each as supported or not supported. Say: 'In Lesson 4 we saw hard surfaces reflect about 85-90% of the sound. Where does the rest of the sound energy go?' (Absorbed, and some transmitted into the second medium.) Do NOT teach Snell's law or calculation yet (Lesson 6).	mud analogy gives a mental model of wave fronts turning, and asking where the analogy fails prevents learners from over-extending it. Returning to the Predict board closes the POE cycle.	the apparent image above the real coin; (3) learner states 'light slows in water' and 'sound speeds up in water'; (4) learner states that refraction needs no bend at 0 degrees (along the normal); (5) learner distinguishes reflection ($i = r$, same medium) from refraction (new medium). Quick exit check: 'In one sentence, why does the straw look broken?' Look for 'light changes speed and direction at the water surface'. Misconception check: 'the straw is physically bent' or 'refraction is the same as reflection' needs re-teaching.
Driving Question Board (DQB) Creation VIDEO: Refraction and frequency Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Bending Light Source: PhET Interactive	(4 minutes) Learners return to the class Driving Question Board, which already holds the driving question 'How do sound waves reflect, and why do some places produce stronger echoes than others?' and questions from Lessons 1-4. Each learner writes one question on a sticky note arising from today's work (for example 'Does the echo change if the cliff is wet?' or 'Why does sound speed up in water but light slows?'). The class sorts new notes into three groups: 'answered now', 'partly answered' and 'to be	Say: 'Look at the board. Which of our questions can we now answer better because of the bent straw and the water waves?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to place a sticky note and justify the placement with evidence. Ask: 'What new question has refraction raised about our four echo sites?' Introduce one deliberate question if none appears: 'If sound moves from air into a wall, would it bend?' Reinforce that Lesson 6 will quantify bending with Snell's law ($n_1 \sin i = n_2 \sin r$) and Lesson 7	Driving Question Board with 'answered / partly answered / to be investigated' sorting: learners see their own questions drive the sequence, take ownership of what remains unknown, and recognise how each lesson adds evidence toward the big driving question.	Observable indicators: (1) new questions refer to refraction, speed change or the echo sites, showing connections to the phenomenon; (2) learners justify a note's placement using evidence from today's observations; (3) at least half the class contributes a note; (4) learners move at least one earlier question to 'partly answered'. Note any learner whose questions show lingering confusion between reflection and refraction.

Simulations (English) Search ARES for similar readings	investigated'. They also move earlier questions, such as 'Where does the sound that does not come back go?', to 'partly answered' and note that the part that is absorbed or transmitted explains some weak echoes.	will explore how waves spread around edges. Keep the original driving question visible. Make sure at least one question remains about why the Kisumu hall sounds muddy (to be resolved by interference and reverberation in later lessons).		
Model Building Phase  VIDEO: Refraction in a glass of water Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Refraction: Light Entering Water (PLIX) Source: CK-12 Search ARES for similar readings	(4 minutes) Amina leads the group in revising the class echo model drawn in earlier lessons (shout, ray to the cliff, reflected ray back to the ear, $d = vt/2$, reflecting percentage by surface). Groups add a new layer: at a boundary, the wave energy splits into (1) a reflected wave that stays in the first medium (this gives the echo) and (2) a refracted wave that enters the second medium with a new speed and direction. They draw a boundary with the normal, an incident ray, a reflected ray with $i = r$, and a refracted ray bending toward the normal when the wave slows or away from it when the wave speeds up. They label the speeds: light 3.0×10^8 m/s in air and 2.26×10^8 m/s in water; sound 340 m/s in air and 1500 m/s in water. They also add a note: 'A bigger share reflected means a stronger echo; a bigger share absorbed or transmitted means a weaker echo', linking it to the surface table (cliff rock 90%, painted concrete 85%, wooden board 60%, curtains 15%, thick grass 10%). Each group states one limitation: 'The model does not yet explain why the Kisumu hall sounds muddy or the quantity of bending (Lesson 6).'	Say: 'Our model so far shows the sound going to the wall and coming back. Today we learned that a wave can also go into a new medium. How should we change the model?' WAIT TIME of 10-15 seconds. Cold-call 3 groups to show one revision, and ask other groups to compare: 'What is the same and different in your diagrams?' Check angles are measured from the normal. Press for the link to the phenomenon: 'At Hell's Gate, did the shout refract?' (Sound that stayed in air only reflected; refraction would occur if it entered a different medium such as water or rock.) Ask groups to keep their revised model for the next lesson and to add one prediction: 'If light bends toward the normal in water, what could we calculate next?' Close by pointing to the board: 'Reflection sends the wave back, refraction sends it on in a new direction; both are part of what a wave does at a boundary.'	Cumulative model revision with a stated limitation: adding refraction to the existing echo model shows learners how a model grows with new evidence rather than being replaced. Naming a limitation keeps the model honest and prepares the next lessons, which will add Snell's law calculations, diffraction and interference.	Observable indicators: (1) the model shows the normal, incident ray, reflected ray with $i = r$ and a refracted ray; (2) the refracted ray bends toward the normal for a slower second medium and away for a faster one; (3) speeds are labelled with the correct values; (4) the model separates the reflected share (echo) from the refracted/absorbed share; (5) learners state a sensible limitation. Rubric: 3 = all five indicators present; 2 = three or four; 1 = two or fewer, which requires a short conference before Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners separate the idea that the straw looks bent (an optical effect at the boundary) from the idea that the straw is physically bent? Which learners still said 'the straw breaks'? (2) How many learners correctly stated that light slows down in water while sound speeds up (340 m/s to 1500 m/s)? Did this surprise them, and how did they react? (3) Did anyone confuse refraction with reflection when linking back to Hell's Gate? Which questions or sketches showed this? (4) Was the 10-15 second WAIT TIME used each time, and did it produce deeper answers from the learners I cold-called? (5) Did the stations run on time, and was glassware handled safely? (6) Which group's model best showed the reflected and refracted split, and could it be used as the class model? (7) What adjustments are needed before Lesson 6, when learners will calculate bending with Snell's law ($n_1 \sin i = n_2 \sin r$) using $n = 1.00, 1.33$ and 1.50 , for learners who still cannot measure angles from the normal?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a straw standing in water looks bent or broken at the water surface, a coin in a cup appears when water is poured in, and a Lake Victoria pool of real depth 1.33 m looks only 1.00 m deep. In the ripple tank videos, water waves moving from deep to shallow water changed direction and had shorter wavelengths. We saw no bend when we looked straight down along the normal.
What did I learn?	We learned that refraction is the change in direction of a wave when it passes from one medium to another because its speed changes. Light travels at 3.0×10^8 m/s in air, 2.26×10^8 m/s in water and 2.0×10^8 m/s in glass, so it slows down and bends toward the normal when entering water or glass. Sound travels at 340 m/s in air and about 1500 m/s in water, so it speeds up. Reflection keeps the wave in the same medium ($i = r$), while refraction sends it into the second medium.
How does this explain the phenomenon?	The straw looks broken and the pool looks shallower (1.00 m instead of 1.33 m) because light changes speed and direction at the water surface, so rays from the bottom bend away from the normal and our eyes trace them back to a higher apparent position. This adds to the echo model: at a boundary, part of the wave is reflected (the echo we hear at Hell's Gate after 0.50 s and Ngong Hills after 1.00 s) and the rest is absorbed or goes into the second medium and refracts. The more energy that is reflected (cliff rock 90%) the stronger the echo, and the more that is absorbed (grass 10%) the weaker the echo.

LESSON 6 (40 minutes (single lesson; extend to 80 minutes if practical ray tracing is done)): Snell's Law, Refractive Index and Lenses: Bending Waves in Our Echo Model

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate refraction using $n_1 \sin i = n_2 \sin r$, draw ray diagrams for converging and diverging lenses, and link lens uses to spectacles, cameras and microscopes. They then decide what refraction adds to the echo model (the sound wave can reflect back, and it can also bend into a new medium) and what it leaves unexplained.
Knowledge	Refractive index n has no units (air 1.00, water 1.33, glass 1.50). Snell's law is $n_1 \sin i = n_2 \sin r$, with i and r measured from the normal. Light entering a medium with a higher n bends towards the normal because its speed drops (Lesson 5: speed change causes bending). A converging lens brings parallel rays to a focal point, and a diverging lens spreads them as if from a focal point. Uses: spectacles, cameras, microscopes.
Skills	Calculating with Snell's law ($\sin 45 = 0.707$: water $r = 32$ degrees, glass $r = 28$ degrees); measuring angles from the normal (Lesson 3); drawing ray diagrams for lenses; using models (NGSS SEP: Developing and Using Models); analysing and interpreting data; using mathematical thinking; constructing explanations; revising a class model.
Attitudes	Curiosity about how spectacles and cameras bend light; precision in measuring and drawing; respect for peers' ideas in group work; appreciation of physics in everyday life, including eye care in Kenyan communities. Core competencies: critical thinking and problem solving; communication and collaboration; learning to learn. Values: responsibility, respect. PCI: health (eye care and vision screening).
Key Inquiry Question	How do waves behave when they meet obstacles or change medium? How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 3 and 4 showed that sound reflects from a surface, giving the Hell's Gate echo (0.50 s) and the Ngong Hills echo (1.00 s). Lesson 5 showed that waves bend when their speed changes. Lesson 6 makes this bending quantitative with Snell's law and applies it to lenses. Learners must decide which parts of the echo model need refraction and which do not. The Hell's Gate echo is reflection, not refraction. Refraction explains how part of a wave goes into a new medium instead of coming back, and how lenses focus light. Learners add a clearly labelled 'wave enters new medium' branch to the class model and record new DQB questions, such as whether a hall wall that absorbs sound is letting some of it enter the material.
Safety Notes	Use low-power torches, laser pointers (if used) must be class 2 and never pointed at eyes. Handle glass blocks and lenses by the edges and keep them away from table edges. Never look at the Sun through a lens. Use water in a transparent container on a stable tray and wipe up spills immediately. Do not look into the light from a torch at close range. Hearing is not a concern in this lesson.

B. LESSON OVERVIEW

Mr. Mwangi starts by recalling the Hell's Gate trip. The learners' shout went to the cliff, reflected and returned after 0.50 s. Then he asks what would happen if the wave met a boundary and went into a different medium instead. Amina has noticed that her spectacles bend the view of the board. Learners predict how a light ray bends from air into water and into glass at 45 degrees. They then trace rays through glass blocks and water with torches and protractors, measuring angles from the normal as in Lesson 3. They test Snell's law, $n_1 \sin i = n_2 \sin r$: with $i = 45$ degrees and $\sin 45 = 0.707$, water ($n = 1.33$) gives $\sin r = 0.532$ and $r = 32$ degrees, and glass ($n = 1.50$) gives $\sin r = 0.471$ and $r = 28$ degrees. They then draw ray diagrams for a converging lens (hand lens or spectacle lens) and a diverging lens and connect them to spectacles, cameras and microscopes. In the Explain phase they use light speed in each medium (3.0×10^8 m/s in air, 2.26×10^8 m/s in water, 2.0×10^8 m/s in glass) to explain why n is higher where light is slower. They add the new idea to the DQB and class model: reflection (Lessons 3 and 4) sends the wave back to cause the echo, while refraction sends part of the wave into a new medium. Refraction does not by itself explain the echo at Hell's Gate, and learners must say so explicitly.

Reflection is what returns the sound.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Refraction and Snell's law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners see a pencil in a glass of water and a hand lens held over printed text. Individually they write for 3 minutes: (1) 'A ray of light in air hits the surface of water at 45 degrees from the normal. Will it bend towards or away from the normal in the water? Will it bend more or less in glass?' Sketch a prediction on a ray diagram with the normal drawn. (2) 'Wanjiru's shout reflected from the Hell's Gate cliff and came back after 0.50 s. Would a wave that goes into a different material behave the same way?' Learners then share in pairs. Kiprono says glass should bend light the most because it is hardest. Otieno thinks the bend depends on how slowly the light travels there.	Hold up the glass of water with a pencil and say: 'Remember Hell's Gate: the shout hit the cliff and came back. Today the wave does not come back, it goes in. What do you predict?' Say: 'Draw the normal first, as we did with the mirror in Lesson 3, then show your ray.' Give WAIT TIME of 10-15 seconds in silence after the question before any pair talk. Cold-call 4 learners (not volunteers) to state 'towards' or 'away' and a reason. Record all predictions in two columns on the board (towards / away) without saying which is correct. Ask: 'Who changed their mind after hearing a partner? What was the reason?' Do not confirm answers.	Predict-Share-Record: committing to a written prediction before instruction exposes initial ideas (for example, 'harder means more bending') so that later evidence can confirm or challenge them. Sharing with a partner lets learners test their reasoning socially before the observations.	Check the written predictions. Look for: (a) the normal drawn before the ray; (b) a ray drawn bent in a specific direction with an angle estimate; (c) a reason using speed or 'denser'. Tally the class: about how many say 'towards the normal' for water. Note learners who measure from the surface rather than from the normal and re-teach the Lesson 3 convention. Record at least 3 distinct reasons on the board for use in the Explain phase.
Observe Phase  VIDEO: Snell's law example 1 Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar readings	In groups of four (Wanjiru, Otieno, Kiprono, Amina style roles: measurer, recorder, drawer, checker) learners do two stations of about 10 minutes each. Station 1: Refraction. Place a rectangular glass block on white paper, shine a thin torch beam or ray box at $i = 45$ degrees from the normal, mark the entry and exit points, remove the block and draw the ray inside. Measure r with a protractor from the normal. Repeat with a transparent container of water. Expected: water r about 32 degrees and glass r about 28 degrees (accept 2-3 degrees of measurement error). Calculate	Circulate and say: 'Show me the normal. Where are you measuring your angle from?' Ask: 'Your water angle is smaller than 45 degrees. Which way did the ray bend, and does your measurement match the prediction on the board?' Use WAIT TIME of 10-15 seconds after each question before helping. Call on 3 groups at random to report their r for water and glass; write all values on the board and ask: 'Why are the numbers close but not identical?' (measurement error: line thickness, protractor reading). For lenses ask: 'Where do the rays meet, and what would you place there in a camera?' Remind	Data table and Claim-Evidence-Reasoning: groups record measured i and r , calculate the Snell's law value and compare. They state a claim ('light bends towards the normal when entering a medium with a higher n '), cite their evidence and give a reason. Seeing that the same law gives 32 degrees and 28 degrees across different groups builds confidence in the model, and the contrast with the mirror (where the ray returns) separates reflection from refraction.	Observable indicators: (a) angles measured from the normal, not the surface; (b) measured r within about 3 degrees of 32 degrees (water) and 28 degrees (glass); (c) correct sin calculation shown, for example $\sin r = 0.707/1.33 = 0.532$; (d) lens diagram with the focal point labelled and arrows on rays; (e) the converging lens rays meet while diverging rays spread. Use a quick checklist per group. Ask each group to explain one reason their values differ slightly from the calculation. Groups with r larger than i for water need to recheck the normal and the direction of the bend.

	using Snell's law with $n(\text{air}) = 1.00$, $n(\text{water}) = 1.33$ and $n(\text{glass}) = 1.50$: $\sin r = 0.707/1.33 = 0.532$ so $r = 32$ degrees; $\sin r = 0.707/1.50 = 0.471$ so $r = 28$ degrees. Compare the measured and calculated values in a table. Station 2: Lenses. Use a converging (convex) lens and a diverging (concave) lens with parallel torch rays or a ray box. Draw the rays: the converging lens brings parallel rays to a focal point; the diverging lens spreads them as if from a focal point behind the lens. Use a hand lens to focus a distant window onto a paper screen and measure the lens-to-screen distance as the approximate focal length. Alternative if no apparatus: PhET Bending Light and Geometric Optics simulations, with the same angle of 45 degrees.	learners never to look at the Sun through a lens. Ask one group: 'The shout at Hell's Gate came back. Did any of your rays come back out of the block towards the torch? Why is that different from the mirror in Lesson 3?'		
Explain Phase  VIDEO: Snell's law example 2 Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Whole class. Learners first compare their data with the earlier predictions: light bent towards the normal in both water and glass, and bent more in glass (28 degrees) than water (32 degrees). Then they build the explanation using speeds: light travels at 3.0×10^8 m/s in air, 2.26×10^8 m/s in water ($3.0 \times 10^8 / 1.33$) and 2.0×10^8 m/s in glass ($3.0 \times 10^8 / 1.50$). So $n = c/v$, and a higher n means slower light and more bending towards the normal (Lesson 5: speed change causes bending). Learners then solve two practice problems in pairs: (1) A ray in air hits glass at $i = 45$ degrees. Find r. (Answer: 28 degrees.) (2) Find the speed of light in glass. (Answer: 2.0×10^8 m/s.) Next they connect lenses to uses: a converging lens in	Ask: 'Light is slower in glass than in water. Which bends more? Use the speed numbers in your answer.' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. Write $n = c/v$ and the three speeds on the board. Say: 'Show me your working: $n_1 \sin i = n_2 \sin r$, with $n_1 = 1.00$ and $\sin 45 = 0.707$.' Walk around while pairs solve problem 1 and problem 2 and select two different methods to display. Ask: 'A Kisumu learner has blurry vision at the back of the class. Which lens would you suggest, converging or diverging, and what does it do to the rays?' Then pose the link: 'At Hell's Gate the sound came back. Is that refraction or reflection? How do you know?' Cold-call 2 learners and correct the misconception that refraction causes an echo. Say: 'Refraction	Explanatory model with mathematical reasoning: learners link observation (angles), mathematics (Snell's law, $n = c/v$) and mechanism (speed change) in a single explanation, and apply it to lenses and to spectacles, cameras and microscopes. A Compare-and-Contrast table of reflection (wave returns, angle $r = i$, echo) and refraction (wave enters new medium, speed changes, direction bends) prevents confusing the two behaviours.	Look for: (a) correct use of Snell's law with $\sin 45 = 0.707$ and $r = 28$ degrees for glass; (b) speed of light in glass 2.0×10^8 m/s; (c) explanation that higher n means slower light and bending towards the normal; (d) correct pairing of converging lens with long sight and diverging with short sight; (e) in the exit sentence, learners say reflection returns the wave (the echo) while refraction takes the wave into another medium. Use the exit sentences as evidence; flag any learner who says 'the echo is caused by refraction' for a follow-up conversation next lesson.

	spectacles corrects long sight, a diverging lens corrects short sight (as in a learner who cannot read the board from the back of a Nairobi classroom), a camera lens focuses an image on the sensor, and a microscope uses lenses to magnify a leaf cell from a sukuma wiki leaf. Finally, learners apply the idea to the phenomenon: at Hell's Gate the returning shout is a reflected wave, so it does not need Snell's law. Refraction describes the part of a wave that goes into the new medium. Learners write one sentence each: 'Reflection and refraction are different because...'	describes the part of the wave that goes into the new medium. The echo comes from the part that is reflected.'		
Driving Question Board (DQB) Creation  VIDEO: Refraction and Snell's law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners revisit the class DQB: 'How do sound waves reflect, and why do some places produce stronger echoes than others?' In groups they write new questions on sticky notes that arose today, for example: 'If light bends going into water, does sound bend too?' (sound speed in water is about 1500 m/s versus 340 m/s in air), 'When the Kisumu hall walls absorb sound, is some of it entering the material?', and 'Does a curved surface focus an echo like a lens focuses light?' They place notes under headings: Answered (for example, how to calculate refraction with Snell's law), Partly answered, New. They also mark which earlier questions are now answered, such as 'Why does the shout come back at Hell's Gate? (reflection, Lessons 3 and 4)'.	Say: 'Write one question that today's lesson has made you wonder about our four echo sites.' Give WAIT TIME of 10-15 seconds before writing. Cold-call 3 groups to read one note and say which cluster it fits. Ask: 'Which of our questions can we now answer with evidence, and which still need new ideas such as diffraction in Lesson 7?' Move answered notes to the Answered column and tick the evidence next to them. Point out the question on sound bending and say: 'Keep that one visible; we will return to what changes speed for sound in water.' Make sure at least one note links to site C (the Kisumu hall) and one to site D (Nairobi open grounds).	Question clustering and gap analysis: sorting learner questions into Answered / Partly answered / New shows progress and what evidence is still missing. It keeps the inquiry learner-driven and links each lesson back to the anchoring phenomenon.	Check that questions are testable and connected to the phenomenon, not just 'what is a lens?'. Count how many groups produce at least one question linking refraction to sound or to the echo sites. Note learners who can state why an earlier question is now answered, citing evidence such as 'r = 28 degrees in glass' or 'reflection gives the echo'. Collect sticky notes to plan Lesson 7 on diffraction.
Model Building Phase  VIDEO:	Each group revises the class echo model on a poster (Amina leads the drawing). The model so far: sound travels from the shouting	Say: 'Add to the model: what happens to the wave when it meets a boundary? Show both the part that comes back and the part	Cumulative model revision: learners add the new boundary behaviour (reflection plus refraction) to the existing echo	Look for models that: (a) show both reflected and refracted rays at a boundary with the normal; (b) keep $i = r$ for the reflected ray; (c)

Snell's law example 1 Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English) Search ARES for similar readings	learner as a wave, reflects from a large hard surface obeying the law of reflection, and returns after $t = 2d/340$ (Hell's Gate 0.50 s, Ngong 1.00 s, Kisumu hall 0.07 s with reverberation, Nairobi grounds no echo). Today's addition: draw a boundary and show two things happening to a wave that hits it: a reflected ray (angle $r = i$) and a refracted ray that enters the new medium and bends according to $n_1 \sin i = n_2 \sin r$. Add a labelled mini-diagram of light going from air into glass at 45 degrees to 28 degrees, and a lens diagram. Add a key box: 'Reflection returns the wave (echo). Refraction bends the wave that enters the new medium.' Add a note: 'Soft surfaces such as grass at the Nairobi grounds (reflect only about 10%) let more energy enter or be absorbed, so little returns.' Learners then complete a summary row: what we observed, what we learned, what we can explain.	that goes in.' Give WAIT TIME of 10-15 seconds, then circulate. Ask: 'Which part of your model explains the Hell's Gate echo at 0.50 s? Which part does not?' Cold-call 3 groups to explain a single feature of their model. Challenge: 'Does your model tell us why Ngong Hills gives a clear echo and the Kisumu hall does not? What is still missing?' (answer: the 17 m condition and reverberation come from earlier lessons; the effect of overlapping waves comes in Lesson 8). Ask learners to mark the model with a colour for 'new today' and a question mark for 'not yet explained'.	model, mark what is explained and what is not, and keep the model consistent with the numerical evidence (340 m/s, 0.50 s, 1.00 s, 0.07 s). Using colour for new ideas makes progress across lessons visible.	show the refracted ray bent towards the normal in a higher-n medium (28 degrees in glass); (d) do not attribute the echo to refraction; (e) include a question mark for something not yet explained. Use a 3-level rubric (not yet, partly, fully) and record which groups need help separating reflection from refraction.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners measure angles from the normal, as in Lesson 3, or from the surface? (2) Could they explain why glass ($n = 1.50$, $r = 28$ degrees) bends light more than water ($n = 1.33$, $r = 32$ degrees) using speeds? (3) Did any learners say refraction causes the echo, and did the compare-and-contrast table correct this? (4) Did the lens diagrams show the focal point clearly, and could learners link them to spectacles, cameras and microscopes? (5) Which DQB questions need to be carried into Lesson 7 (diffraction)? (6) Did all four group members take part in measuring, drawing and explaining? Plan extra support for learners who struggle with sine calculations, for example by using a calculator and a sine table.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Light bent towards the normal when it entered water (r about 32 degrees) and glass (r about 28 degrees) from air at $i = 45$ degrees; a converging lens brought parallel rays together at a focal point and a diverging lens spread them out.
What did I learn?	Snell's law $n_1 \sin i = n_2 \sin r$ with $n = 1.00$ (air), 1.33 (water) and 1.50 (glass); a higher n means slower light (2.26×10^8 m/s in water, 2.0×10^8 m/s in glass) and more bending towards the normal; converging and diverging lenses are used in spectacles, cameras and microscopes.

How does this explain the phenomenon?

At a boundary a wave can reflect and also refract. The Hell's Gate echo (0.50 s) and the Ngong Hills echo (1.00 s) come from reflection, which sends the sound back. Refraction describes the part of the wave that enters a new medium and bends because its speed changes. Refraction alone does not explain the echo, so the muddy Kisumu hall sound and the missing echo on the Nairobi grounds still need other wave ideas.

LESSON 7 (40 minutes (one double lesson may be used if the simulation needs more time)): Diffraction of Waves: Why Sound Turns Corners but Light Does Not

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how waves spread through gaps and around edges, so that they can explain why the sound of a shout is heard around a building or through a doorway while light seems to travel straight. This adds a new wave behaviour to the class echo model: the sound that leaves the shouting learner does not travel only in one straight beam. It spreads, and that spreading helps explain why the sound is weaker in open places such as the Nairobi grounds and why it fills the Kisumu hall.
Knowledge	Diffraction is the spreading of waves around edges and through gaps. Diffraction is greatest when the gap width is comparable to the wavelength and is small when the gap is much wider than the wavelength. Wavelength is found from $v = f \times \text{wavelength}$ ($\lambda = v/f$). A 340 Hz sound has $\lambda = 340/340 = 1.0$ m. A 1000 Hz sound has $\lambda = 340/1000 = 0.34$ m. Green light has λ of about 5.5×10^{-7} m. A classroom doorway is 1.0 m wide, comparable to the 340 Hz sound wavelength (strong spreading) and enormously larger than the light wavelength (almost no visible bending). Diffraction does not change the frequency or wavelength of the wave.
Skills	Planning and carrying out investigations (NGSS SEP 3): vary gap width and wavelength in the PhET Wave Interference simulation. Analysing and interpreting data (SEP 4): record gap-to-wavelength ratios and the amount of spreading. Using mathematics (SEP 5): calculate $\lambda = v/f$ for sound and compare it with the 1.0 m doorway. Developing and using models (SEP 2): add diffraction to the class echo model. Constructing explanations (SEP 6): explain why sound turns corners and light does not.
Attitudes	Curiosity about everyday hearing, such as hearing a voice from the next classroom. Collaboration and respect for peers' ideas during the predict-observe-explain routine. Honesty when reporting simulation results that differ from predictions. Responsibility when sharing roles at the computer or tablet. Appreciation of how physics explains daily life in Kenya.
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life? Lesson focus: When a wave meets an edge or a gap, how much does it spread, and what controls the amount?
Purpose in Storyline	Lesson 2 gave learners the ray model, in which light travels in straight lines. Lesson 4 gave them echo measurements at sites A to D and the idea that hard surfaces reflect strongly. Lesson 7 tests the limit of the ray model: Otieno and Wanjiru can hear Mr. Mwangi calling from around a corner even though they cannot see him. Diffraction explains this with wavelength. It prepares Lesson 8 (interference), in which diffracted and reflected waves overlap and meet, and it adds one more reason why a shout on the open Nairobi grounds spreads out and weakens. This lesson does not re-teach reflection or the echo formula; it uses them only as background evidence.
Safety Notes	Learners work with tablets, computers or a teacher projector, so keep cables tidy and clear of walkways. If a ripple tank is used, wipe up spilt water at once to prevent slips and keep water away from mains sockets. Do not use loud sounds for the doorway demonstration; speak or clap at normal level (about 60 dB, ordinary conversation). Learners with hearing sensitivity may stand slightly farther from the door. Never look directly at the sun or a bright torch beam when comparing light and sound.

B. LESSON OVERVIEW

Back on campus after the Hell's Gate and Ngong trip, Wanjiru notices that she can hear Mr. Mwangi's voice from behind the classroom block even though she cannot see him. Otieno says: 'In Kisumu, I can hear the assembly from around the corner of the hall too.' Yet when a torch is shone through the same doorway, the light makes a sharp patch with a clear edge. Learners first predict what happens to waves meeting a gap and whether sound and light will behave differently. They then use the PhET Wave Interference simulation (with a ripple tank if available) to change the gap width and wavelength and observe how much the waves spread. They calculate

the wavelengths of 340 Hz and 1000 Hz sounds (1.0 m and 0.34 m) and compare them with the 1.0 m doorway and with the wavelength of light (5.5×10^{-7} m). In the Explain phase they use a claim-evidence-reasoning frame to explain why sound turns corners more than light. Learners add new questions to the Driving Question Board and revise the class model to show that sound spreads as well as reflecting. They connect diffraction back to the phenomenon: the shout does not leave as one narrow beam, and spreading over a large open area helps explain why the Nairobi grounds give no echo.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview Source: CK-12  Search ARES for similar videos  HTML: Diffraction (Flexbook) Source: CK-12  Search ARES for similar readings	Learners gather round the classroom door, which is 1.0 m wide. Mr. Mwangi stands out of sight around the corner outside and speaks, then a learner shines a torch through the doorway onto the floor. Learners hear him clearly but cannot see him, and the torch makes a sharp-edged patch of light. In pairs, then individually in their notebooks, they draw a prediction diagram: a wave (sound and light) approaching a 1.0 m gap, showing what the wave looks like on the other side. They answer: 'Will a wave pass straight through a gap, or spread out? Will sound and light behave the same way? Why?' They link it to the trip: Wanjiru's shout at Hell's Gate left her as a wave, so what happens to the wave as it moves outward in open ground?	Say: 'Close your eyes. Where is Mr. Mwangi? Point to him. Now open your eyes. Can you see him?' Then ask: 'Why can we hear someone around a corner but not see them?' WAIT TIME 10-15 seconds in silence before taking any answer. Ask learners to draw before they talk. Cold-call 4 pairs to share one prediction each and write all predictions on the board without judging. Say: 'I am not looking for the right answer yet. I want your best thinking.' Ask: 'What would you need to measure to know which prediction is better?' Use probing: 'Wanjiru said the voice spreads out. What do you think she means?' Leave the unresolved predictions visible for the whole lesson.	Predict-Observe-Explain (POE) with individual drawn predictions: committing to a diagram before instruction exposes learners' initial ideas, such as 'sound goes in straight lines like light' or 'sound is just louder, so it gets round'. Their predictions are revisited in the Explain phase so that they can see their own thinking change.	Check prediction diagrams for: (1) waves drawn continuing as a straight beam (ray-model misconception), (2) waves drawn spreading after the gap, (3) any mention of wavelength or size of gap. Record on a tally on the board how many learners predicted sound and light would behave differently. Look for learners who refer to the earlier ray model of light travelling in straight lines (Lesson 2) or to echoes as sound bouncing back (Lesson 4). Note learners who give no reason as needing support during the Observe phase.
Observe Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview Source: CK-12  Search ARES for similar videos  EXERCISE:	In groups of 3-4 (roles: operator, recorder, timekeeper/checker, reporter), learners open the PhET Wave Interference simulation (water or sound view) or use a ripple tank if one is available. Task A: send plane waves toward a barrier with a gap. Keep the frequency fixed and change the gap width from wide to narrow, recording the amount of spreading on the far side (none, slight, strong). Task B: keep the gap the	Circulate and ask: 'What did you change? What stayed the same? What do you see on the far side of the gap?' Say: 'Change one thing at a time, and write down what you see before you move on.' At each group ask the reporter: 'Tell me the gap and the wavelength. Which is bigger?' Prompt: 'Look at where the waves go behind the barrier. Is it empty or are waves there?' Challenge groups that finish early: 'Put the gap at 3 wavelengths.	Structured data table with controlled variables: changing one variable at a time (gap width, then wavelength) lets learners find a relationship between spreading and the ratio of gap to wavelength rather than relying on the teacher. The ratio is the key idea that unlocks the sound-versus-light puzzle.	Observe whether groups vary only one factor at a time. Check the data table for: correct lambda calculations (1.0 m and 0.34 m), a statement that narrower gaps or longer wavelengths give more spreading, and the gap-to-wavelength comparison for light. Listen for 'comparable to the wavelength' language. A learner can be marked secure when they can point to the simulation and say, 'When the gap is about the

Diffraction Practice Source: CK-12 Search ARES for similar readings	same and change the wavelength (by changing frequency) to see how spreading changes. Task C: observe waves passing an edge (a single barrier end) and watch the wave bend into the 'shadow' region. Learners then calculate wavelengths using $v = f \times \text{wavelength}$: for sound, 340 Hz gives $340/340 = 1.0$ m and 1000 Hz gives $340/1000 = 0.34$ m. They compare the gap with wavelength in a data table: gap 1.0 m with 1.0 m wavelength (ratio 1, strong spreading); gap 1.0 m with 0.34 m wavelength (gap about 3 wavelengths, less spreading); gap 1.0 m with light of 5.5×10^{-7} m (gap about 1.8 million wavelengths, practically no spreading). Learners then sketch what they observed at each setting.	Does the pattern look more like a beam or more like a spreading wave?' WAIT TIME 10-15 seconds after asking: 'What happens to spreading when the gap is much bigger than the wavelength?' Cold-call 3 groups to report one observation each, and write them on the board in a class data table. Check for the common error of confusing wavelength with amplitude. If the simulation is not working, project it from the teacher's computer and let learners direct the settings.		same as the wavelength, the waves spread out most.' Note any group that still draws spreading as a result of a louder wave rather than a wavelength effect.
Explain Phase  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Diffraction (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to their predictions and compare them with the evidence. In pairs they write a claim-evidence-reasoning (CER) explanation answering: 'Why can Wanjiru hear Mr. Mwangi around the corner of the classroom but not see him?' Claim: sound diffracts much more than light. Evidence: the simulation showed strong spreading when the gap was comparable to the wavelength and almost none when the gap was much larger; the 340 Hz sound has $\lambda = 1.0$ m, similar to the 1.0 m doorway, whereas light has $\lambda = 5.5 \times 10^{-7}$ m, far smaller than the gap. Reasoning: diffraction depends on the ratio of wavelength to gap size. They then apply the idea to a new situation: why a low-pitched sound (340 Hz) is heard around a corner better than a high-pitched sound (1000	Say: 'Take out your prediction diagram. Use a different colour pen to fix anything the simulation showed you was wrong.' Provide the CER frame on the board: 'I claim _____. My evidence is _____. This is because _____.' WAIT TIME 10-15 seconds for individual writing before pairs compare. Cold-call 3 pairs to read their reasoning and ask 2 other pairs: 'Do you agree? What would you add?' Push: 'You said wavelength matters. Which wavelength: the one for sound or for light? Show me the numbers.' Ask: 'If the doorway were 10 cm wide, what would a 1.0 m sound wave do?' Ask: 'Does diffraction change the speed or the pitch of the sound?' (No: frequency and wavelength stay the same.) Address the misconception that sound is not a ray: 'Sound can also reflect, as you	Claim-Evidence-Reasoning (CER) with numerical evidence: requiring learners to cite specific wavelengths (1.0 m, 0.34 m, 5.5×10^{-7} m) and the 1.0 m doorway moves them from vague statements ('sound is stronger') to a quantitative reason. Applying the idea to a transfer case (low versus high pitch) checks whether the reasoning generalises.	Use a 3-point check of written CER: (1) claim states sound diffracts more than light, (2) evidence includes at least two wavelengths with the doorway width, (3) reasoning links spreading to gap compared with wavelength. Observable indicators of mastery: learners say '1.0 m wave through a 1.0 m gap spreads strongly', and '5.5×10^{-7} m is tiny compared to the gap so light barely bends'. Exit-ticket question: 'Which spreads more through a 1.0 m doorway: 340 Hz or 1000 Hz sound? Show λ for each.' Correct answer: 340 Hz ($\lambda = 1.0$ m, compared with 0.34 m). Note learners who say high frequency spreads more and re-teach with the simulation next lesson.

	Hz), and why a matatu's bass is heard from far away behind buildings. They connect to the phenomenon: at the Nairobi open grounds the shout spreads out as it leaves, and in the Kisumu hall sound spreads through the doorway into corridors. They revise their original prediction diagram in a different colour.	measured at the stone wall, but now we see it also spreads.' Do not re-teach reflection; link only to what learners already know about echoes.		
Driving Question Board (DQB) Creation  VIDEO: Wave interference <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Diffraction Practice Source: CK-12  Search ARES for similar readings	Learners revisit the DQB (How do sound waves reflect, and why do some places produce stronger echoes than others?). They write new questions on sticky notes that came from today's work, such as: 'Does diffraction help sound fill the Kisumu hall?', 'If sound spreads out, does it become weaker on the open Nairobi grounds?', 'What happens when diffracted waves from two gaps meet?', 'Why do we hear bass from a matatu before the high notes?' They sort their notes under headings such as 'Already answered', 'Partly answered' and 'Next lessons'. They place one sticky note near the existing questions about the Nairobi grounds and the Kisumu hall to show a new link. They update the class list of evidence: 'Lesson 7: waves spread through gaps and around edges; spreading is greatest when gap is comparable to wavelength; sound (1.0 m) spreads more than light (5.5 x 10 ⁻⁷ m).'	Say: 'Look at the board. Which of our questions can we answer better today? Which new question do you have?' Give each learner one sticky note and WAIT TIME 10-15 seconds in silence to write. Cold-call 4 learners to place and read a question and say which earlier lesson (reflection of sound and echo distance, or the ray model of light) it links to. Ask: 'Which question do you think the next lesson on interference will help us answer?' Group similar questions together with the class. Avoid answering new questions straight away: 'Keep it on the board. We will gather evidence.' Point out questions that connect today's gap-and-wavelength idea to the echo sites, especially sites C and D, and mark them as questions for Lesson 8 and Lesson 12.	Driving Question Board sorting: learners categorise questions as answered, partly answered or open, which makes the growth of knowledge visible and gives them ownership of the next steps. Linking questions to evidence from a specific lesson trains them to treat science as a cumulative process.	Check the sticky notes for questions that use the new vocabulary (diffraction, gap, wavelength) and connect to the phenomenon (echo sites C and D). Observable indicators: learners place at least one new question and can name the evidence that partly answers it. Count how many questions are factual (what is diffraction?) versus explanatory (why does the hall sound muddy if sound also spreads?). Note learners who need prompts to move from recall questions to 'why' and 'how' questions.
Model Building Phase  VIDEO: Properties of periodic waves - dubbed(KY) <i>Related topic: not</i>	Learners revise the class echo model. Amina leads by drawing on large paper the sound of a shout moving away from the learner, the waves spreading as they travel, the sound reflecting at the Hell's Gate cliff and returning, and the sound spreading across the open	Say: 'Look at the class model from Lesson 4. What is missing that today's evidence tells us?' WAIT TIME 10-15 seconds, then cold-call 3 learners to suggest one addition. Ask Amina's group to add the doorway diagram to the class model while the other groups	Cumulative model revision with labelled diagrams: adding the new idea (diffraction) to the existing echo model, and marking hypotheses separately from conclusions, helps learners see how each new piece of evidence changes the model. Drawing both	Check models for: correct labelling of wavelength and gap, sound drawn spreading widely through the 1.0 m gap, light drawn as a narrow beam, a statement that diffraction is greatest when the gap is comparable to the wavelength, and a hypothesis (not a claim)

<i>an exact match for this lesson.</i> Source: Khan Academy (Kiswahili) Search ARES for similar videos  HTML: Wave Interference <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Nairobi grounds. In groups, learners add a new part to their own model: a labelled diagram of a sound wave (wavelength 1.0 m) passing through a 1.0 m doorway and spreading widely, next to a light wave (5.5×10^{-7} m) passing through the same doorway as a narrow beam. They write a short caption: 'Diffraction: waves spread around edges and through gaps; most when the gap is comparable to the wavelength.' They also add a note in the margin about how spreading may relate to sites C and D, to be tested later: 'Does spreading help the sound fill the Kisumu hall and weaken in the open Nairobi grounds?' Each learner completes the summary table (observed, learned, explained) in their notebook.	complete their own. Prompt: 'Label the wavelength on your drawing. Which is bigger, the gap or the wave?' Ask: 'Which of our four sites is most affected by sound spreading out, and why do you think so?' (Likely D, the open grounds; accept tentative reasons and mark as a hypothesis, not a conclusion.) Remind learners not to include interference yet. Collect one model from each group and photograph or display them. Say: 'Next lesson, we will see what happens when spreading waves meet and overlap.'	sound and light side by side makes the role of wavelength visible.	about sites C and D. Observable indicators: learners use the phrase 'comparable to the wavelength' in their captions and show $\lambda = 1.0$ m and 0.34 m values correctly. Review notebook summary tables for accurate entries and give feedback to those who confuse diffraction with reflection.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners tie spreading to the ratio of gap to wavelength, or did they remain at 'sound is a wave and light is a ray'? (2) Which prediction diagrams showed that learners think sound goes in a straight beam? (3) Did the wavelength calculations (1.0 m and 0.34 m) cause difficulty with $v = f \times \text{wavelength}$? (4) How well did learners keep diffraction separate from reflection when explaining echoes? (5) Which learners were quiet during the CER task, and what support do they need? (6) Did the simulation work on all devices, and was the ripple tank or projector needed? (7) Which sticky-note questions show readiness for interference in Lesson 8? Record any misconceptions, such as 'high-pitched sound spreads more', and plan a short starter activity next lesson to revisit them using the simulation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that waves passing through a gap spread out on the far side, and that the spreading was greatest when the gap was about the same size as the wavelength. In the simulation, narrow gaps and long wavelengths gave strong spreading, while wide gaps gave a nearly straight beam. At the classroom door, we could hear Mr. Mwangi around the corner but could not see him, and the torch light made a sharp patch.
What did I learn?	We learned that diffraction is the spreading of waves around edges and through gaps. We learned to calculate wavelength using $v = f \times \text{wavelength}$: 340 Hz gives 1.0 m and 1000 Hz gives 0.34 m, while green light is about 5.5×10^{-7} m. A 1.0 m doorway is comparable to the 340 Hz sound wavelength but enormously larger than the wavelength of light. Diffraction does not change frequency or wavelength.
How does this explain the phenomenon?	Sound turns corners more than light because its wavelength (about 1.0 m for 340 Hz) is comparable to the size of doorways and obstacles, so it spreads strongly, while the wavelength of light (5.5×10^{-7} m) is tiny compared to the gap, so light barely bends

	and seems to travel in straight lines. This adds to our echo model: the shout spreads as it travels as well as reflecting, which we will test later against the Kisumu hall and the open Nairobi grounds.
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LESSON 8 (40 minutes (double lesson of 80 minutes if the ripple tank practical is done in full)): Interference of Waves: When Sound Waves Meet in the Kisumu Hall

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain how waves that meet add together (superposition) to give constructive or destructive interference, and use this to explain the muddy overlapping reflections in the Kisumu hall (site C) and the working of noise-cancelling headphones.
Knowledge	Superposition: when two waves meet, the displacements add. Constructive interference happens when the path difference is a whole number of wavelengths (0, 1, 2 ...), giving a larger amplitude and a loud sound. Destructive interference happens when the path difference is an odd number of half wavelengths (0.5, 1.5 ...), giving a smaller amplitude and a quiet sound. For two speakers playing 340 Hz, wavelength = $340/340 = 1.0$ m, so a path difference of 1.0 m gives loud and 0.5 m gives quiet. In the Kisumu hall (12 m to the far wall, delay 0.07 s) the overlapping reflections add and cancel at different places, and the sound is loud, muddy and drawn out. Interference is applied in noise-cancelling headphones.
Skills	Observing and describing ripple patterns; predicting and testing with the PhET Sound simulation and LabXchange Wave Interference; using $v = f \times \text{wavelength}$ to find lambda; calculating path difference and deciding loud or quiet; drawing superposition diagrams; constructing explanations from evidence (NGSS SEP: Planning and Carrying Out Investigations, Developing and Using Models, Constructing Explanations).
Attitudes	Curiosity about how two sounds can make silence; teamwork when sharing observations; respect for others' ideas during discussion; appreciation of how physics protects hearing in Kenyan workplaces, matatus and noisy environments.
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 4 showed that sound reflects from hard surfaces and that the Kisumu hall wall is only 12 m away (0.07 s delay, less than 0.1 s), and Lesson 7 showed that waves spread around edges and through gaps. Both mean that several waves arrive at the same place at the same time. This lesson answers: what happens when they meet? It gives the next piece of the explanation for why site C sounds muddy and sets up Lesson 9, where a wave and its reflection interfere to form stationary waves.
Safety Notes	Keep speaker volume low (well below the 85 dB class limit) and never place speakers close to the ears. Wipe up any spilt water around the ripple tank at once and keep mains electrical equipment away from water. Use a power source with a low voltage for ripple tank motors and lamps. Move carefully around the speaker and tape measure lines to avoid tripping. Check that no learner has hearing sensitivity before the tone test, and stop the tone if anyone is uncomfortable.

B. LESSON OVERVIEW

Mr. Mwangi returns to the DQB question: why is the Kisumu hall muddy and loud while Hell's Gate gives a clean echo? Otieno suggests that many reflections arrive together. Learners predict what happens when two waves meet, then gather evidence from ripples in a ripple tank (or the LabXchange Wave Interference simulation) and from two speakers playing 340 Hz (wavelength 1.0 m) in the PhET Sound simulation and in the classroom. They find loud places where the path difference is 1.0 m and quiet places where it is 0.5 m. In the Explain phase they apply this to the overlapping reflections in the Kisumu hall and explain how noise-cancelling headphones use a wave that is half a wavelength out of step to cancel noise, for example the engine rumble of a bus from Kisumu to Nairobi. Learners add new questions to the DQB and revise the class echo model with a new idea: reflected waves overlap and add, strengthening sound in some places and cancelling it in others.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Constructive and Destructive interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the four echo sites from the DQB (Hell's Gate 0.50 s, Ngong Hills 1.00 s, Kisumu hall 0.07 s with about 1.2 s of lingering sound, Nairobi grounds no echo). Otieno's observation is read aloud: 'In our hall the voice comes from everywhere and gets muddy.' Learners individually write a prediction in their notebooks: 'When two waves meet at the same point, what will happen to the sound or ripple there? Draw what you think the combined wave looks like.' They then predict whether two speakers playing the same note could ever make a quiet spot. Learners share in pairs and then update their prediction cards with a thumbs-up for 'always louder', thumbs-sideways for 'sometimes louder, sometimes quieter', or thumbs-down for 'no change'.	Point to the DQB and say: 'Last lesson we saw waves spread through gaps. Today let us ask: what happens when two waves arrive at the same spot at the same time, like in the Kisumu hall?' Read Otieno's observation. Give 2 minutes of silent writing. Say: 'Draw it. Do not worry about being right; I want your first idea.' Then pose: 'Could two speakers playing the same note ever make a quiet place?' WAIT TIME 10-15 seconds, counting silently. Cold-call 3 learners from different parts of the room (not volunteers) to share predictions and ask 'What made you think that?' Tally thumbs on the board and record the split in a column titled 'Our first ideas'. Do NOT correct predictions; say 'We will test these with evidence.'	Prediction with Evidence Commitment (Predict-Observe-Explain): learners commit to a drawn prediction before seeing data, so that when the evidence conflicts with 'waves always add to make louder', the contrast creates cognitive dissonance and motivates the superposition model. Drawing the combined wave makes the hidden idea visible.	Check notebooks for: (1) a drawn prediction of the combined wave, not just words; (2) whether learners hold the misconception that two waves always make a louder or bigger wave, or that waves 'collide and stop'; (3) number of thumbs for each option, recorded on the board. Indicator of readiness: at least a third of learners suggest 'sometimes louder, sometimes quieter' or can say why. Note names of learners who think that waves bounce off each other when they meet, to revisit during Observe.
Observe Phase  VIDEO: Constructive and Destructive interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) Source: CK-12  Search ARES for similar readings	Station work in groups of four (about 25 minutes). Station 1: ripple tank (or LabXchange: Wave Interference simulation if no tank is available). Two dippers touch the water at the same rate. Learners sketch the pattern and mark where the water surface moves a lot (constructive) and where it stays nearly flat (destructive), noting that lines of calm water radiate outwards. Station 2: PhET Sound simulation with two speakers set to 340 Hz. Learners move the listener to different positions and record the sound level. They measure distances from each speaker with a tape measure on the screen or classroom floor and	Before groups begin, say: 'Your job is to find loud places and quiet places and measure how far each place is from each speaker. Evidence first, explanation later.' Circulate and ask: 'Where on the pattern is the water almost flat? How far is that point from each dipper?' and 'Why do you think the sound dropped here even though both speakers are still on?' Give a warning: 'Keep the volume low and the speaker away from your ears.' When groups find 1.0 m loud and 0.5 m quiet, ask: 'What is the wavelength? How does the path difference compare with it?' WAIT TIME 10-15 seconds after each question before prompting. Check	Data Table Pattern Finding (Gallery of Evidence): learners collect numerical evidence on path difference and loudness and look for a pattern across stations. Converting 1.0 m and 0.5 m into 'whole' and 'half' wavelengths reveals the rule themselves rather than being told. Using several stations (water, sound, drawing) shows that interference is a general wave behaviour, not only a sound effect.	Observable indicators: (1) learners calculate $\lambda = 340/340 = 1.0$ m correctly using $v = f \times \lambda$; (2) the table has path difference 1.0 m marked loud and 0.5 m marked quiet; (3) ripple sketches show alternating bright and calm lines; (4) groups can say that the speakers were still both on in the quiet spot. Teacher checklist: tick each group that links path difference to loudness. Ask struggling groups to compare path difference with the wavelength by dividing ($0.5/1.0 = \text{half a wavelength}$).

	calculate the path difference. Using $v = f \times \lambda$, they confirm $\lambda = 340/340 = 1.0$ m. They fill a table: path difference 1.0 m gives loud, path difference 0.5 m gives quiet. They also try 0 m (equal distances) and 1.5 m and 2.0 m to see the pattern. Station 3: Kiprono's challenge: two small phone speakers (or a single speaker moved and the sound reflected off a wall) play the same tone; learners walk slowly across the room with eyes closed and notice loud and quiet spots. Station 4 (optional): draw two pulses on tracing paper and overlay them in step and half a wavelength out of step. Learners record 'what I saw' and 'what I measured' columns.	at least one pair at every station. After 20 minutes, cold-call 4 groups (one per station) for a 30-second report of their key data: 'Tell me one number and one observation.' Write the data on the board in a single table: path difference, in wavelengths, result.		
Explain Phase  VIDEO: Beat frequency <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Wave Interference Practice Source: CK-12  Search ARES for similar readings	Whole-class sensemaking. Learners use their data to write the rule in their own words: 'When two waves meet, their displacements add (superposition). If crest meets crest (path difference 0, 1.0 m, 2.0 m ... whole wavelengths) the wave is bigger: constructive interference, loud. If crest meets trough (path difference 0.5 m, 1.5 m ... half wavelengths) they cancel: destructive interference, quiet.' Learners draw two waves on the board in step and out of step and add them point by point. Then apply the idea to two situations. (1) Kisumu hall (site C): the speaker is 12 m from the far wall and the delay is 0.07 s, so the echo arrives less than 0.1 s after the original and overlaps it. Walls, floor and roof give many reflected waves that meet. At some places they add (loud), at others they partly cancel, and the changing mix with the 1.2 s lingering sound makes	Ask: 'Looking at your table, what is the link between path difference and what we hear?' WAIT TIME 10-15 seconds, then cold-call 3 learners to complete the sentence 'Loud happens when the path difference is ...' and 'Quiet happens when ...'. Draw two sine waves on the board and add them with a different colour chalk, narrating: 'Crest plus crest gives a bigger crest; crest plus trough gives zero.' Then ask: 'Otieno says the Kisumu hall is muddy. Using interference and the 0.07 s delay, can you explain why?' Use Think-Pair-Share: 1 minute thinking, 2 minutes pairs, then cold-call 4 pairs. Press for evidence: 'Which number tells you the echo overlaps the voice?' (0.07 s is less than 0.1 s). Then show how noise-cancelling headphones work and ask: 'If the noise is 340 Hz, how far out of step must the headphone's wave be?' (half a wavelength, 0.5	Claim-Evidence-Reasoning with Model Application: learners state a claim (e.g. 'the hall is muddy because reflections overlap and interfere'), cite evidence (0.07 s delay, path difference table, 1.2 s lingering) and give reasoning (superposition). Applying the same rule to a new context, the headphones, tests transfer and shows that the same wave idea explains both a natural phenomenon and a technology.	Indicators: (1) learners state both conditions correctly (whole wavelengths loud, half wavelengths quiet) using numbers from the data; (2) the worked example answer is 'path difference 0.5 m, half a wavelength, destructive, quiet'; (3) learners link the 0.07 s delay to the overlap in site C and do not claim that the Kisumu hall has no reflection; (4) learners explain headphones as 'a second wave half a wavelength out of step' and not as 'blocking' the sound. Use a three-question exit slip: path difference 1.0 m, path difference 1.5 m, and one sentence on headphones. Mark: correct, partly correct, not yet.

	the voice loud, muddy and drawn out. Learners link this to Lesson 4 where the delay was less than 0.1 s. (2) Noise-cancelling headphones: a tiny microphone picks up the noise, for example a matatu with 95 dB music or a bus engine. The electronics produce a wave of the same frequency but half a wavelength out of step (crest where the noise has a trough), so when the two meet at the ear they cancel, and the noise is much quieter. Learners complete a short worked example: speakers at 340 Hz, listener 5.0 m from one speaker and 5.5 m from the other. Path difference 0.5 m is half a wavelength, so quiet. They also decide what would happen at path difference 2.0 m (loud).	m). Correct the misconception 'the sound energy disappears' by saying: 'The waves still meet; the displacements cancel at that point, and energy is redistributed to the loud places.' Assign the worked example (5.0 m and 5.5 m) and walk around to check.		
Driving Question Board (DQB) Creation  VIDEO: More on single slit interference <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners revisit the class DQB and, in groups, write new questions on sticky notes arising from today's evidence, for example: 'Why does the Kisumu hall sound loud in some seats and quiet in others?', 'Can interference make a quiet spot at Hell's Gate if we shout from two places?', 'How does an ambulance siren on Waiyaki Way sound different as it passes?', 'What happens when a wave and its reflection interfere on a nyatiti string?' Learners sort the notes into 'answered today', 'partly answered' and 'for later lessons'. They place the notes beside the original DQB question: How do sound waves reflect, and why do some places produce stronger echoes than others?	Say: 'Write one question that today's evidence made you wonder. Start with how or why.' Allow 3 minutes. Circulate and ask 'What did you see or hear that made you ask this?' Cold-call 3 groups to read one question each and ask the class: 'Which lesson do you think will help us answer it?' Gently steer the question about a wave and its reflection to Lesson 9 and the ambulance question to Lesson 10, saying: 'Hold that thought; we will meet it soon.' Move answered questions (e.g. 'Why is the Kisumu hall muddy?') to the 'answered, partly' column and ask a learner to write the evidence beside it. WAIT TIME 10-15 seconds before accepting responses to the sorting question.	Question Sorting on the DQB (Student-Generated Questions): turning today's evidence into new questions keeps learners as owners of the inquiry. Sorting into answered, partly answered and future shows progress and links each lesson to the unit's driving question, making coherence visible.	Indicators: each group writes at least one question that uses interference vocabulary correctly; at least two questions connect to the phenomenon (Kisumu hall, Hell's Gate, Nairobi grounds); learners can justify placing a question in the answered, partly or later column. Record questions that reveal a misconception (e.g. 'Where does the sound go in the quiet spot?') and address these at the start of Lesson 9.
Model Building	Learners revise the class echo model started in Lesson 1, adding	Say: 'Open your model from Lesson 4. What is missing that	Cumulative Model Revision with Gallery Walk: learners add	Observable indicators: (1) the hall diagram shows multiple reflected

Phase  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) Source: CK-12  Search ARES for similar readings	interference. In groups (Amina's drawing role rotates among members) they draw a labelled diagram of the Kisumu hall (site C): a speaker 12 m from the far wall, the direct sound, reflected sound from the wall, roof and floor, and the places where the waves meet. They add labels: 'delay 0.07 s, less than 0.1 s: echo merges with voice', 'many reflections add (constructive) and partly cancel (destructive)', 'sound lingers about 1.2 s: reverberation'. Beside it they draw a small panel for Hell's Gate (85 m, 0.50 s, a single clean reflection, so little overlap). A third panel shows a noise-cancelling headphone: noise wave, anti-noise wave half a wavelength out of step, and the near-silent result. Each group adds an arrow saying 'what is new in our model today' and writes a one-sentence explanation of why Hell's Gate echoes are clear but Kisumu hall sound is muddy.	explains what the waves do when they meet?' Hand out large paper. Circulate asking: 'Where in your hall diagram do waves meet? What decides whether they add or cancel?' and 'Why does Hell's Gate give a clean echo and not a muddy one?' WAIT TIME 10-15 seconds before following up. Remind groups to keep the numbers exactly: 12 m, 0.07 s, 0.1 s, about 1.2 s. After 12 minutes, run a gallery walk: each group leaves one member at the model and the others visit two other groups. Cold-call 3 visitors: 'What is one idea you saw in another group's model that we should add to ours?' Then complete the class model on the wall together and have learners copy the new interference box in their notebooks. Close by asking: 'What is still unexplained?' to point towards stationary waves in Lesson 9.	interference to the existing echo model, so earlier ideas (reflection, delay, surfaces, diffraction) and the new idea of superposition fit together in one explanation. Peer critique in the gallery walk exposes gaps and strengthens the reasoning.	paths meeting, not only one echo; (2) the labels use 0.07 s and 0.1 s correctly; (3) both constructive and destructive interference are named and linked to path difference of whole and half wavelengths; (4) the headphone panel shows a wave half a wavelength out of step; (5) the model distinguishes echo (site A, B) from reverberation (site C). Use a 4-point checklist per group and note groups that need support before Lesson 9.
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners move from 'two waves always add to a louder wave' to 'path difference decides loud or quiet'? Which evidence (ripple tank, PhET, walking test) convinced them most? Did anyone suggest the sound 'disappears' at quiet spots, and how did I respond? Were the numbers 340 Hz, 1.0 m and 0.5 m used accurately? Did learners link the 0.07 s delay in the Kisumu hall to interference without re-teaching reflection from Lesson 4 or diffraction from Lesson 7? How equitably did cold-calling include quieter learners? What do I need to adjust, such as giving more time with the tape measure or simpler wave drawings, before Lesson 9 on stationary waves?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the ripple tank and in the PhET Sound simulation with two speakers at 340 Hz, we observed alternating regions of large movement or loud sound and regions of almost no movement or quiet sound. At path difference 1.0 m ($\lambda = 1.0$ m) the sound was loud and at 0.5 m it was quiet.
What did I learn?	Waves that meet add their displacements (superposition). Whole-wavelength path differences give constructive interference (loud) and half-wavelength path differences give destructive interference (quiet). Noise-cancelling headphones use a wave half a

	wavelength out of step with the noise.
How does this explain the phenomenon?	In the Kisumu hall (site C) the far wall is 12 m away, the delay is 0.07 s (less than 0.1 s), so reflections overlap the voice and each other. They add in some places and cancel in others, and the many hard surfaces keep the sound lingering for about 1.2 s, so it is loud, muddy and drawn out. At Hell's Gate there is a single clean reflection after 0.50 s, so the echo is distinct.

LESSON 9 (40 minutes (double lesson of 80 minutes recommended if the string or air-column experiment is done by every group)): Stationary Waves and Resonance: Nodes, Antinodes and the Pitch of the Nyatiti and Flute

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners explain how a wave and its reflection interfere in a string or air column to form a stationary wave with nodes and antinodes, and to link the set frequencies $f = v/2L$ to resonance and to the pitch of the nyatiti and flute. This extends the echo phenomenon: reflection does not only send sound back to the ear, it can also trap and reinforce waves between two ends.
Knowledge	Learners will: (a) state that a stationary wave forms when a wave and its reflection, travelling in opposite directions, interfere (builds on Lesson 3: reflection, and Lesson 8: superposition giving constructive and destructive interference); (b) identify nodes as points of zero amplitude and antinodes as points of maximum amplitude; (c) use $v = f \times \text{wavelength}$ and $\text{wavelength} = 2L$ for the fundamental to calculate $f = v/2L$; (d) calculate the nyatiti fundamental (0.60 m string, 120 m/s: wavelength 1.20 m, $f = 100$ Hz) and second harmonic (200 Hz, 3 nodes including the ends, 2 antinodes); (e) calculate the flute fundamental (open tube 0.34 m, wavelength 0.68 m, $f = 340/0.68 = 500$ Hz); (f) describe resonance as large-amplitude vibration when a driving frequency matches a natural frequency; (g) state that pitch depends on frequency, so a shorter string or tube gives a higher pitch.
Skills	Science and Engineering Practices: Developing and using models (drawing and revising stationary-wave diagrams); Planning and carrying out investigations (varying string length or tube length and measuring frequency); Analysing and interpreting data (comparing predicted and measured frequencies); Using mathematics ($v = f \times \text{wavelength}$, $f = v/2L$); Constructing explanations (why only certain frequencies make stable patterns). Core competencies: critical thinking and problem solving, communication and collaboration, digital literacy (PhET Standing Waves on a String).
Attitudes	Learners will appreciate the physics behind Kenyan musical heritage such as the Luo nyatiti, show curiosity about links between echoes and musical instruments, work responsibly and respectfully in groups, and value evidence and careful measurement. Values: responsibility, respect, unity. PCIs: Cultural heritage (valuing indigenous instruments), Safety in using equipment.
Key Inquiry Question	How do waves behave when they meet obstacles or change medium? How are the properties of waves applied in day-to-day life? (Reflection at the fixed ends of a string or tube produces stationary waves, which are applied in musical instruments such as the nyatiti and flute.)
Purpose in Storyline	Lessons 3 and 4 showed that sound reflects from a surface and returns as an echo. Lesson 8 showed that waves that meet add by superposition. Lesson 9 combines these ideas: when a wave is reflected back along the same path, the incident and reflected waves overlap and make a stationary pattern. This explains why the Kisumu hall can ring or boom at some notes (resonance), prepares for the Doppler effect in Lesson 10 by fixing the link between frequency, wavelength and pitch, and adds a resonance piece to the final model in Lesson 12.
Safety Notes	Keep strings at low tension so they cannot snap and hit eyes; wear safety glasses or keep faces away from the string line. Do not blow hard into tubes or share flute mouthpieces without wiping them with a clean cloth. Keep speaker or signal-generator volume below 70 dB and limit loud pure tones to a few seconds (safe limit in class 85 dB). Keep electrical equipment away from water. Tie back long hair near clamps. Handle the nyatiti gently as a cultural object and ask the owner or resource person before touching.

B. LESSON OVERVIEW

Mr. Mwangi opens by recalling the trip: the learners' shouts bounced off the Hell's Gate cliff and came back after 0.50 s. He asks what happens when a wave is reflected

and trapped between two fixed ends, like the 0.60 m string on a nyatiti. Learners first predict what a vibrating string looks like and what note it will give. They then gather evidence in the PhET Standing Waves on a String simulation and a hands-on string or air-column experiment (a long rope or elastic cord fixed at one end, or a tube of varying length with a tuning fork), identifying nodes and antinodes and recording harmonics. In the Explain phase they use $v = f \times \text{wavelength}$ and $\text{wavelength} = 2L$ to calculate the nyatiti (100 Hz and 200 Hz) and flute (500 Hz) frequencies and use superposition to explain why only certain patterns survive (resonance). They add questions and evidence to the Driving Question Board (DQB) and revise the class model so that reflection both sends sound back as an echo and, when trapped between two ends, builds stationary waves.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Standing waves on strings Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: properties waves stationary node	(8 minutes) Wanjiru recalls shouting at the Hell's Gate cliff and hearing her voice return after 0.50 s. Mr. Mwangi shows a nyatiti (or a picture/video) with a vibrating length of 0.60 m and plucks one string. Learners individually sketch in their notebooks what they think the plucked string looks like at its fastest vibration (a snapshot or a blurred shape) and predict: (1) How many points on the string do not move at all? (2) What will happen to the pitch if the string is made shorter, as when the player presses it? (3) Will a flute, which has no strings, also make notes by a similar process? Learners then share in pairs, using the sentence frame: 'I think the string will look like ... because the wave ...' They also link to the echo: 'When a wave reaches the fixed end of the string, I think it ...'	Say: 'Remember the cliff at Hell's Gate. The sound hit the rock and came back. Now picture a wave on this nyatiti string hitting the fixed end. What do you think it does?' Pluck the string once and ask learners to watch closely. Give 2 minutes of silent individual drawing, then say: 'Draw what you think the shape looks like at the moment of widest swing. Mark any place you think stays still.' WAIT TIME: 10-15 seconds after asking 'What happens to the wave at the fixed end?' before taking any answer. Cold-call 4 learners (including Otieno's group and Kiprono's group) to share their sketches without telling them if they are right or wrong. Record 3 distinct predictions on the board labelled P1, P2, P3. Ask: 'What would you need to see to decide which prediction is correct?' Do not correct misconceptions such as 'the wave travels along the string and disappears' or 'all parts move equally'.	Predict-Observe-Explain (POE) with initial model drawing: Learners commit to a prediction before seeing evidence, which exposes their existing ideas (for example, that a wave on a string simply travels away like a shout away from the learner). Linking it to reflection (Lesson 3) and interference (Lesson 8) activates prior knowledge so that later evidence can be compared with their own ideas. Making a sketch (Developing and using models) makes thinking visible for the teacher and for peers.	Observable indicators: (a) at least 80% of learners have a labelled sketch showing a string shape and marking where they think the string stays still; (b) learners use reflection language such as 'the wave bounces back from the end' when asked about the fixed end; (c) record on a checklist which learners predict that shorter length gives higher pitch. Look for misconceptions: 'the wave is lost at the end', 'the whole string moves the same amount', 'pitch depends on how hard it is plucked' (loudness and pitch confused). Note the names of learners with these ideas to revisit during the Observe phase.
Observe Phase  VIDEO: Standing waves on strings Source: Khan Academy (English	(Part of 20 minutes in total for observing) Station A (PhET: Standing Waves on a String, or a phone/projector demonstration if computers are scarce): groups of 3-4 run the simulation, set the	Set up stations before the lesson and test the simulation. Say: 'Your job is to find the frequencies where the string shape stops moving along and just stays in place, with some points that never move.'	Data table and pattern-seeking (Analysing and interpreting data): Learners collect nodes, antinodes and frequency into a shared table so the pattern emerges for themselves (each higher pattern	Observable indicators: (a) each group's table correctly shows 2 nodes and 1 antinode for the fundamental and 3 nodes and 2 antinodes for the second harmonic; (b) learners correctly

- US curriculum) Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: properties waves stationary node	string to fixed ends, change the frequency slowly, and watch for the frequencies at which a clear stable pattern forms. They record in a table: number of loops, number of nodes (including ends), number of antinodes, and relative frequency (f, $2f$, $3f$). They observe that between those frequencies the pattern is messy and small. Station B (hands-on): either (i) a 3-4 m rope or elastic cord fixed at one end and shaken by hand at the other, or a 0.60 m taut string or wire on a sonometer/nyatiti; or (ii) an air column: a tube or measuring cylinder with water, a tuning fork (for example 500 Hz or 512 Hz) and the water level adjusted until the sound suddenly becomes loud (resonance). Learners mark nodes with chalk or tape and antinodes with a different mark, measure length L with a tape measure, and compare fundamental and second harmonic. They also measure the pitch change when they press the string to make it shorter (like a nyatiti player) or use a shorter tube (like covering flute holes). Kiprono times the vibrations with a stopwatch where possible, Amina draws the patterns, Otieno reads measurements, Wanjiru checks the table and asks 'Why only these frequencies?'	Circulate and ask each group: 'Show me a point that does not move. What do we call it? Show me a point with the biggest swing.' Use the language: node = zero amplitude, antinode = maximum amplitude. Prompt: 'Count the nodes including the two ends. Count the antinodes. What do you notice as the frequency goes up?' WAIT TIME: 10-15 seconds after asking 'What do you see happening between the loops?' Cold-call 3 groups to report their node and antinode counts for the first and second patterns. Challenge groups that finish early: 'Predict the pattern for three times the lowest frequency, then test it.' Ask: 'Which of your observations looks like the pattern when a wave and its reflection overlap, like Lesson 8?' Safety: remind about low string tension, safety glasses and low volume. Collect data on a class chart: fundamental (1 loop, 2 nodes, 1 antinode); second harmonic (2 loops, 3 nodes, 2 antinodes); third harmonic (3 loops, 4 nodes, 3 antinodes). Show the nyatiti string case: $L = 0.60$ m, fundamental pattern one loop, second harmonic two loops with 3 nodes including the ends and 2 antinodes.	has one more loop and a frequency that is a whole-number multiple of the lowest). Modelling with physical and simulated systems lets them see the wave and its reflection superpose. Noticing and wondering: learners record 'I notice ...' and 'I wonder ...' statements, for example 'I wonder why the Kisumu hall sometimes booms on certain notes', connecting back to the echo phenomenon.	label a node and an antinode on their sketch; (c) learners state that a shorter string or tube gives a higher pitch, based on their measurement; (d) groups find the resonance point in the air column (loud sound) and measure L within about 1 cm of the expected value; (e) quick thumbs check: 'Hold up one finger for node, two fingers for antinode' when the teacher points to positions on a displayed wave. Note groups that confuse nodes with the ends only or think nodes are the places where the string moves most.
Explain Phase VIDEO: Wave properties Related topic: not an exact match for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos	(Part of 20 minutes, continuing the lesson) In their groups, learners use their data to build a causal explanation in three steps. Step 1 (Why a pattern forms): the wave travelling along the string reflects at the fixed end (reflection, Lesson 3) and meets the incoming wave; the two waves superpose (Lesson 8): where they arrive in step the displacement doubles (antinodes,	Say: 'Let's build the explanation together. First, what two waves are on the string at the same time?' WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 3 learners to name the incident wave and the reflected wave, then 2 learners to describe what happens at a node and at an antinode using the words 'cancel' and 'add'. Ask: 'Why can the string not vibrate at	Claim-Evidence-Reasoning (CER) with mathematical modelling: Learners state a claim about the frequency, support it with measurements from the experiment and simulation, and give reasoning using superposition, reflection at the ends and $v = f \times \text{wavelength}$. This forces them to link observations, mathematics and mechanism. Talk	Observable indicators: (a) learners correctly calculate nyatiti fundamental 100 Hz, second harmonic 200 Hz and flute 500 Hz, with correct units and working shown ($\text{wavelength} = 2L$ first, then $f = v/\text{wavelength}$); (b) CER statements mention reflection at the ends AND interference (superposition) as the cause of stationary waves; (c) learners

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: properties waves stationary node	constructive interference) and where they arrive opposite they cancel (nodes, destructive interference), so the pattern stays in place instead of travelling. Step 2 (Why only certain frequencies): a stable pattern forms only when a whole number of half-wavelengths fits the length, so for the fundamental wavelength = $2L$ and $f = v/2L$. Step 3 (Calculations and pitch): Nyatiti: $L = 0.60\text{ m}$, $v = 120\text{ m/s}$, wavelength = $2 \times 0.60 = 1.20\text{ m}$, $f = 120/1.20 = 100\text{ Hz}$; second harmonic has wavelength 0.60 m, $f = 120/0.60 = 200\text{ Hz}$ (3 nodes including the ends, 2 antinodes). Flute modelled as an open tube $L = 0.34\text{ m}$: wavelength = $2L = 0.68\text{ m}$, $f = 340/0.68 = 500\text{ Hz}$ (using $v = 340\text{ m/s}$). Learners compare their experiment with calculations and explain pitch: the shorter the string or tube, the higher the frequency and the higher the pitch; loudness depends on amplitude, not frequency. They define resonance: when a driving frequency matches a natural frequency (100 Hz, 200 Hz for the nyatiti string), the vibration grows large, as when a tuning fork makes the air column loud. They connect to the phenomenon: reflection from the hall walls and roof in Kisumu can make some notes boom, and a closed space acts like a tube or string whose natural frequencies are strengthened. Learners write a CER statement (Claim, Evidence, Reasoning): 'The nyatiti string of length 0.60 m gives 100 Hz because ...'.	just any frequency and still hold a steady pattern?' Draw the fundamental and second harmonic on the board with L marked, and ask learners to say aloud how many half-wavelengths fit in L each time. Model the calculation for the nyatiti on the board: 'wavelength = $2L = 1.20\text{ m}$, then $f = v$ divided by wavelength = 120 divided by $1.20 = 100\text{ Hz}$.' Then ask groups to calculate the second harmonic and the flute, with the flute done on mini-whiteboards and shown on the count of three. Cold-call 4 groups for their answers (200 Hz and 500 Hz) and ask them to show the units. Ask: 'If Nyatiti player presses the string halfway along, L becomes 0.30 m. Before calculating, will the pitch go up or down? Then calculate: wavelength 0.60 m and $f = 200\text{ Hz}$.' Address the misconception that nodes are places where there is no wave at all: 'The wave is still there, but the two waves cancel at that point.' Stress that this lesson builds on reflection and interference and does not re-teach them. Connect to the phenomenon: 'In Kisumu hall, hard walls reflect sound back and forth. Why might some notes sound extra loud there?' Accept answers about resonance of the hall.	moves (revoicing, 'add on', 'who agrees or disagrees and why?') build collective explanation. Using the same wave equation for the nyatiti, flute and echo problems shows that one set of ideas explains many situations, supporting transfer.	correctly state that pitch depends on frequency and that a shorter length gives higher pitch; (d) learners define resonance with the idea of matching frequencies and larger amplitude; (e) exit question on a slip: 'Name one node-antinode difference and give the fundamental frequency of the nyatiti string.' Aim for at least 75% correct; plan a short re-teach with a diagram for those who use L instead of $2L$ as the wavelength.
Driving Question Board (DQB) Creation	(5 minutes) Learners return to the Driving Question Board: 'How do sound waves reflect, and why do some places produce stronger	Say: 'Look at our DQB. Which questions can we now answer or partly answer after today?' WAIT TIME: 10-15 seconds before	Driving Question Board as a living record of understanding: Learners connect new evidence to the driving question, see their growing	Observable indicators: (a) each group posts at least one evidence note and one new question; (b) at least half of the new questions

 VIDEO: Question 5: 2015 AP Physics 1 free response <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: properties waves stationary node	echoes than others?' Each group writes on sticky notes: (1) one new piece of evidence from today (for example 'a wave and its reflection on a 0.60 m string make a stationary wave with 2 nodes at the ends and 1 antinode in the middle, giving 100 Hz'); (2) one question they can now answer from earlier lessons (for example how a reflected wave can be trapped and reinforced); (3) one new question (for example 'Does the Kisumu hall have natural frequencies that make some notes boom?', 'Why does a flute sound different from a nyatiti at the same pitch?', 'Can reflected sound at Hell's Gate ever form a stationary wave?'). They place notes under categories such as Reflection, Surfaces, Stationary waves and resonance, and Still to find out.	taking responses. Cold-call 3 learners to read a sticky note aloud and say which lesson evidence supports it. Ask: 'Which of our earlier questions about the muddy Kisumu hall does resonance help with?' Move answered questions to an 'Answered/Partly answered' area and mark partly answered ones with a star. Add a teacher-prepared question if needed: 'Why do some notes boom in a hall with hard walls and a roof?' Highlight links to the next lesson: 'Next time we will ask what happens when the source itself is moving, like the ambulance on Waiyaki Way.' Photograph the DQB for records and for learners who are absent.	understanding and own the questions that drive the unit. Sorting questions into 'answered', 'partly answered' and 'new' encourages metacognition, and explicitly linking evidence to a lesson (Lesson 3 reflection, Lesson 8 interference, Lesson 9 stationary waves) builds a coherent storyline rather than a set of separate topics.	refer to resonance, nodes or stationary waves in connection with the echo sites, for example the Kisumu hall; (c) learners can state which lesson supports each answered question; (d) record questions that show misconceptions (for example 'does the sound stop at a node?') for the next lesson. Listen for use of the words node, antinode, resonance and frequency.
Model Building Phase  VIDEO: AP Physics 1 review of Waves and Harmonic motion <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms:	(7 minutes) Amina leads each group in revising the class echo model drawn in earlier lessons (shout leaves learner, travels as a wave, reflects from a surface following the law of reflection, returns after $t = 2d/340$). Groups add a new panel showing a wave trapped between two fixed ends (a nyatiti string 0.60 m or a flute tube 0.34 m): incident wave and reflected wave in opposite directions, nodes (N) and antinodes (A) labelled, and wavelength = $2L$ for the fundamental with $f = v/2L$. They add a short caption: 'Reflection at the ends plus interference gives a stationary wave. Only certain frequencies fit. Resonance happens when the driving frequency matches one of them.' They also add a 'What the model cannot yet explain' box (for	Say: 'Our model so far explains how a shout goes out and comes back as an echo. What new piece does today's evidence add?' WAIT TIME: 10-15 seconds before groups start drawing. Provide a model checklist on the board: (1) source and wave path, (2) reflection at a boundary, (3) two waves overlapping, (4) nodes and antinodes labelled, (5) frequency or pitch link with numbers, (6) a limitation. Circulate and ask: 'Where on your drawing is the reflected wave? Where do the two waves cancel? What would change if the string were shorter?' Cold-call 3 groups to explain their new panel in under 30 seconds each; ask the class to give one 'glow' (what is right) and one 'grow' (what to improve). Ask: 'Does your model still explain the echo at Hell's Gate with a 0.50 s delay?	Cumulative model revision (Developing and using models): Learners extend a single class model rather than starting afresh, so each lesson adds a mechanism. Peer critique with 'glow and grow' and a limitations box models how scientists test and refine ideas. Linking the string and tube to the hall shows that models can transfer across contexts.	Observable indicators: (a) models include both the incident and reflected waves and label at least one node and one antinode correctly; (b) the fundamental diagram shows wavelength = $2L$ with L marked; (c) frequencies for the nyatiti (100 Hz and 200 Hz) and flute (500 Hz) are correct on the model; (d) a limitation is listed that honestly identifies something not yet explained about the Kisumu hall; (e) peers' feedback uses evidence from the investigation. Use a 3-2-1 exit ticket if time allows: 3 key terms, 2 equations, 1 question left.

properties waves stationary node	example why the Kisumu hall sounds loud and muddy; connecting resonance and reverberation will be refined in Lesson 12). A volunteer from each of two groups compares models and checks that numbers match the established data: nyatiti 100 Hz and 200 Hz, flute 500 Hz.	Does the new part explain the Kisumu hall completely?' Encourage honest limits: 'We have a piece of the Kisumu puzzle, not all of it.' Keep the class model on the wall for Lesson 10.		
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D. TEACHER REFLECTION

Before the next lesson, consider: (1) Did learners connect stationary waves to reflection (Lesson 3) and interference (Lesson 8), or did they treat today as an isolated topic? Check CER statements for both ideas. (2) Which learners confused nodes with the ends only, or used L instead of $2L$ as the wavelength? Plan a quick correction with a diagram. (3) Did the PhET simulation and the hands-on station give consistent results, and where did measured frequencies differ from calculated ones (tension, end effects, measurement error)? Use that discussion to teach about reliable data. (4) Was the nyatiti used respectfully and did the lesson value Kenyan heritage and any local musicians who could visit as resource persons? (5) How well did the wait time work, and did cold-calling bring out quiet learners? (6) Are learners ready for the Doppler effect in Lesson 10, where frequency and pitch change because of motion? Ensure they are secure with $v = f \times \text{wavelength}$ and the idea that pitch depends on frequency. (7) Record any new DQB questions about the Kisumu hall to revisit in Lessons 10 to 12.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a plucked or shaken string fixed at both ends (and an air column in a tube) forms steady patterns at particular frequencies only. In these patterns some points never move (nodes) and some swing the most (antinodes). The fundamental has 2 nodes (the ends) and 1 antinode; the second harmonic has 3 nodes and 2 antinodes. A shorter string or tube gave a higher pitch, and the air column became loud at a particular length (resonance).
What did I learn?	We learned that a stationary wave forms when a wave and its reflection travel in opposite directions and interfere: they add at antinodes and cancel at nodes. For the fundamental, wavelength = $2L$, so $f = v/2L$. Pitch depends on frequency, and resonance happens when a driving frequency matches a natural frequency of the string or air column.
How does this explain the phenomenon?	We can explain that the nyatiti string 0.60 m long with wave speed 120 m/s has wavelength 1.20 m and fundamental $f = 120/1.20 = 100$ Hz, with a second harmonic of 200 Hz, and that a flute modelled as a 0.34 m open tube has wavelength 0.68 m and $f = 340/0.68 = 500$ Hz. Reflection at the ends plus interference gives stationary waves, and in an echo situation such as the Kisumu hall reflected sound can also build up at some frequencies, which adds to our model of why some places sound stronger than others.

LESSON 10 (40 minutes): Doppler Effect: Why the Ambulance Siren Changes Pitch as It Passes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain that when a sound source and an observer move relative to each other, the observed frequency changes: higher when they approach and lower when they recede. They link this to $v = f \times \text{wavelength}$ and apply it to the ambulance siren, radar and astronomy, and ask what it adds to the echo story.
Knowledge	By the end of the lesson the learner should be able to: a) state that relative motion of source and observer changes the observed frequency (higher when approaching, lower when receding); b) use $v = f \times \text{wavelength}$ to explain that waves bunch up ahead of a moving source and stretch out behind it; c) interpret the ambulance data: 700 Hz emitted, 744 Hz approaching, 661 Hz receding at 20 m/s; d) describe uses of the Doppler effect in radar and astronomy.
Skills	Developing and using models (wavefront drawings and a human-model walk-through); analysing and interpreting data (comparing 700, 744 and 661 Hz); constructing explanations; using mathematics ($v = f \times \text{wavelength}$); obtaining and communicating information (YouTube Doppler demonstration).
Attitudes	Curiosity about everyday sounds such as sirens and matatus; respect for classmates' ideas during discussion; responsibility in group roles; appreciation of how Physics is used in road safety, weather and space science. Core competencies: communication and collaboration, critical thinking, digital literacy. Values: responsibility, respect.
Key Inquiry Question	How do waves behave when they meet obstacles or change medium? How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 7 to 9 gave learners frequency, wavelength and pitch linked by $v = f \times \text{wavelength}$. Lesson 10 adds a new wave behaviour: the frequency heard changes when the source or observer is moving. Echoes at Hell's Gate, Ngong Hills, the Kisumu hall and the Nairobi grounds involve a reflecting wall, but the Doppler effect shows that motion alone changes pitch, and a moving reflector also returns a shifted frequency. This is the idea behind radar and prepares Lesson 11 on communication and medicine. Learners add a Doppler panel to the class model and a new question to the DQB.
Safety Notes	Do the human-model walk-through in a clear open space away from desks. Learners walk, never run. For any outdoor siren observation, stay well away from the road, especially Waiyaki Way style busy roads. Keep any speaker or phone tone at a low, comfortable volume, well below the 85 dB class limit. Never look at the Sun when discussing astronomy.

B. LESSON OVERVIEW

Mr. Mwangi recalls the Hell's Gate trip: on the road back to Nairobi an ambulance passed the bus on Waiyaki Way, and its siren seemed to drop in pitch as it went by. Wanjiru asks whether the siren itself changed. Learners predict, watch a YouTube Doppler demonstration, then model the moving source with wavefront drawings and a human wave walk. They use $v = f \times \text{wavelength}$ with the siren data (700 Hz emitted, 744 Hz approaching, 661 Hz receding at 20 m/s) to explain the change, and connect the idea to radar speed guns and the redshift of stars. They then update the Driving Question Board and class model, noting that sound can return changed in pitch as well as delayed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Doppler effect formula for observed frequency Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Doppler Effect (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the bus ride back from Hell's Gate when an ambulance passed on Waiyaki Way. Mr. Mwangi plays or imitates a siren rising then falling in pitch. Individually, learners write a prediction in their notebooks: 'Does the siren itself change its note as it passes, or does something else cause the change we hear? Sketch what you think the sound waves look like in front of and behind the ambulance.' They share with a partner for 2 minutes, then a few share with the class.	Say: 'On the way back from Hell's Gate, an ambulance passed us on Waiyaki Way. Kiprono, you were sitting by the window. What did you notice about the siren?' Ask: 'The driver says the siren always makes 700 Hz. So why did we hear different notes?' Give WAIT TIME of 10-15 seconds before taking answers. Then say: 'Write your prediction and draw the waves in front of and behind the ambulance.' Cold-call 4 learners, including at least one who has not spoken today, to read their predictions. Record all ideas on the board without judging, for example 'siren changes', 'wind', 'waves squeezed'.	Predict-Observe-Explain (prediction stage): committing to a prediction in writing and drawing surfaces prior ideas, such as the siren really changes its note, so that later evidence can confirm or challenge them. Linking to the echo phenomenon shows that learners already know sound is a wave that travels and returns.	Look for: (1) learners who draw evenly spaced waves on both sides, showing the common misconception; (2) learners who mention the ambulance moving or waves bunching, which shows readiness; (3) learners who confuse pitch with loudness, since the siren gets louder as it approaches. Tally the number of learners in each group on the board. Use this to decide how much support to give during modelling.
Observe Phase  VIDEO: Doppler effect formula for observed frequency - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Doppler Effect Practice Source: CK-12  Search ARES for similar readings	Learners watch a short YouTube Doppler demonstration (a siren or moving sound source passing a stationary observer, with wavefront animation). Before it plays, groups of four are given a recording sheet with three columns: 'Approaching', 'Passing', 'Receding'. They record what happens to the pitch and what happens to the spacing of the wavefronts in the animation. Then they study the ambulance data in a table: emitted 700 Hz, approaching 744 Hz, receding 661 Hz, speed 20 m/s. Learners calculate the wavelength of the siren sound at rest using $v = f \times \text{wavelength}$: $340/700 = 0.486 \text{ m}$, and notice that the observed frequencies are above and below 700 Hz.	Say: 'While you watch, focus only on two things: what the pitch does and how the wavefronts are spaced. Do not write anything until the clip ends.' Play the clip twice. After the second viewing give WAIT TIME of 10 seconds for silent recording. Say: 'Turn to your group. Compare your columns. Where did the wavefronts look closer together?' Circulate and ask each group: 'Which is higher, 744 Hz or 700 Hz? Is that approaching or receding?' Then project the data table and ask: 'Using $v = f \times \text{wavelength}$, what is the wavelength of the 700 Hz siren when the ambulance is parked?' Cold-call 3 learners for the working. Ask: 'If the waves are squeezed closer together while the speed of sound stays 340 m/s, what must happen to the frequency?'	Structured evidence table with Think-Pair-Share: recording observations in fixed categories (approaching, passing, receding) helps learners separate what they see and hear from what they infer. Pairing the animation with numerical data (700, 744, 661 Hz) builds the link between the visual wavefront model and measured frequency.	Check recording sheets for: approaching = higher pitch with wavefronts closer together; receding = lower pitch with wavefronts further apart. Check that learners compute $340/700 = 0.486 \text{ m}$ correctly and state that 744 Hz is greater than 700 Hz. Thumbs-up check: 'Does the siren itself change?' Expect most learners to say no. Note groups that still say the ambulance changes its siren.
Explain Phase	Learners build the explanation in	Say: 'Amina is our siren. Amina,	Embodied modelling followed by	Look for CER statements that

 VIDEO: Doppler effect formula for observed frequency Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Doppler Effect (Flexbook) Source: CK-12  Search ARES for similar readings	two steps. First, a human-model walk-through in an open space: one learner (Amina) acts as the siren and walks steadily while clapping once per second. A second learner (Otieno) stands still ahead as the observer and a third (Wanjiru) stands behind. Other learners count claps heard per ten seconds at each position. The class sees that Otieno receives claps more often and Wanjiru less often, because each clap starts from a position closer to or further from them. Second, groups draw wavefronts from a moving source on paper: circles starting from successive positions, bunched up in front and stretched behind. They then use the data: approaching $700 \times 340 / (340 - 20) = 744$ Hz, receding $700 \times 340 / (340 + 20) = 661$ Hz. They write a claim-evidence-reasoning statement explaining why pitch changes although the siren emits 700 Hz. Finally, Mr. Mwangi introduces uses: a radar speed gun sends radio waves at a moving matatu and measures the shifted frequency of the reflected wave to find its speed, and astronomers see light from distant galaxies shifted to lower frequency (redshift) when they move away.	walk slowly and clap once every second. Everyone else, count claps from where you stand.' After the walk give WAIT TIME of 10-15 seconds, then ask: 'Otieno, in front of Amina, did you get more claps or fewer than Wanjiru behind? Why?' Cold-call 4 learners to explain in their own words. Say: 'Draw circle one where the siren was at time zero, circle two where it was one second later. What happens to the gap in front?' Circulate and prompt: 'Remember, the speed of sound is still 340 m/s. If waves arrive closer together, what does this do to frequency?' Show the two calculations and ask: 'Why is the bottom number 340 minus 20 for approaching and 340 plus 20 for receding?' Wait 10 seconds. Link to the phenomenon: 'Now think of the echo from the Hell's Gate cliff. If a bus was driving towards the cliff while hooting, what would the returning echo sound like?' Cold-call 2 learners. For radar say: 'The radar gun does the same job as the echo: it sends a wave, receives it back, and compares frequency instead of time.' Correct misconceptions: pitch changes are not the same as loudness changes.	Claim-Evidence-Reasoning: acting out the moving source makes the abstract idea of crowded wavefronts concrete, and the CER structure forces learners to connect the claim (siren appears higher approaching) to evidence (744 Hz and 661 Hz versus 700 Hz) and to the reasoning from $v = f \times \text{wavelength}$. Applying the idea to radar and redshift transfers the model to new situations.	include: claim (observed frequency depends on relative motion), evidence (744, 700 and 661 Hz), and reasoning (wavefronts bunched ahead and stretched behind, speed of sound unchanged). Check wavefront drawings for closer circles in front of the source and wider circles behind. Check calculations: $700 \times 340 / 320 = 744$ Hz and $700 \times 340 / 360 = 661$ Hz. Success criterion: at least 80 percent of groups state that the siren does not change, but the number of waves reaching the ear per second does. Note learners who say the speed of sound changes, and revisit $v = 340$ m/s.
Driving Question Board (DQB) Creation  VIDEO: Doppler effect formula for observed frequency - dubbed(KY) Source: Khan Academy (Kiswahili)	Learners return to the class Driving Question Board. Groups check which earlier questions the Doppler evidence helps with, for example 'what happens to a wave when it meets a moving surface?' They write new questions on sticky notes such as 'Does an echo from a moving wall change pitch?', 'How does a speed gun on a Nairobi highway work?', 'How do we know stars are moving away?' They	Say: 'Look at our Driving Question Board. Which question from earlier lessons does the Doppler effect help with, even a little?' Give WAIT TIME of 10-15 seconds. Then: 'Write one new question on a sticky note and one thing we still do not understand about echoes at the four sites.' Cold-call 3 groups to read notes aloud and place them. Ask: 'Does Doppler explain why Kisumu hall is muddy?' Guide:	Driving Question Board questioning: public tracking of what is answered and what is new keeps the whole unit connected to the anchoring phenomenon and shows learners that science grows through questions. Sorting questions by theme builds the habit of separating different wave behaviours.	Check that each group contributes at least one new question and can say which category it belongs to. Look for questions that link Doppler to echoes (moving reflector, moving observer) as a sign of deeper connection. Note learners who place Doppler questions under 'interference' and clarify in the next lesson.

 Search ARES for similar videos  EXERCISE: Doppler Effect Practice Source: CK-12  Search ARES for similar readings	place notes on the board under categories (reflection, frequency, applications) and mark any older question they can now partly answer.	'No, that was overlapping reflections. Doppler is about motion, not overlap. So what is different?' Prompt Kiprono's group to link to timing: 'How is the echo delay different from the Doppler pitch change?'		
Model Building Phase  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Position Relative to Motion (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revise the class echo model. Amina leads a group in adding a new Doppler panel: a moving source with bunched wavefronts ahead and stretched wavefronts behind, labelled 'approaching: higher f, shorter wavelength; receding: lower f, longer wavelength'. Each group adds a note to the existing four-site echo model (Hell's Gate 0.50 s, Ngong 1.00 s, Kisumu hall 0.07 s, Nairobi grounds no echo): 'A moving reflector or source changes the pitch of what returns; speed of sound stays 340 m/s.' Learners then complete the summary table prompts in their notebooks: what we observed, what we learned, how it helps explain echoes.	Say: 'Amina, bring your wavefront diagram to the board. Everyone else, tell us what must be added to our class model.' Give WAIT TIME of 10-15 seconds. Ask: 'Which parts of our echo model stay the same and which parts are new?' Cold-call 4 learners: one to state the Doppler rule, one to give the 744 Hz and 661 Hz values, one to name a use in radar, and one to name a use in astronomy. Say: 'At Hell's Gate the cliff does not move, so the echo has the same pitch as the shout. What would change if the cliff or the shouter moved?' Close with: 'Next lesson we will see how waves are used in FM radio and ultrasound, which both depend on frequency and on echo timing.'	Cumulative model revision: adding a Doppler panel to the class model shows that the same wave ideas ($v = f \times \text{wavelength}$, pitch, reflection) explain different behaviours. Explicitly stating what stays the same in the Hell's Gate echo and what would change with motion helps learners transfer the model.	Check that the revised model shows wavefront spacing with labelled higher and lower frequency and that the speed of sound is kept at 340 m/s. Exit check: each learner writes one sentence beginning 'The siren seems higher when approaching because...' and one use of the Doppler effect. Success indicator: sentences mention waves reaching the observer more often, not that the siren changed.

D. TEACHER REFLECTION

Which learners still believe the siren itself changes its note, and what evidence from the wavefront drawings or the human-model walk helped them change their mind? Did learners keep the speed of sound at 340 m/s while explaining frequency change? Was the claim-evidence-reasoning structure used with the numbers 700, 744 and 661 Hz? Which DQB questions need to be carried into Lesson 11 on communication and medicine?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We watched and acted out a moving sound source. The pitch of the ambulance siren sounded higher as it approached (744 Hz) and lower as it receded (661 Hz), although the siren emits 700 Hz.
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What did I learn?	Relative motion of source and observer changes the observed frequency: waves bunch up ahead of a moving source (higher frequency) and stretch out behind it (lower frequency). The speed of sound stays 340 m/s and $v = f \times \text{wavelength}$ still holds. This is used in radar to measure speed and in astronomy to tell whether stars and galaxies are moving towards or away from us.
How does this explain the phenomenon?	At Hell's Gate and Ngong Hills the sound returns from a fixed cliff, so its pitch matches the shout and only the delay (0.50 s and 1.00 s) tells us the distance. If the source, observer or reflecting surface were moving, the returned sound would also change in frequency, which is how radar guns find the speed of a matatu. Doppler explains pitch change, not why the Kisumu hall is muddy or why the Nairobi grounds have no echo.

LESSON 11 (40 minutes (double period 80 minutes recommended if the ultrasound timing task is completed in class)): Waves in Communication and Medicine: FM Radio and Ultrasound as Echo Applications

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply what they know about echoes and frequency to two real technologies. They explain how an FM radio station carries a voice by changing the frequency of a radio carrier wave. They then show how an ultrasound scanner at a county hospital finds the depth of an organ boundary by timing a reflected pulse, in the same way that learners at Hell's Gate timed their echo. Both applications show that the wave behaviours in this unit are used in everyday life.
Knowledge	By the end of the lesson the learner should be able to: (a) state that radio waves are electromagnetic waves that travel at 3.0×10^8 m/s and need no medium; (b) calculate the wavelength of the 91.0 MHz FM carrier using $v = f \times \text{wavelength}$ (3.30 m); (c) explain that in FM the sound signal changes the frequency of the carrier, not its amplitude; (d) explain that ultrasound is sound above the range of human hearing, and that a 3.5 MHz probe in soft tissue (1540 m/s) has a wavelength of 0.44 mm; (e) use $d = vt/2$ to relate echo time to boundary depth, including the 7.7 cm boundary returning an echo after 100 microseconds.
Skills	NGSS Science and Engineering Practices: Developing and Using Models (draw a labelled model of an FM carrier and of a pulse-echo scan); Using Mathematics and Computational Thinking (apply $v = f \times \text{wavelength}$ and $d = vt/2$ with correct units and powers of ten); Analyzing and Interpreting Data (compare Hell's Gate echo timing with ultrasound timing); Constructing Explanations (link technology to the echo phenomenon); Obtaining, Evaluating and Communicating Information (use the Khan Academy resource on waves applications and communicate findings to peers).
Attitudes	Core competencies: Digital literacy (learners use the Khan Academy resource on radio and FM modulation); Critical thinking and problem solving (learners reason from echo timing to depth); Communication and collaboration (learners explain to a partner and the class). Values: Responsibility (handling devices and the offline resources carefully, and taking turns); Respect (listening to peers' explanations). PCI: Health education (learners appreciate that ultrasound is a safe imaging method used in county hospitals, for example in antenatal care).
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 1 to 10 gathered evidence that sound reflects, that echo delay follows $t = 2d/v$, and that frequency and wavelength are linked by $v = f \times \text{wavelength}$. In Lesson 11 learners see that the same ideas run radio and medical imaging: an ultrasound scanner is a Hell's Gate echo on a tiny scale, and an FM station uses frequency, the property studied in the Doppler lesson, to carry a voice. The lesson adds two entries to the DQB and the class model and prepares learners for Lesson 12, where they bring all wave behaviours together to give the final explanation of the four echo sites and examine noise.
Safety Notes	Keep all sound levels from phones, radios and speakers below the 85 dB class limit, and keep the volume moderate when the FM radio is played. Do not use any real ultrasound device on learners, because the scanner is studied only through data and diagrams. Handle phones, radios and batteries carefully, and keep electrical devices away from water. Learners should not put earphones in at high volume.

B. LESSON OVERVIEW

Wanjiru asks Mr. Mwangi, 'If sound needs air to travel, how does my voice get from the radio station in Nairobi to the radio in Kericho?' The lesson uses this question to link the Hell's Gate echo to two technologies. In the Predict Phase learners predict how a voice travels to an FM radio and how a doctor could see an organ with sound. In the Observe Phase they use the Khan Academy resource on waves applications (radio, radar, Doppler) and a short data task to see how an FM carrier at 91.0 MHz

has a wavelength of 3.30 m and how a 3.5 MHz ultrasound pulse returns from a boundary 7.7 cm deep after 100 microseconds. In the Explain Phase they use $d = vt/2$ and $v = f \times \text{wavelength}$ to explain both technologies and compare them to the echo timing of Hell's Gate (0.50 s) and Ngong Hills (1.00 s). They then add questions to the DQB and extend the class model with a 'technology' branch showing that the same wave ideas work at very different scales.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Electromagnetic waves and the electromagnetic spectrum Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Radio Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. Mr. Mwangi tells the story: Wanjiru listens to an FM station at 91.0 MHz in Kericho while Kiprono's mother has an ultrasound scan at a county hospital. Wanjiru asks, 'Is the radio using echoes like our shouting at Hell's Gate?' Individually, learners write and sketch predictions for two questions. (1) How does a voice from a studio in Nairobi reach a radio far away, and what do you think a 'station frequency' of 91.0 MHz means? (2) How can a doctor use sound that we cannot hear to find where an organ boundary is inside the body? Learners write one sentence for each and mark how sure they are (1 to 3). They then share with a partner and give a possible link to the Hell's Gate echo (85 m, 0.50 s).	Say: 'Today we leave the cliffs and walk into a radio studio and a hospital. Before I teach anything, I want your best thinking.' Display the two questions. Give 3 minutes of silent writing. WAIT TIME: 10 to 15 seconds of silence after posing each question before any talking. Say: 'Wanjiru's radio plays a voice that was in a studio kilometres away. Does the sound travel through the air all the way? Write what you think.' Cold-call 4 learners (two per question) with: 'Tell us your prediction and one reason, even if you are not sure.' Record predictions on a chart under the headings 'Radio' and 'Ultrasound'. Do not correct. Say: 'I notice many of you think the voice itself travels as sound. We will test that today.' Ask: 'What do the shout at Hell's Gate and a hospital scan have in common?' Cold-call 2 more learners and note any who mention reflection or delay.	Predict-Observe-Explain (POE) with a Prediction Chart: Writing and sharing predictions first surfaces the common misconception that radio carries ordinary sound through air, and the idea that sound at Hell's Gate cannot go through the body. Learners have a record to compare their later evidence with, which makes the change in their thinking visible and gives them a reason to want to explain.	Look for: (1) every learner has written a prediction and a confidence rating; (2) at least 6 of the learners can name a link to echo or reflection; (3) the misconceptions to monitor are 'radio waves are sound waves' and 'ultrasound is just very loud sound'. Tally them on the board. Use a quick show of fingers (1 = not sure, 3 = very sure) to gauge confidence and record the number of 3s to revisit at the end of the lesson.
Observe Phase  VIDEO: Mechanical and Electromagnetic Waves - Overview Source: CK-12  Search ARES for similar videos  EXERCISE:	Part A (about 8 minutes): FM radio. Learners use the Khan Academy resource on waves applications (radio, radar, Doppler), or the printed offline extract, and the teacher's FM radio tuned to a station at moderate volume. Task: (1) record that radio waves are electromagnetic waves, travelling at 3.0×10^8 m/s, and need no medium (build on	Say: 'Scientists, you are data collectors now. Write down numbers with units.' Hold up the FM radio and say: 'This station is 91.0 MHz. M means mega, which is one million hertz. How many waves pass each second?' WAIT TIME: 10 to 15 seconds before taking answers. Circulate and ask: 'Which speed do you use for radio, 340 m/s or 3.0×10^8 m/s? Why?'	Data table comparison and Claim-Evidence notes: Putting three echo situations (a cliff, a valley and an organ boundary) in one table makes learners see that the same relation $d = vt/2$ works across very different scales, and that only the wave speed and the time change. Recording numbers and units first, before explanation, trains learners to base claims on evidence.	Check the worksheets for: wavelength 3.30 m written with units; wavelength 0.44 mm (or 0.00044 m); $t = 100$ microseconds (1.0×10^{-4} s); use of 1540 m/s in tissue and 3.0×10^8 m/s for radio; the statement that in FM the carrier frequency changes while amplitude stays constant. Observable indicator: at least 3 of the 4 groups correctly fill in the

Radio Waves Practice Source: CK-12 Search ARES for similar readings	Lessons 2 and 5); (2) calculate the wavelength of the 91.0 MHz carrier: $\lambda = v/f = 3.0 \times 10^8 / 91.0 \times 10^6 = 3.30 \text{ m}$; (3) look at the drawn graphs of a carrier wave and a voice signal and note that in FM the carrier's frequency is squeezed and stretched by the voice while the height (amplitude) stays the same. Part B (about 10 minutes): Ultrasound. Learners look at the pulse-echo diagram and data card: probe 3.5 MHz, speed in soft tissue 1540 m/s. They calculate the wavelength: $1540 / 3.5 \times 10^6 = 0.00044 \text{ m}$ (0.44 mm). They then use $d = vt/2$ to find the time for an echo from a boundary 7.7 cm (0.077 m) deep: $t = 2 \times 0.077 / 1540 = 1.0 \times 10^{-4} \text{ s}$ (100 microseconds). Groups complete a comparison table of Hell's Gate (85 m, 0.50 s), Ngong Hills (170 m, 1.00 s) and the 7.7 cm boundary (100 microseconds), recording the speed, the distance, the time and what is reflecting.	Cold-call 3 learners. Point to the carrier graph: 'What stays the same here, and what changes?' Cold-call 2 learners. In Part B ask: 'Why do we divide by 2 in $d = vt/2$ just like at Hell's Gate?' WAIT TIME: 10 to 15 seconds. Cold-call 3 learners; look for 'the pulse goes to the boundary and back'. Check powers of ten for groups struggling with 10^6 and 10^{-4}: 'Show me how you entered 3.5×10^6 on your calculator.' Remind learners that the ultrasound speed is 1540 m/s, not 340 m/s. Do not explain the full modulation yet; let the data speak.		table row for the 7.7 cm boundary. Note the groups that divide by 2 correctly and those that forget. Ask any group that has not finished to explain the first row verbally.
Explain Phase  VIDEO: Electromagnetic waves and the electromagnetic spectrum Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Radio Waves Source: PhET Search ARES for similar readings	Learners meet in their groups of four with the four character roles: Wanjiru (asks questions), Otieno (checks the evidence), Kiprono (checks the timing and calculations) and Amina (draws the model). Each group writes a short Claim-Evidence-Reasoning explanation for two statements. (1) 'An FM station sends a voice by changing the frequency of a carrier wave, and the radio wave does not need air.' (2) 'An ultrasound scanner finds the depth of a tissue boundary using the same echo idea as the Hell's Gate cliff.' They explain why the 7.7 cm echo returns in 100 microseconds when the Hell's Gate echo returns in 0.50 s, using the speeds and	Say: 'Use evidence from your table. A claim is not enough, I want a number and a reason.' Display the sentence frames: 'I claim that ... because the data show ... which means ...'. Move among groups and ask: 'If the carrier frequency is 91.0 MHz and I speak into the microphone, what exactly changes in the wave?' WAIT TIME: 10 to 15 seconds. Cold-call 3 groups to share; ask the others to agree or disagree and say why. Ask: 'What is the same between the cliff at Hell's Gate and the organ boundary? What is different?' Cold-call 4 learners. Push with: 'Which wave needs a medium, which does not, and how do we know?' Cold-call 2 learners	Claim-Evidence-Reasoning (CER) with peer critique: Building a claim from data and then reasoning with wave ideas (echo timing, frequency, medium) pushes learners to connect the technology to the same physics as the phenomenon. Peer critique makes weak reasoning visible, such as 'radio is just loud sound', and lets learners repair it themselves before the teacher confirms the science.	Listen for and record: (1) learners say 'frequency of the carrier changes, amplitude stays the same' in FM; (2) learners state that radio waves are electromagnetic and need no medium; (3) learners explain 'divide by 2 because the pulse goes there and back'; (4) learners connect the short wavelength (0.44 mm) to seeing small detail. Success criterion: at least 75% of groups give a CER that includes a number, a unit and a link to the echo or frequency idea. Mark learners who still confuse radio waves and sound waves for a short follow-up conversation before the next lesson.

	distances in their table. They also answer the Lesson 10 link: the Doppler lesson showed frequency changing with motion, and FM shows frequency changing because the sound signal changes it. Finally, each group answers: why can a doctor not use the Hell's Gate shout to scan a body? (The wavelength of audible sound is too long to show small detail, and ultrasound has a wavelength of 0.44 mm, a size suited to small boundaries.) Groups share their reasoning with another group.	and connect to Lessons 2 and 5 (electromagnetic waves, light needing no medium). Correct only after learners have tried: radio is not sound; the voice is converted to an electrical signal that changes the carrier frequency, and a radio converts it back to sound. Ask: 'Why does the scanner need to send short pulses and then wait?' WAIT TIME: 10 to 15 seconds, so learners link this to the 0.1 s echo idea (separate pulses and echoes). Close with: 'So what do the cliff, the Kisumu hall, the radio and the scanner have in common?'		
Driving Question Board (DQB) Creation  VIDEO: Mechanical and Electromagnetic Waves - Overview Source: CK-12  Search ARES for similar videos  HTML: Radio Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the DQB (How do sound waves reflect, and why do some places produce stronger echoes than others?). Each group writes on sticky notes: (a) one question that this lesson has answered (for example, 'How is the echo idea used in medicine?') and sticks it in the 'answered' area with the evidence; (b) one new question that has appeared (for example, 'Could a hall be designed so that the muddy echo disappears?', 'Does ultrasound reflect more from some tissues than others like the surface table?', 'Why do radios in the Rift Valley sometimes lose a station?'). They place the notes and then read the notes of other groups. The class sorts them into the headings 'Reflection', 'Frequency and waves', 'Applications', and 'Noise and design for Lesson 12'.	Say: 'Our board is a record of our thinking. Today, what can we move from questions to answers?' Give 4 minutes to write. Ask: 'Which old question does the ultrasound example help answer? Show me the evidence.' WAIT TIME: 10 to 15 seconds. Cold-call 4 groups to place a note and read it. Ask: 'Does ultrasound reflect equally from every tissue? What did the surface table from Lesson 4 say about different surfaces?' Cold-call 2 learners. Point at the question on the muddy echo in the Kisumu hall: 'Which of today's ideas can help? We will look at that in Lesson 12.' Do not answer new questions, but mark them as 'to investigate'. Say: 'Wanjiru, which of your questions has changed since Hell's Gate?' to model that questions evolve.	Driving Question Board (DQB) tracking and sorting: Writing and categorising their own questions gives learners ownership of the unit and shows how each lesson fits the larger puzzle. Moving questions to 'answered' with evidence shows learning progress, and the unanswered ones guide the next investigation.	Count and review the notes: at least 4 new questions from different groups; at least 2 notes moved to 'answered' with a cited piece of evidence (for example, 'the 100 microsecond calculation'); at least 1 question that links radio or ultrasound back to echoes or reflection from different surfaces. Look for questions that use the key terms (frequency, wavelength, reflection, medium). Note any group that only copies earlier questions and prompt them with 'What puzzled you today?'.
Model Building Phase  VIDEO: Properties of periodic waves	Amina leads each group in revising the class model of the echo. Groups add a 'Technology' branch to the model with two labelled sketches. Sketch 1: a radio station tower sending a	Say: 'Models are tools to explain and predict. Make sure every arrow and label is evidence-based.' Display the class model and ask: 'Where does the ultrasound pulse go, and where	Cumulative Model Building and Revision: Extending the class echo model with technology branches asks learners to apply the same mechanism to new contexts, which shows transfer of understanding.	Check each model for: (1) FM sketch showing frequency variation with constant amplitude, the 3.30 m wavelength and no medium; (2) ultrasound sketch with the 7.7 cm boundary, 1540 m/s

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Radio Waves Practice Source: CK-12 Search ARES for similar readings	carrier at 91.0 MHz (wavelength 3.30 m) with a note 'voice changes the frequency, not the height; radio waves do not need a medium; speed 3.0×10^8 m/s'. Sketch 2: an ultrasound probe (3.5 MHz, wavelength 0.44 mm, speed 1540 m/s) sending a pulse to a boundary 7.7 cm deep and receiving an echo after 100 microseconds, using $d = vt/2$. Each sketch has an arrow back to the central model of Hell's Gate (0.50 s), Ngong Hills (1.00 s), the Kisumu hall (0.07 s, reverberation) and the Nairobi open grounds (no echo). Groups add a sentence: 'The same ideas of reflection, timing and frequency work at every scale.' They complete the summary table (what we observed, what we learned, how we explain it) and note one thing in the model still unexplained, to be handled in Lesson 12.	does it come back from? Point on your sketch.' WAIT TIME: 10 to 15 seconds. Cold-call 3 groups to explain a sketch to the class. Ask: 'Does the ultrasound echo model use the same formula as the cliff? What changes?' Cold-call 2 learners (speed and time). Ask: 'Which part of the model explains why the Kisumu hall is muddy, and which part do we still need?' Cold-call 2 learners and record 'Lesson 12: noise and absorbing treatment'. Check that radio waves are not drawn as sound and the FM sketch shows the frequency changing. Finish with an exit slip: 'Write one similarity and one difference between the Hell's Gate echo and an ultrasound echo.' Collect the slips.	Labelled arrows to the four sites force them to connect the wave ideas to the original phenomenon, and the 'still unexplained' note maintains a purpose for the final lesson.	and 100 microseconds; (3) $d = vt/2$ written with 'divide by 2 because there and back'; (4) at least one arrow linking back to a Hell's Gate, Ngong, Kisumu or Nairobi site. Exit slip success: at least 80% of learners state a correct similarity (both use reflection and timing, $d = vt/2$) and a correct difference (speed, scale or wave type). Learners who miss these get a short re-teach at the start of Lesson 12.
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D. TEACHER REFLECTION

After the lesson ask yourself: Did learners distinguish radio waves (electromagnetic, no medium, 3.0×10^8 m/s) from sound waves (mechanical, need a medium, 340 m/s in air and 1540 m/s in tissue)? Which learners still say that radio carries the sound itself? Could they use $d = vt/2$ correctly, including the division by 2 and the powers of ten for 100 microseconds? Did the table linking Hell's Gate, Ngong Hills and the 7.7 cm boundary help them see the same idea at different scales? Which groups needed the most prompting to give reasoning rather than only a claim? Did the DQB questions show curiosity about the Kisumu hall, which will be useful in Lesson 12? Note adjustments for pacing (the calculations with powers of ten may need extra time) and plan support for learners who need it before the final lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the 91.0 MHz FM carrier is a radio wave of wavelength 3.30 m travelling at 3.0×10^8 m/s, and that in FM the voice changes the frequency of the carrier while its amplitude stays constant. We observed that a 3.5 MHz ultrasound pulse in soft tissue (1540 m/s) has a wavelength of 0.44 mm and that an organ boundary 7.7 cm deep returns an echo after 100 microseconds.
What did I learn?	We learned that radio waves are electromagnetic waves that need no medium, that FM carries sound by changing the carrier frequency, and that ultrasound imaging uses reflected pulses and echo timing with $d = vt/2$ to locate tissue boundaries. We learned that the echo idea from Hell's Gate (0.50 s) and Ngong Hills (1.00 s) works at both large and tiny scales.
How does this explain the	We explain that the same wave ideas explain the phenomenon and the technologies: a wave travels, meets a boundary, reflects,

phenomenon?	and returns after a time equal to $2d/v$, so the delay tells us the distance. In the radio, the frequency of the wave carries the information. This shows how wave properties are applied in day-to-day life, and we will use all these ideas in Lesson 12 to give the final explanation of the four echo sites.
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LESSON 12 (80 minutes (2 x 40-minute periods)): Noise Pollution and the Final Model: Why Do Some Places Echo and Others Do Not?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners analyse sources and health effects of noise, design noise-reduction treatments for the Kisumu hall, and use all wave ideas from Lessons 1 to 11 to build and present a final model explaining echoes at Hell's Gate, Ngong Hills, the Kisumu hall and the Nairobi open grounds.
Knowledge	Noise levels in decibels (whisper 30 dB, quiet classroom 50 dB, conversation 60 dB, Kisumu hall during assembly 85 dB, matatu with loud music 95 dB; safe limit used in class 85 dB for up to 8 hours). Hard smooth surfaces reflect most sound energy (cliff rock 90%, painted concrete 85%, wooden board 60%) while soft surfaces absorb it (curtains 15%, thick grass or foam mattress 10% reflected). Echo means a distinct reflected sound at least 0.1 s later (reflector at least 17 m away), and $d = vt/2$ with $v = 340$ m/s. Reverberation means overlapping reflections that merge and prolong the sound. All four echo sites A to D can be explained by reflection, delay, distance and absorption. Other wave behaviours (refraction, diffraction, interference, resonance, Doppler) follow the same wave ideas.
Skills	Developing and using models (NGSS SEP 2): compare the Lesson 1 model with the final model. Constructing explanations (SEP 6): write a claim-evidence-reasoning final explanation. Analysing and interpreting data (SEP 4): use the decibel list, echo table and surface table. Designing solutions (SEP 6 engineering): propose absorbing treatment for the Kisumu hall. Communication and collaboration: present and critique models.
Attitudes	Responsibility: care for hearing health and the community's noise environment. Respect: listen to and fairly critique peers' models and designs. Curiosity and appreciation of the usefulness of Physics in day-to-day life.
Key Inquiry Question	1. How do waves behave when they meet obstacles or change medium? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	This is the final lesson. Learners apply what they learned across Lessons 1 to 11, especially echo distance and surfaces (Lesson 4), interference (Lesson 8), resonance (Lesson 9) and reflection (Lesson 3), to a real problem (noise in the Kisumu hall). They then close the Driving Question Board (DQB) and compare their Lesson 1 first model with a final model that fully answers: How do sound waves reflect, and why do some places produce stronger echoes than others?
Safety Notes	Do not play loud sounds in class; use a phone or speaker at normal volume only. If a sound level app is used, measure only ordinary classroom sounds, never produce loud noise on purpose. Keep movement around the room orderly while learners post cards on the DQB and carry foam, cloth and wooden board samples.

B. LESSON OVERVIEW

Mr. Mwangi opens with Otieno's question from the trip: why was the Kisumu hall so loud and muggy when it was only 12 m from the far wall? Learners first predict which noise sources are dangerous using the decibel list, then analyse sources and health effects, test how much different materials absorb sound (using the surface table: cliff rock 90%, painted concrete 85%, wooden board 60%, curtains 15%, foam or grass 10% reflected) and design an absorbing treatment for the Kisumu hall. In the second half, learners revisit the Driving Question Board, sort evidence into answered, partly answered and new questions, and build a final model of all four echo sites (A Hell's Gate 0.50 s, B Ngong 1.00 s, C Kisumu 0.07 s with about 1.2 s reverberation, D Nairobi 1.76 s but too faint). Amina's Lesson 1 model is compared with the final model, and every learner writes a Final Explanation using claim, evidence and reasoning. The Final Explanation: waves are disturbances that transfer energy without transferring matter; sound reflects from hard smooth surfaces following the law of reflection, returns after $t = 2d/v$, and is heard as an echo only when the reflector is at least 17 m away and reflects enough energy.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Wave Interference <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners look at five cards showing: whisper, quiet classroom, ordinary conversation, Kisumu hall during assembly, and matatu with loud music. Individually, in their notebooks, they predict (a) the order from quietest to loudest, (b) which sounds could damage hearing, and (c) why the Kisumu hall during assembly is so loud even though it is a school. They write a number guess in dB for each card without looking at any list. Then in pairs they predict: 'If we hung curtains and put foam on the Kisumu hall walls, what would happen to the muddy sound and why?' Predictions are connected to Otieno's observation at the Kisumu hall (site C) where the voice was loud, muddy and drawn out.	Say: 'Remember Otieno in the Kisumu hall? The wall was only 12 m away, yet the sound was loud and muddy. Today we decide how to fix that and then finish our big answer.' Hand out five cards and say: 'Rank them and guess the decibels. Do not talk yet, write for 3 minutes.' After writing, WAIT TIME 10-15 seconds before asking for any answer. Cold-call 4 learners with: 'Which source do you think is most harmful? What is your reason?' Cold-call 2 more: 'What will curtains and foam do to the sound in the hall?' Record the predictions on the board without judging. Ask: 'What do you already know from Lesson 4 about hard and soft surfaces that could help you predict?' WAIT TIME 10 seconds.	Predict-Observe-Explain (POE) with individual commitment: writing a prediction before seeing data makes learners' prior ideas visible, so they notice later where evidence confirms or changes their thinking. Linking the prediction to site C connects noise to reflection and reverberation from earlier lessons.	Observe whether learners (1) rank matatu with loud music above conversation, (2) mention loudness or amplitude and not only pitch, (3) say that soft materials will reduce the reflected sound. Tally on a clipboard how many learners predict that soft materials absorb sound (target: more than half). Note the misconception that thicker walls always stop noise, or that echoes and noise are the same thing, for follow-up in the Explain Phase.
Observe Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML:	Part 1 (decibel data): In groups of four (one Wanjiru-style questioner, one Kiprono-style timekeeper, one Amina-style recorder, one Otieno-style reporter), learners get the decibel list: whisper 30 dB, quiet classroom 50 dB, ordinary conversation 60 dB, Kisumu hall during assembly 85 dB, matatu with loud music 95 dB, class safe limit 85 dB for up to 8 hours. They draw a bar chart, mark the 85 dB line, identify which sources are at or above it, and list noise sources they meet daily (matatus along Kisumu roads, market	Say: 'You are acoustic engineers for a Kisumu school. First, let us read the noise data, then test materials.' Circulate and ask each group: 'Which source is above the 85 dB limit? What does that mean for a matatu driver or a conductor who works all day?' Push for the idea that dose depends on loudness and time: 'Why is 85 dB allowed for 8 hours but 95 dB is risky?' WAIT TIME 10-15 seconds after each question. For the absorption test, say: 'Keep the distance and volume the same. Why is that important?' (fair test,	Data analysis with a fair-test investigation (SEP 3 and 4): learners combine numerical decibel data with a hands-on comparison of materials, then check their results against the surface table. Comparing hard and soft surfaces turns the abstract idea of 'percentage of energy reflected' into something they can hear.	Check that bar charts have the 85 dB line and correctly identify the Kisumu hall (85 dB, at the limit) and matatu with loud music (95 dB, above the limit) as hazards. Look for groups that keep distance and volume constant. Listen for explanations that use the idea of energy being absorbed (not just 'cloth blocks sound'). Check that learners calculate 0.07 s correctly and state that it is less than 0.1 s. Indicator of success: at least 80% of groups rank wooden board above cloth and foam in reflection, matching the surface table.

Wave Interference (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	loudspeakers, generators, boda boda horns, school assemblies). They read a short card on health effects: hearing damage when exposure is above the limit, poor concentration, headaches, stress and disturbed sleep. Part 2 (absorption investigation): Using the surface table (approximate share of sound energy reflected), groups examine samples (wooden board, cloth or curtain, foam mattress piece, and a smooth painted board or wall surface). They hold a phone playing a steady ordinary-volume tone or a ticking clock near each sample at the same distance, listen to or measure the sound reflected, and rank the samples from strongest reflection to weakest. They compare their ranking with the table values: cliff rock 90%, painted concrete 85%, wooden board 60%, curtains/cloth 15%, thick grass/foam mattress 10%. Part 3 (calculation link): groups recompute for site C: round trip 24 m, $t = 2d/v = 24/340 = 0.07$ s, which is less than 0.1 s, so the echo merges with the voice, and reflections from hard walls and roof keep the sound lingering about 1.2 s (reverberation).	control variables). Cold-call 3 groups to report their ranking: 'Which material reflected most? Which least? What evidence did you hear or measure?' Then ask: 'In site C, why is a 0.07 s delay not heard as a separate echo?' WAIT TIME 10 seconds, cold-call 2 learners. Remind learners to keep all sounds at ordinary classroom volume.		
Explain Phase  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos	Learners design a noise-reduction treatment for the Kisumu hall in their groups. They must (1) state the problem using evidence (12 m to the far wall, 0.07 s delay, reverberation about 1.2 s, 85 dB at assembly), (2) choose two or three materials and where to place them (curtains at windows, foam or acoustic panels on the back wall, a carpet or mats on the floor, wooden panels or ceiling tiles), (3) justify each choice using the	Say: 'Design like an engineer: every choice needs evidence from our tables.' Provide a design template with the five required parts. Circulate and ask: 'Why curtains and not another wooden board? What percentage of sound does each reflect?' WAIT TIME 10-15 seconds. Ask: 'If you remove 85% reflection from the walls, what happens to the reverberation time of about 1.2 s?' Cold-call 3 groups for their reasoning. Say:	Engineering design with Claim-Evidence-Reasoning (CER): learners propose a solution (claim), cite surface and decibel data (evidence) and explain with wave ideas of reflection, absorption and overlap (reasoning). Star-and-wish peer critique builds understanding by asking learners to evaluate other designs against the same evidence.	Use a four-point checklist: (1) problem stated with correct numbers (12 m, 0.07 s, about 1.2 s, 85 dB), (2) materials justified with percentages, (3) reverberation explained as repeated reflections reduced by absorption, (4) at least one link to interference, resonance or reflection. Look for the misconception that absorbing material removes the 0.07 s delay itself. Exit signal: learners hold up 1 to 4 fingers for how many

 EXERCISE: Wave Interference Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	surface table (for example curtains reflect only 15%, foam 10%, compared to painted concrete 85%), (4) explain how the treatment reduces reverberation by reducing repeated reflections and sound energy, and (5) propose one barrier or planting solution for the outside noise from a nearby road (a thick wall, a row of trees and shrubs, or moving the hall windows away from the matatu stage), explaining that barriers reflect or absorb sound before it reaches the hall. They then extend to other wave ideas. Using two sentences each, they link: interference (Lesson 8), since the overlapping reflections in the Kisumu hall meet and add; resonance (Lesson 9), since some hall frequencies can be amplified when the room or objects vibrate at their natural frequency; and reflection (Lesson 3), since the law of reflection explains the direction of the returning sound. Each group presents its design in a 2-minute gallery-style share, and the class gives one star (strength) and one wish (improvement).	'Absorbing material reduces reverberation. It does not remove the sound from the speaker, so the hall is still audible, only clearer.' Challenge a common misconception: 'Does absorbing material make a place quieter and also remove echo?' Clarify: in a hall of 12 m the echo was never a separate echo (0.07 s), it was reverberation. During the gallery share, cold-call 4 learners from different groups to give a star and a wish. Highlight any design that links absorption to reducing reflected energy. Make links explicit: 'Lesson 8 told us waves overlap and add. Lesson 3 told us reflection follows the law. Lesson 4 told us hard surfaces reflect strongly.'		checklist items their design meets. Target: at least 75% show 3 or more.
Driving Question Board (DQB) Creation  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos	Final DQB review. The DQB, which has grown from Lesson 1 with questions such as Wanjiru's 'Why does the voice come back?', Otieno's 'Why is the hall echo muddy?' and Kiprono's 'How can we measure the delay?', is revisited. Groups receive their question cards and sort them into three columns: Answered (with evidence and lesson number), Partly Answered, and New Questions. Examples: 'Why is there a delay?' is answered by Lesson 4: $t = 2d/v$, so site A is $170/340 = 0.50$ s and site B is	Say: 'Today we close our board. For each question, show the evidence that answers it and name the lesson.' Give each group 8 to 10 cards. WAIT TIME 10-15 seconds before groups begin sorting. Circulate and ask: 'Which lesson gave you this answer? What number or observation supports it?' Cold-call 5 learners to place one card and defend its column. Challenge: 'Which card is still only partly answered, and what evidence is missing?' Ask Wanjiru-type questioners to read their original question and say whether	Question sorting with evidence mapping on the Driving Question Board: linking each question to a specific lesson and data point makes progress visible and shows learners that understanding is built from accumulated evidence. Adding new questions promotes continued inquiry beyond the unit.	Check that learners cite a lesson and a piece of evidence for at least 80% of 'Answered' cards. Listen for accurate use of $d = vt/2$, the 17 m minimum distance, 0.1 s rule, surface percentages and the difference between echo and reverberation. Note whether new questions are testable and connected to waves (for example about noise control, animals or technology). Look for learners who still place the Kisumu hall under 'echo' rather than 'reverberation' and address this immediately.

 HTML: Reflection of Mechanical Waves (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	340/340 = 1.00 s. 'Why no echo at Nairobi grounds (site D)?' is answered by spreading, absorption by grass and bushes, and the 300 m wall giving a 1.76 s delay but a very faint sound. 'Why does sound turn corners but light does not?' is answered by diffraction (Lesson 7). Each learner writes one new question for further inquiry, such as 'How do bats use echoes?', 'How does noise-cancelling work in a matatu?' or 'Can hearing damage from noise be reversed?' The class reads the main driving question aloud and decides how fully it is now answered.	they are satisfied. Ask each learner to add one new question card. Say: 'Good scientists finish with new questions.' Photograph or note the final DQB for the class record.		
Model Building Phase  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Wave Interference <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners build the Final Model and Final Explanation. Step 1: Each group takes Amina's Lesson 1 first model (a simple sketch of a shout going to a cliff and back with the question 'Why does the voice come back?') and lists what it explained and what it missed. Step 2: Groups draw a final model on a large sheet showing all four sites A to D. Each site shows the learner, the straight ray path to the reflector with the angle of incidence i equal to the angle of reflection r measured from the normal, distance d, round trip $2d$ and the delay $t = 2d/340$. Site A Hell's Gate: 85 m, 170 m round trip, 0.50 s, clear echo, cliff rock reflects about 90%. Site B Ngong Hills: 170 m, 340 m round trip, 1.00 s, clear echo. Site C Kisumu hall: 12 m, 24 m round trip, 0.07 s, less than 0.1 s, so the echo merges with the voice, and hard walls and roof keep reflecting, giving reverberation of about 1.2 s, loud, muddy and drawn out, with an 85 dB assembly level. Site D Nairobi	Say: 'Pull out our first model from Lesson 1. What could it not explain?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Give model criteria: (1) all four sites with numbers, (2) the law of reflection with the normal, (3) delay equals $2d/v$, (4) the 17 m and 0.1 s condition, (5) surface absorption and spreading, (6) reverberation versus echo, (7) at least four other wave behaviours. While groups work, ask: 'Why is site D silent even though its delay is the longest? Why is site C not an echo even though the walls reflect well?' WAIT TIME 10 seconds. Cold-call 2 learners per question. Challenge: 'Does a longer delay mean a stronger echo?' (No: strength depends on the energy reflected, which depends on the surface and spreading.) Lead the gallery walk: groups leave one star and one question sticky note on two other models. Then bring the class together and ask 5 cold-called learners to state the biggest	Model comparison and revision (SEP 2) with Claim-Evidence-Reasoning (SEP 6): learners contrast the Lesson 1 model with the final model to see how evidence, calculation and wave ideas have strengthened their understanding. Writing an individual final explanation consolidates learning and applies the same principles to all four sites and to other wave behaviours.	Use a rubric with four levels. Level 4: all four sites correctly modelled with $d = vt/2$ numbers; explains echo versus reverberation; links strength of echo to reflection percentage, absorption and wave spreading; includes the law of reflection and at least four other wave behaviours; proposes noise reduction with evidence. Level 3: three of these. Level 2: describes reflection and delay but ignores absorption or the 17 m condition. Level 1: describes only that sound bounces. Look for misconceptions: 'longer delay means stronger echo', 'echo and reverberation are the same', 'sound travels faster in a hall', 'grass stops sound because it is soft only, without energy absorption'. Target: at least 80% at Level 3 or 4. Peer swap suggestions are recorded as evidence of revision.

	open grounds: nearest large hard wall (classroom block) 300 m away, 600 m round trip, 1.76 s, but the sound spreads over a huge area and the grass and bushes absorb and scatter it (grass about 10% reflected), so the returning sound is too faint, no echo. Step 3: Each group adds a panel linking other wave behaviours to the same ideas: refraction (waves change direction when speed changes), diffraction (sound with wavelength 1.0 m at 340 Hz spreads through a 1.0 m doorway), interference (path difference 1.0 m gives loud, 0.5 m gives quiet), resonance (nyatiti 100 Hz, flute 500 Hz), Doppler (ambulance 744 Hz approaching, 661 Hz receding) and applications (FM 91.0 MHz with wavelength 3.30 m, ultrasound 3.5 MHz with 100 microsecond echo). Step 4: Gallery walk and comparison of the Lesson 1 model with the final model. Step 5: Each learner writes an individual Final Explanation using claim, evidence and reasoning, beginning: 'Waves are disturbances that transfer energy from one point to another without transferring matter.' It must answer: How do sound waves reflect, and why do some places produce stronger echoes than others? and finish with one real-life use of waves and one way to reduce noise.	change between the Lesson 1 model and the final model. Finish with: 'Write your Final Explanation alone, in about 10 sentences, then swap with a partner for one improvement.' Collect explanations as the summative evidence for the sub-strand.		
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D. TEACHER REFLECTION

After the lesson, review the Final Explanations and the revised models. Did learners use $d = vt/2$ correctly for all four sites (0.50 s, 1.00 s, 0.07 s, 1.76 s) and state that 0.07 s is less than 0.1 s so site C is reverberation not a separate echo? Did they explain site D by spreading and absorption rather than distance alone? Which learners still confuse loudness (amplitude, dB) with pitch (frequency)? Did the noise-reduction designs cite the surface table with evidence? Which DQB questions remain unanswered, and how will you use them in future lessons or a school project on a quieter school? Consider what worked in the model comparison with the Lesson 1

sketch, which scaffolds helped quieter learners, and how to link the noise health message to the community, such as matatu noise on Kisumu and Nairobi roads. Note which groups struggled with time management between design and model building and plan how to adjust the two-period schedule.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that noise levels range from 30 dB (whisper) to 95 dB (matatu with loud music), with 85 dB as the class safe limit for up to 8 hours. We also observed that hard materials such as painted concrete (85% reflected) and wooden board (60%) reflect sound strongly, while curtains (15%) and foam or grass (10%) reflect very little. At the four sites, we saw that Hell's Gate (0.50 s) and Ngong Hills (1.00 s) give clear echoes, the Kisumu hall (0.07 s) gives a loud muddy sound, and the Nairobi open grounds give no echo.
What did I learn?	We learned that noise above safe limits can damage hearing and cause stress and poor concentration, and that absorbing materials and barriers reduce noise and reverberation. We learned that an echo needs a reflector at least 17 m away (delay of at least 0.1 s) and a surface that reflects enough energy, that $d = vt/2$ with $v = 340$ m/s, and that reverberation is overlapping reflections that merge and prolong the sound. We also learned how reflection, refraction, diffraction, interference, resonance and the Doppler effect all follow the same wave ideas.
How does this explain the phenomenon?	Sound is a mechanical wave that leaves the shouting learner, reflects from a large hard smooth surface following the law of reflection ($i = r$ from the normal) and returns after $t = 2d/v$. At Hell's Gate and Ngong Hills the surface is far away (more than 17 m) and hard, so a strong separate echo is heard. In the Kisumu hall the wall is only 12 m away (0.07 s) with many hard surfaces, so reflections overlap the voice, causing loud, muddy reverberation of about 1.2 s, which curtains and foam can reduce by absorbing energy. On the open Nairobi grounds the sound spreads out and is absorbed by grass, and the only big wall is 300 m away (1.76 s), so the returning sound is too faint to hear.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.